Global **Ballast Water** Management Programme

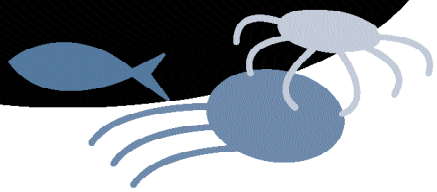
GLOBALLAST MONOGRAPH SERIES NO.16

Phytoplankton
Atlas

DECEMBER 2004

Sepetiba Bay, Brazil

D.R. Tenenbaum, M.C. Villac, S.C. Viana, M. Matos, M. Hatherly, I.V. Lima & M. Menezes



GEF UNDP IMO

GloBallast Monograph Series No.16

Phytoplankton Atlas of Sepetiba Bay, Rio de Janeiro, Brazil

Tenenbaum, D.R.[1], Villac, M.C.[2], Viana, S.C.[3], Matos, M.[4], Hatherly, M.[5], Lima, I.V.[6], Menezes, M.[7]

[1] denise@biologia.ufrj.br; [2] mcvillac@uol.com.br; [3] scviana@yahoo.com.br; [4] ema.klassen-vasconcellos@yahoo.com.br; [5] mabuvas@globo.com.br; [6] isabel.vdl@yahoo.com.br; [7] mmenezes@superig.com.br

© International Maritime Organization
ISSN 1680-3078

Published in December 2004 by the
Programme Coordination Unit
Global Ballast Water Management Programme
International Maritime Organization
4 Albert Embankment, London SE1 7SR, UK
Tel: +44 (0)20 7587 3251
Fax: +44 (0)20 7587 3261
Email: anagnostaki@imo.org
Web: http://globallast.imo.org

The correct citation of this report is:
Terenbaum, D.R., Villac, M.C., Viana, S.C., Matos, M., Hatherly, M., Lima, I.V., Menezes, M. 2004.
Phytoplankton Atlas of Sepetiba Bay, Rio de Janeiro, Brazil. GloBallast Monograph Series No.16 IMO London.

The Global Ballast Water Management Programme (GloBallast) is a cooperative initiative of the Global Environment Facility (GEF), United Nations Development Programme (UNDP) and International Maritime Organization (IMO) to assist developing countries to reduce the transfer of harmful organisms in ships' ballast water.

The GloBallast Monograph Series is published to disseminate information about and results from the programme, as part of the programme's global information clearing-house function.
The opinions expressed in this document are not necessarily those of GEF, UNDP or IMO.

Acknowledgements

The Phytoplankton Atlas of Sepetiba Bay, Brazil, is an outcome of the Global Ballast Water Management Programme (GloBallast). This programme is an initiative of the International Maritime Organization (IMO), with funding provided by the Global Environment Facility (GEF) through the United Nations Development Programme (UNDP). Its goal is to reduce the transfer of harmful marine species in ships' ballast water by assisting developing countries to implement existing IMO voluntary guidelines on ballast water management (IMO Assembly Resolution A.868(20)), and to prepare for the new International Convention for the Control and Management of Ships' Ballast Water and Sediments adopted by the Diplomatic Conference in February 2004.

The programme aims to provide technical assistance, capacity building and institutional strengthening to remove barriers to ballast water management arrangements in developing countries through six initial demonstration sites: Sepetiba (Brazil), Dalian (China), Mumbai (India), Khang Island (Iran), Saldanha (South Africa) and Odessa (Ukraine). These sites are intended to be representative of the six main developing regions of the world: South America, Asia/Pacific, South Asia, ROPME Sea area & Red Sea, Africa and Eastern Europe. As the programme proceeds, it is intended to replicate these initial demonstration sites throughout each region. The GloBallast Programme will carry on its activities from March 2000 to December 2004.

One of the activities implemented by GloBallast at each demonstration site is the assessment of local marine biodiversity in and around the port, across all habitat types and for all taxonomic groups. This survey, based on standardized methodologies, provides baseline data on the presence and distribution of native and invasive aquatic species. This allows any existing invasion to be monitored and managed, any new invasion to be detected and responded to, and also assists port States to notify outbreaks of harmful species.

In Brazil, the Lead Agency for the GloBallast Programme is the Secretaria de Qualidade Ambiental nos Assentamentos Humanos of the Ministério do Meio Ambiente (SQA-MMA) and the coordination of the port survey is carried out by the Instituto de Estudos do Mar Almirante Paulo Moreira (IEAPM). In this regard, we are indebted to Robson José Calixto (SQA-MMA) and Flavio da Costa Fernandes (IEAPM) for their support. Special thanks are also due to Alexandre de Carvalho Leal Neto, the Country Focal Point Assistant, for his sterling efforts to provide us the means to carry out this task.

The survey of the phytoplankton, carried out in November 2001 and April 2002, was land-based in the city of Itacuruçá which was, for our sampling purposes, strategically located on the coast of Sepetiba Bay, State of Rio de Janeiro. We are thankful to the Port Captaincy of Itacuruçá and the Yacht Club of Itacuruçá for kindly providing fast motor boats, at no cost, for our sampling. The phytoplankton group also used the laboratory and housing facilities of the marine station of the Universidade Federal Rural do Rio de Janeiro (UFRRJ) in Itacuruçá. Sample analysis was done at the Instituto de Biologia (IB) and the Museu Nacional (MN), both in the Universidade Federal do Rio de Janeiro (UFRJ).

The opportunity to create this Phytoplankton Atlas was made possible by the dedication and initial work of Simone de Castro Viana and Maria Gouveia Matos, who used part of this data set in their undergraduate final research reports. The countless hours they spent doing microscopy work and photographing organisms is at the core of this publication.

Part of the light microscopy work was done at the Laboratório de Captação de Imagens/Projeto Banco Multimídia (IB/UFRJ) and at the Laboratório de Neurogênese (IBCCF/UFRJ). We are grateful to the Laboratório de Ultra-estrutura Celular Hertha Meyer (Instituto de Biofísica/UFRJ) that kindly allowed the use of their scanning electron microscope with the skillful assistance of Noemia Rodrigues.

Although the port baseline survey is mostly a qualitative assessment, we took the opportunity to evaluate some environmental variables (water temperature, salinity, nutrients) with the support and resources provided by Rodolfo Paranhos (Laboratório de Hidrobiologia, IB/UFRJ).

The layout, formatting and image enhancing of the figures and plates were done by Carlos Artêncio (CR2 Design). The line drawings were made by Benjamin Paul Shaw and the maps were plotted by Gustavo de M. Figueiredo. These are "final touches" that require a great deal of skill — they add precision and beauty to this work.

This work was funded by GEF through the Globallast Programme.

CONTENTS

Acknowledgements i

1 Introduction 1

2 Methods 3

3 Results and Discussion 5

References 7

Plates 8

Morphometrics Guide 9

DIVISION BACILLARIOPHYTA

Class Coscinodiscophyceae

Order Thalassiosirales

Family Thalassiosiraceae

Thalassiosira punctigera Plate 1 12

Thalassiosira rotula Plate 2 13

Family Skeletonemataceae

Detonula pumila Plate 3 14

Skeletonema costatum Plate 4 15

Family Stephanodiscaceae

Cyclotella litoralis Plate 5 16

Cyclotella stylorum Plate 6 17

Order Melosirales

Family Stephanopyxidaceae

Stephanopyxis turris Plate 7 18

Order Paraliales

Family Paraliaceae

Paralia sulcata Plate 8 19

Order Coscinodiscales

Family Coscinodiscaceae

Coscinodiscus gigas Plate 9 20

Coscinodiscus granii Plate 10 21

Coscinodiscus cf. *asteromphalus* Plate 11 22

Coscinodiscus wailesii Plate 12 23

Family Heliopeltaceae

Actinoptychus senarius Plate 13 24

Order Asterolamprales

Family Asterolampraceae

Asteromphalus cf. *flabellatus* Plate 14 25

Order Triceratiales

Family Triceratiaceae

Odontella mobiliensis Plate 15 26

Odontella sinensis Plate 16 27

Triceratium favus Plate 17 28

Order Hemiaulales

Family Hemiaulaceae

Cerataulina pelagica Plate 18 29

Climacodium frauenfeldianum Plate 19 30

Eucampia cornuta Plate 20 31

Eucampia zodiacus Plate 21 32

Hemiaulus hauckii Plate 22 33

Hemiaulus membranaceus Plate 23 34

Hemiaulus sinensis Plate 24 35

Order Lithodesmiales
Family Lithodesmiaceae
Ditylum brightwellii Plate 25....36
Helicotheca tamesis Plate 26....37
Lithodesmium undulatum Plate 27....38
Order Rhizosoleniales
Family Rhizosoleniaceae
Dactyliosolen fragilissimus Plate 28....39
Guinardia delicatula Plate 29....40
Guinardia flaccida Plate 30....41
Guinardia striata Plate 31....42
Proboscia alata Plate 32....43
Pseudosolenia calcar-avis Plate 33....44
Rhizosolenia cf. *castracanei* Plate 34....45
Rhizosolenia hebetata Plate 35....46
Rhizosolenia hyalina Plate 36....47
Rhizosolenia pungens Plate 37....48
Rhizosolenia robusta Plate 38....49
Rhizosolenia setigera Plate 39....50
Rhizosolenia cf. *stylitformis* Plate 40....51
Order Chaetocerales
Family Chaetocerotaceae
Bacteriastrum cf. *delicatulum* Plate 41....52
Bacteriastrum hyalinum Plate 42....53
Chaetoceros aequatorialis Plate 43....54
Chaetoceros affinis Plate 44....55
Chaetoceros cf. *brevis* Plate 45....56
Chaetoceros coarctatus Plate 46....57
Chaetoceros cf. *concavicornis* Plate 47....58
Chaetoceros curvisetus Plate 48....59
Chaetoceros danicus Plate 49....60
Chaetoceros decipiens Plate 50....61
Chaetoceros didymus Plate 51....62
Chaetoceros peruvianus Plate 52....63
Chaetoceros sp.1 Plate 53....64
Order Leptocylindrales
Family Leptocylindraceae
Leptocylindrus danicus Plate 54....65
Leptocylindrus minimus Plate 55....66
Class Fragilariophyceae
Order Fragilariales
Family Fragilariaceae
Asterionellopsis glacialis Plate 56....67
Order Rhaphoneidales
Family Rhaphoneidaceae
Delphineis sp. Plate 57....68
Order Thalassionematales
Family Thalassionemataceae
Licolomia pacifica Plate 58....69
Thalassionema nitzschioides Plate 59....70
Thalassionemataceae 1 Plate 60....71

Class Bacillariophycaeae
Order Naviculales
Family Naviculaceae
Haslea wawrikae Plate 61....72
Haslea sp. 1 Plate 62....73
Meuniera membranacea Plate 63....74
Family Pleurosigmataceae
Pleurosigma angulatum Plate 64....75
Pleurosigma diverse-striatum Plate 64....75
Pleurosigma cf. *strigosum* Plate 64....75
Pleurosigma sp. 1 Plate 65....76
Pleurosigma sp. 2 Plate 65....76
Pleurosigma sp. 3 Plate 65....76
Pleurosigma sp. 4 Plate 66....77
Pleurosigmataceae 1 Plate 66....77
Pleurosigmataceae 2 Plate 66....77
Donkinia sp. Plate 67....78
Gyrosigma balticum Plate 67....78
Order Thalassiosiphysales
Family Catenulaceae
Amphora cf. *arenaria* Plate 68....79
Order Bacillariales
Family Bacillariaceae
Bacillaria paxillifera Plate 69....80
Cylindrotheca closterium Plate 70....81
Fragilariopsis delidus Plate 71....82
Nitzschia constricta Plate 72....83
Nitzschia lorenziana var. *subtilis* Plate 73....84
Nitzschia morphotype *nitzschiella* Plate 74....85
Pseudo-nitzschia "delicatissima" sp.1 Plate 75....86
Pseudo-nitzschia "delicatissima" sp.2 Plate 75....86
Pseudo-nitzschia "seriata" sp.1 Plate 76....87
Pseudo-nitzschia "seriata" sp.2 Plate 76....87
Pseudo-nitzschia "seriata" sp.3 Plate 76....87
Pseudo-nitzschia pungens Plate 77....88

DIVISION DINOFLAGELLATA
Class Dinophyceae
Order Gonyaulacales
Family Gonyaulacaceae
Gonyaulax spinifera Plate 78....92
Family Ceratiaceae
Ceratium breve Plate 79....93
Ceratium furca Plate 80....94
Ceratium fusus Plate 81....95
Ceratium gibberum Plate 82....96
Ceratium horridum Plate 83....97
Ceratium macilence Plate 84....98
Ceratium cf. *trichoceros* Plate 85....99
Family Goniodomaceae
Alexandrium cf. *tamarense* Plate 86....100

Order Peridiniales
Family Peridiniaceae
Scrippsiella spinifera Plate 87 101
Scrippsiella trochoidea Plate 88 102
Family Congruentidiaceae
Protoperidinium divergens Plate 89 103
Protoperidinium leonis Plate 90 104
Protoperidinium mariaelebourae Plate 91 105
Protoperidinium oblongum Plate 92 106
Protoperidinium pentagonum Plate 93 107
Protoperidinium cf. *pentagonum* Plate 94 108
Protoperidinium sinulum Plate 95 109
Protoperidinium steinii Plate 96 110
Protoperidinium symmetricum Plate 97 111
Order Dinophysiales
Family Dinophysiaceae
Dinophysis acuminata Plate 98 112
Dinophysis caudata Plate 99 113
Order Prorocentrales
Family Prorocentraceae
Prorocentrum gracile Plate 100 114
Prorocentrum mexicanum Plate 101 115
Class Noctiluciphyceae
Order Noctilucales
Family Noctilucaceae
Noctiluca scintillans Plate 102 116
DIVISION CHROMOPHYTA
Class Dictyochophyceae
Order Dictyochales
Family Dictyochaceae
Dictyocha fibula Plate 103 120

Appendix 1: Complete list of the phytoplankton taxa found in Sepetiba Bay 122

1. Introduction

The knowledge about the occurrence, distribution and abundance of organisms in port areas is a prerequisite for any attempt to manage the introduction and spread of non-indigenous species by shipping activities. To this end, a standardized set of survey methods that can provide a consistent basis on which to assess the status of introduced species of individual ports was developed under the coordination of the CSIRO Center for Research on Introduced Marine Pests (CRIMP) (Hewitt & Martin, 2001). The study that led to this Phytoplankton Atlas of Sepetiba Bay is part of a baseline port survey carried out based on this set of procedures, here referred to as the CRIMP Protocol.

Sepetiba Bay is located in the State of Rio de Janeiro, Brazil (**Figure 1**). Its unique geomorphology is created by the presence of the 40km-long Restinga da Marambaia, a sandbar that separates the bay from the Atlantic Ocean. The northern margins of the bay have small sandy beaches separated by rocky headlands, as well as mangrove vegetation that renders this a very productive ecosystem. The climate in the area is tropical and humid: rainy in the summer and drier in the winter (**Figure 2**).

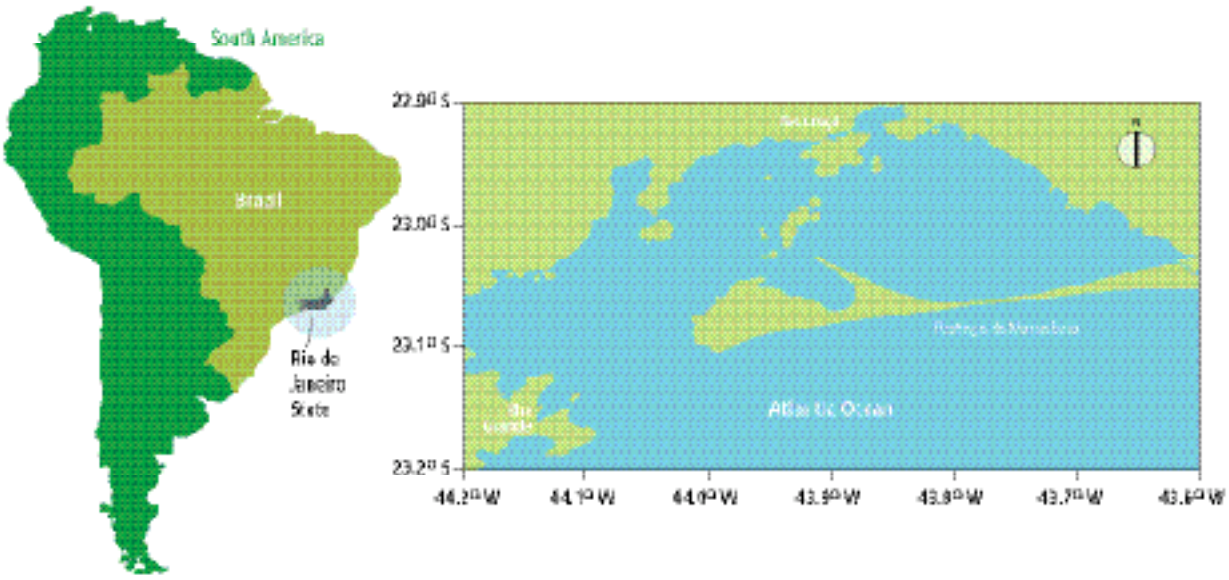


Figure 1. Sepetiba Bay, State of Rio de Janeiro, Brazil.

On the northeastern portion, rivers and channels that flow into the bay drain a watershed of about 2,654 km² that includes agricultural, industrial and urban areas. Marine water contribution is most effective from the western portion of the bay, where a wide and deep channel is found between Ilha Grande and the edge of the Restinga da Marambaia. This channel leads to the Port of Sepetiba, located on the northeastern portion of the bay (43° 50'W and 22° 56'S). Except for this navigation channel, Sepetiba Bay is a fairly shallow system, that is, most of the ca. 300 km² of the bay area is less than 10 meters deep.

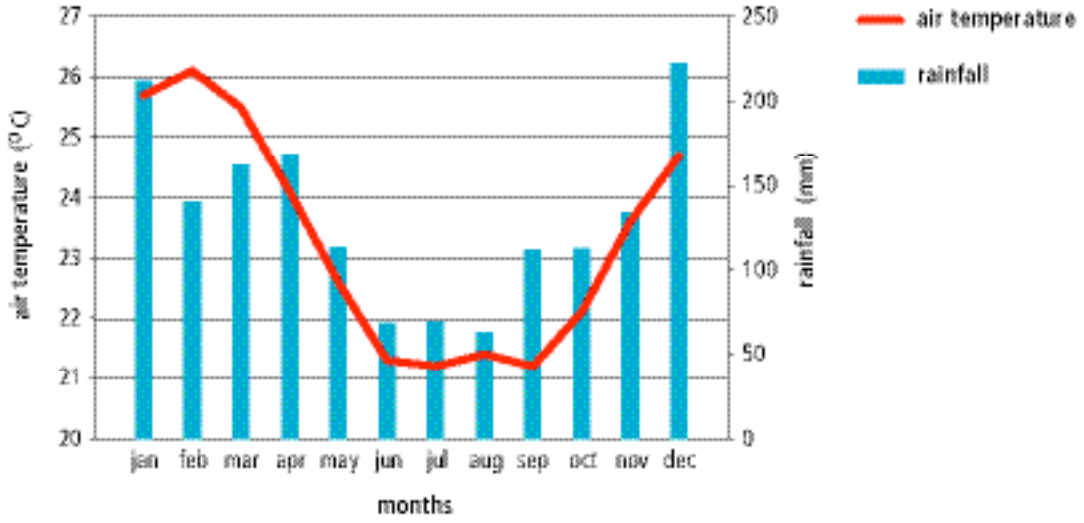


Figure 2. Air temperature and rainfall in Sepetiba Bay area: montly averages for 1960-1990 time series (Instituto Nacional de Meteorologia, INMET/ Brazil).

Sepetiba Bay can, therefore, be divided into two zones according to its estuarine nature: an innerbay area to the east, shallower and less saline (28, in average), and an outerbay area to the west, with more transparent waters and higher salinity (33, in average) (Cruz Filho *et al*., 2000). A chain of small islands creates a natural divider between the innerbay and the outerbay. Mean spring tide range is 1.1m and mean neap tide range is 0.3 m.

For those who are familiar with phytoplankton work, it is known that to produce a complete species list in a short period of time is an impossible task. Several of the specimens are still under study. This atlas presents a species list of 246 taxa, 103 of which come with illustrations, some of these with information on their distribution in the study area. It may serve as an identification guide to those interested in the phytoplankton of Sepetiba Bay and elsewhere.

2. Methods

The CRIMP Protocol applied to our study area determined the establishment of a total of 44 sampling sites that comprised all habitat types (water, hard substrate, soft bottom) and taxonomic groups (represented in the plankton, microalgae cysts, seaweeds, zoobenthos, and fish). For the phytoplankton, sampling was carried out at 13 sites distributed along two major areas of the bay: in the outer bay, where marine influence is more effective, and in the inner bay which is subject to direct freshwater inflow (**Figure 3**). These sites were selected so as to represent the environmental gradients found in Sepetiba Bay.

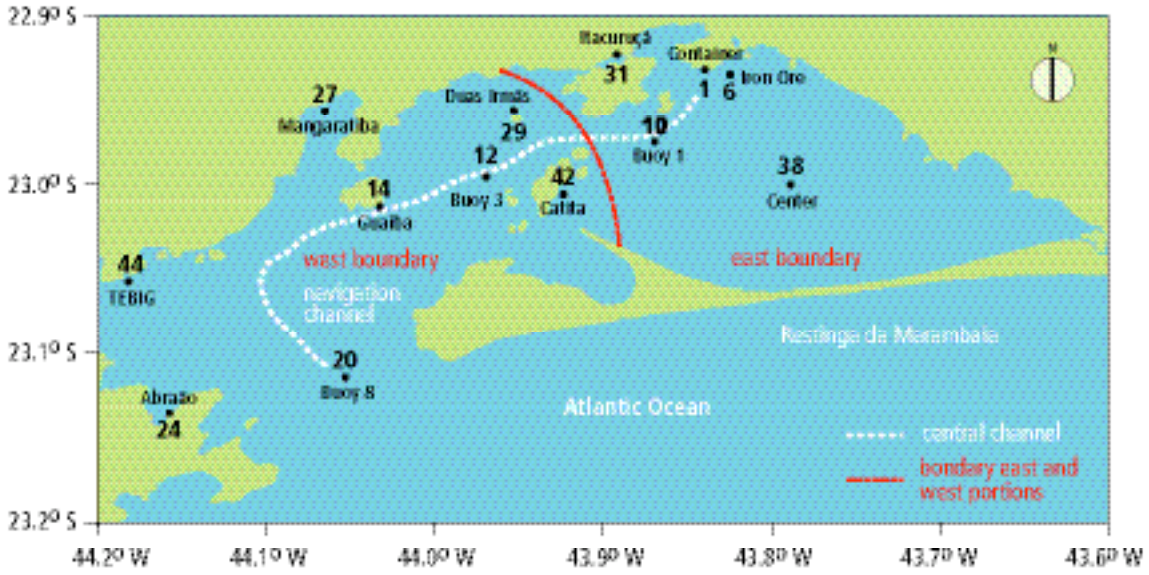


Figure 3. Phytoplankton sampling sites.

The CRIMP Protocol requires a descriptive coding to identify each site, which also has corresponding arabic numbers (**Table 1**). For sake of simplicity, theses sites are hereafter referred to by their numbers only.

Table 1. List of sampling sites.

SITE Nº	SITE CODE	SITE NAME	LOCAL DEPTH (m)
1	BRRJSBIBCOT1	Container Terminal 1	15.0
6	BRRJSBIBORET	Iron Ore Terminal	7.5
10	BRRJSBIBBUO1	Channel Buoy 1	15.5
12	BRRJSBIBBUO3	Channel Buoy 3	16.5
14	BRRJSBOBGUPI	Guaíba Pier	15.0
20	BRRJSBOBBUO8	Channel Buoy 8	15.0
24	BRRJSBOBABRO	Abraão Bay	10.0
27	BRRJSBOBMAPI	Mangaratiba Pier	2.5
29	BRRJSBIBIRRO	Duas Irmãs Island	5.6
31	BRRJSBIBHIPI	Itacuruçá Pier / Historical Piles	5.0
38	BRRJSBIBSECE	Sepetiba Bay (Center)	6.5
42	BRRJSBIBCARO	Catita Beach	14.0
44	BRRJSBOBTBG1	TEBIG Oil Terminal 1	10.0

The period chosen for the port survey was November 2001 (representative of spring), when biodiversity was expected to be at its highest point during the annual cycle. This survey was complemented by another field trip carried out in April 2002 (representative of autumn). Both field trips were done during two consecutive days, outerbay on the first and innerbay on the second. The methods recommended in the CRIMP Protocol were also complemented by the use of an additional type of sampling gear. Our goal was to provide the most accurate picture possible of the biodiversity of the phytoplankton of Sepetiba Bay to serve as a record for this baseline port survey.

The recommendations of the CRIMP Protocol for the phytoplankton study (Hewitt & Martin, 2001) were followed very closely (**Table 2**). It emphasizes qualitative analysis of samples collected with net hauls. It was complemented with another method that allows for a better evaluation of the quantitative attributes of the populations (**Table 2**). Objectives, advantages and disadvantages of both types of sampling, and the sample analysis that followed, is thoroughly discussed in Souria (1978). The effect of the sampling methods applied on our diagnosis of the local phytoplankton biodiversity will be presented in the "results & discussion" section.

The equipment used for the analysis were an upright Zeiss Axiophot light microscope with Media Cybernetics Image-Pro Plus 4 image analysis software, an upright Olympus BH2 light microscope, an inverted Olympus IX 70 light microscope and a Jeol JSM-5310 Scanning Electron Microscope.

Table 2. Methods used to sample and to analyse the phytoplankton.

	CRIMP PROTOCOL	COMPLEMENTATION
Sampler	20µm-mesh net	Van Dorn bottle
Depth	Vertical tow: 0.5m from the bottom to the surface	Two depths: surface and 0.5m from the bottom
Fixative	Formaldehyde buffered with borax (final concentration of 2%)	Formaldehyde buffered with borax (final concentration of 2%)
Analysis (qualitative)	1) water mounts 2) acid-clean permanent mounts for diatoms (Hasle & Fryxell, 1970), observation in phase contrast 3) water mounts with hipoclorito of sodium to help to distinguish and/or dissociate dinoflagellate plates (Boltovskoy,1973), observation in phase contrast 4) water-washed permanent mounts with calcofluor to help distinguish dinoflagellate plates (Fritz & Triemer, 1985), observation with epifluorescence using UV-light 5) Scanning Electron Microscopy of acid-clean or water-washed material (sputtered with gold-palladium)	Settling technique (Utermohl, 1958): identification during cell count, with final magnification of 200X or 400X, using bright field and phase contrast
Analysis (quantitative)	Relative abundance in 1,5mL of sample settled in Utermohl's counting chambers: cells were counted until the curve determined by the number of new taxa by the number of new counting fields became stable (Uehlinger, 1964); coordenates of counting fields were pre-determined at random	Absolute abundance using the settling technique (Utermohl, 1958): counting of at least 300 settling units (one cell–chain of cells), which allowed the count of 100 cells of the best represented taxon (counting error of 20% according to Lund *et al*., [1958], and a 95% chance of finding a species that represents 1% of the community according to Shaw [1964])

3. Results and Discussion

The species list of Sepetiba Bay (spring/2001 and autumn/2002) included a total of 246 taxa, of which 150 were diatoms (Bacillariophyta), 86 dinoflagellates (Dinoflagellata), 5 silicoflagellates (Chromophyta, Dictyochophyceae), 3 coccolithophorids (Chromophyta, Primnesiophyceae), 1 euglenophycean (Euglenophyta) and 1 ebriidean (Zoomastigophora, Ebriidea). Of the 246 taxa found, 155 were identified to species level to date (Appendix 1).

The taxa found were organized according to the following systems of classification: Round *et al.* (1990) for the diatoms, Fensome *et al.* (1993) for the dinoflagellates, Throndsen (1997) for the silicoflagellates and the ebriideae, Leedale (1967) for the euglenophycean, and Heimdal (1997) for the coccolithophorids.

The in-depth discussion of phytoplankton distributional patterns and their ecological implications is beyond the scope of this publication. It is important, however, to understand that the environmental settings where the phytoplankton populations were found presented space-time gradients.

The sampling periods were quite different in terms of temperature and salinity, which reflected well the seasonal variations of rainfall and air temperature: late spring had cooler and less saline waters, while autumn had higher salinity and warmer waters (**Figure 4**). The data presented here are a composite of the whole water column, although one can expect to find vertical stratification of the phytoplankton populations in systems of estuarine nature such as Sepetiba Bay.

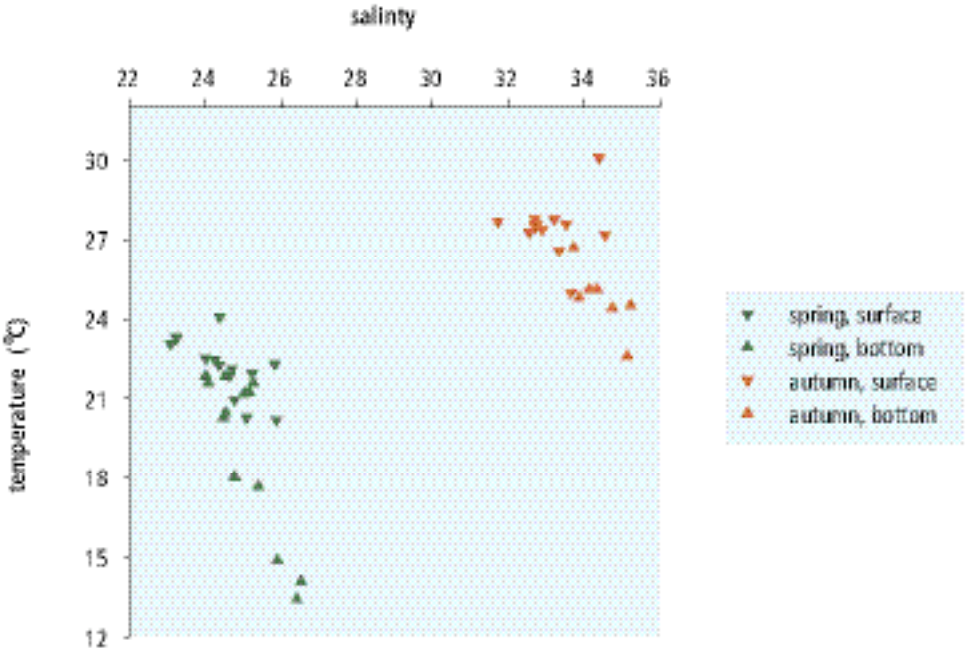


Figure 4. T-S diagram of Sepetiba Bay for 19-20 November 2001 and 25-26 April 2002.

Another indicator of seasonal differences can be established by analyzing nutrient concentrations (**Table 3**). Spring had more eutrophic waters than autumn, probably due to increasing river input and overall runoff that take place during the beginning of the rainy season (see **Figure 2**). The wide range found for all nutrients within one single field trip reveals spatial gradients that can be associated with the estuarine nature of Sepetiba Bay, as can be also confirmed by the wide range found for surface salinity on both field trips (**Figure 4**). As already explained, the eastern portion of the Bay is subjected to direct river discharge, some of them with organic pollution, while the western portion receives the contribution of more saline and cleaner waters.

Table 3. Characterization of Sepetiba Bay waters in terms of nutrient concentrations for 19-20 November 2001 and 25-26 April 2002.

		Phosphate (µM)			Total Nitrogen (µM)			Silicate (µM)		
Spring	range	0.04	–	3.23	0.08	–	7.28	0.60	–	14.83
	mean	0.61	±	0.60	1.60	±	1.90	6.45	±	3.98
Autumn	range	0.06	–	1.10	0.13	–	4.54	nd	–	8.04
	mean	0.38	±	0.29	1.59	±	1.30	2.92	±	2.23

We believe that the sampling sites selected have included different environmental conditions that can be found in the study area, and their respective phytoplankton populations. Therefore, we feel confident that the sampling sites chosen, although apparently restricted in number, provided a species list that can serve as a good baseline for future studies.

It is interesting to note that, except for a few occasions, the net captured a higher number of taxa than the bottle at any given sampling site (**Figure 5**). However, when all taxa found at all sampling sites are combined, the bottle was able to capture a higher number of taxa than the net (**Figure 6a**). The distribution of taxa is not consistent in Sepetiba Bay. There are gradients in terms of temperature, salinity and nutrients that determine the heterogeneous distribution of the phytoplankton populations according to their preferences and tolerances. The net concentrates material during the hauls, but it is selective. It was probably consistently selective against some organisms, mostly flagellates, at some sites. It is beyond the scope of this study to discuss ecological and community structure aspects. From a qualitative point of view, however, it became clear that the use of the bottle greatly enhanced our perception of the phytoplankton biodiversity of Sepetiba Bay.

The second field trip, carried out in April 2002, also complemented our understanding of the biodiversity of Sepetiba Bay (**Figure 6b**). The difference, however, was not as great as the one found in the comparison made between sampling methods. This comparison exercise indicates that, for management purposes, if the study area presents horizontal gradients, using the bottle as an additional sampling gear can be more cost-effective than adding another field trip at a different season of the year.

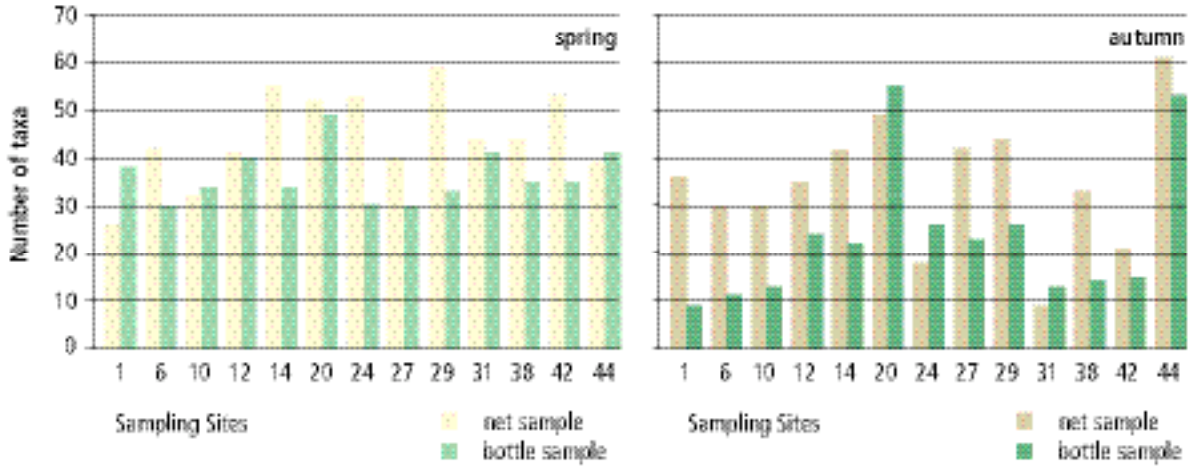


Figure 5. Number of taxa found in each sampling site during the spring and the autumn field trips.

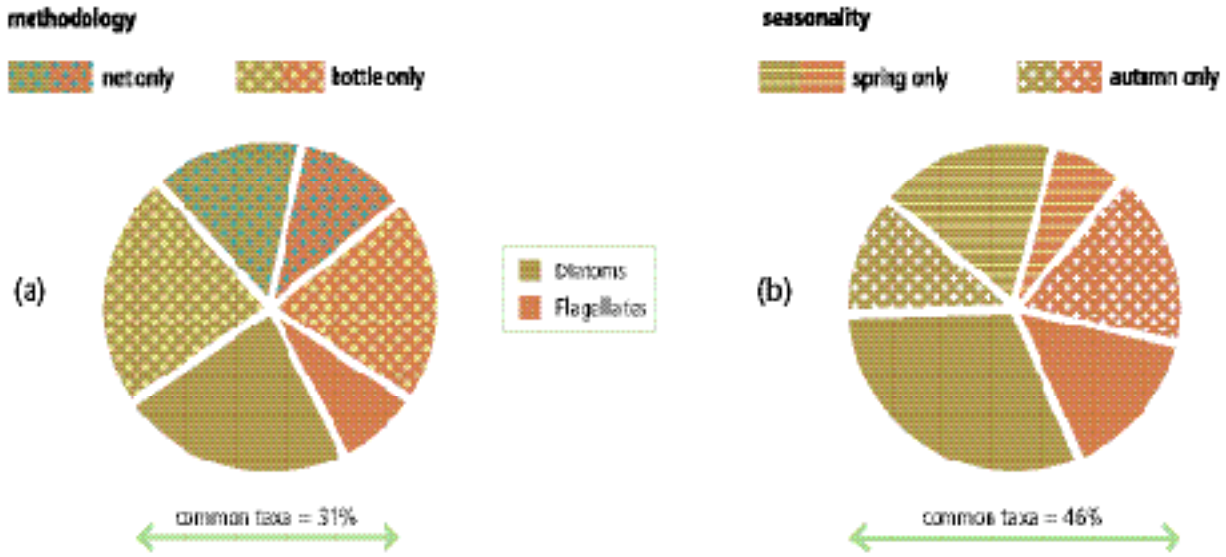


Figure 6. Comparison of the number of taxa found: (a) net versus bottle and (b) spring versus autumn.

REFERENCES

Bolovshoy, A. (1995). Taxonomia y morfologia de los dinoflagelados: métodos de trabajo. In Alveal, K., Ferrario, M.E., Oliveira, E.C. & Sar, E. (eds.). *Manual de Métodos Ficológicos*, p. 55-82. Editora Anibal Pinto S.A., Concepcion, 863pp.

Cruz Filho, A. G.; Araujo, F. G.; Azevedo, M. C. C.; Santos, A. C. A.(2000). Distribuição de populações de peixes demersais na Baía de Sepetiba, RJ. *Acta Biologica Leopoldensia*, São Leopoldo, RS, v. 22, n. 2, p. 227-257.

Fensome, R.A., Taylor, F.J.R., Norris, G., Sarjeant, W.A.S., Wharton, D.I. & Williams, G.L. (1993). *A Classification of Living and Fossil Dinoflagellates*. Sheridan Press, Hanover. 350pp.

Fritz, L. & Triemer, R.E. (1985). A rapid simple technique utilizing Calcofluor white M2R for the visualization of dinoflagellate thecal plates. *J. Phycol.*, 21(4): 662-664.

Hasle, G.R. & Fryxell, G.A. (1970). Diatoms: cleaning and mounting for light and electron microscopy. *Trans. Amer. Soc.*, 89 (4): 469-474.

Heimdal, B.R. (1997). Modern Coccolithophorids. In Tomas, C.R. (ed.), *Identifying Marine Phytoplankton*, p. 731-833. Academic Press, San Diego, 858 pp.

Hewitt, C.L. & Martin, R.B. (2001). *Revised protocols for baseline port surveys for introduced marine species - survey design, sampling protocols and specimen handling*. Technical Report 22, Center for Research on Introduced Marine Pests - CSIRO Marine Research, Tasmania, 46 pp.

Leedale, G.F. (1967). *Euglenoid flagellates*. Prentice - Hall, Englewood Cliffs, New York, 242 pp.

Lund, L.W.G., Kipling, C. & Le Cren, E.D. (1958). The inverted microscope method of estimating algal numbers and the statistical basis of estimations by counting. *Hydrobiologia* 11: 143-170.

Round, F.E., Crawford, R.M. & Mann, D.G. (1990). *The Diatoms: Biology and Morphology of the Genera*. The Bath Press, United Kingdom. 747 pp.

Shaw, A.B. (1964). Time in Stratigraphy. McGraw-Hill Book Company, New York. 365 pp.

Sournia, A. (1978). *Phytoplankton manual*. Paris, Unesco. 337 pp.

Throndsen, J. (1997). The planktonic marine flagellates. In Tomas, C.R. (ed.), *Identifying Marine Phytoplankton*, p. 591-729. Academic Press, San Diego, 858 pp.

Uehlinger, V. (1964). Étude statistique des méthodes de dénombrement planctonique. *Archives des Sciences*, 17(2): 121-230.

Utermöhl, H. (1958). Perfeccionamento del Método Cuantitativo del Fitoplancton. Associación Internacional de Limnologia Teórica y Aplicada - Comité de métodos limnologicos, comunicación, 9: 1-39.

PLATES

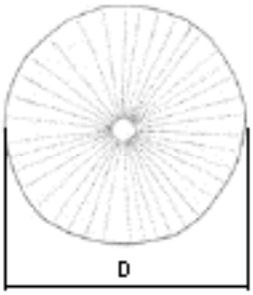
The taxa considered in this section were those collected with the net hauls.

Maps of their distribution in the Bay have been included only for those taxa that were considered the most important in terms of their relative abundance (over 50% of the average concentration in any given sample). The complete list of taxa can be found in Appendix 1.

The legends of the figures indicate the type of microscopy used to illustrate the taxa: light microscopy (LM), scanning electron microscopy (SEM) or fluorescence microscopy (FM).

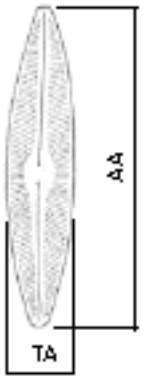
MORPHOMETRICS GUIDE

DIATOMS



valve view

girdle view



valve view

girdle view

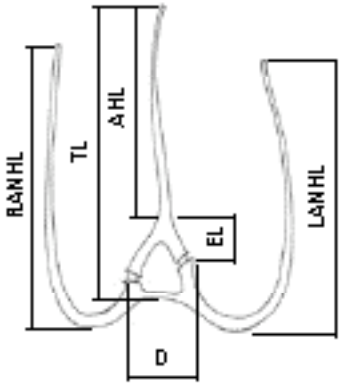
Centric Diatoms

D Diameter
PA Pervalvar axis

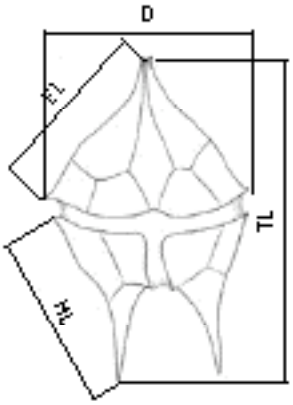
Pennate Diatoms

TA Transapical axis
PA Pervalvar axis
AA Apical axis

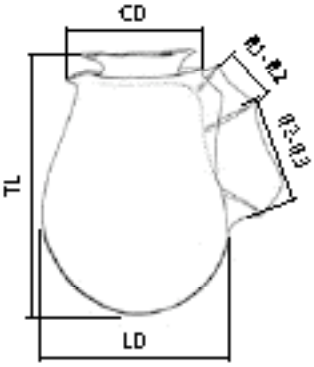
DINOFLAGELLATES



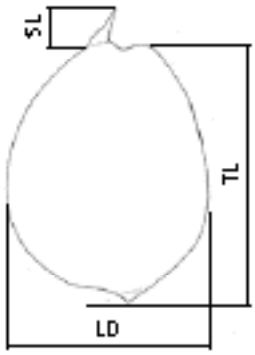
ventral view



ventral view



dorsal view



valve view

Ceratium

TL Total length
D Diameter
EL Epicone length
AHL Apical horn length
RANHL Right antapical horn length
LANHL Left antapical horn length

Protoperidinium

TL Total length
D Diameter
EL Epitheca length
HL Hypotheca length

Dinophysis

TL Total length
CD Cingulum diameter
LD Larger diameter
R1-R2 Space between ribs
R2-R3 Space between ribs

Prorocentrum

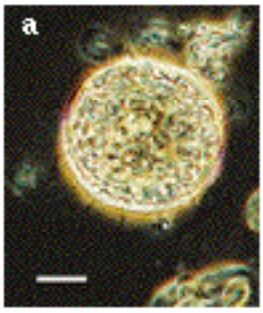
TL Total length
LD Larger diameter
SL Spine length

DIATOMS

Plate 1

Division Bacillariophyta
Class Coscinodiscophyceae
Order Thalassiosirales
Family Thalassiosiraceae

Thalassiosira punctigera (Castracane) Hasle,1983



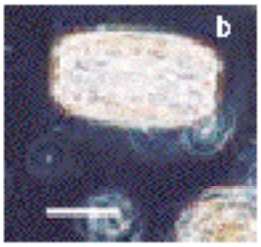
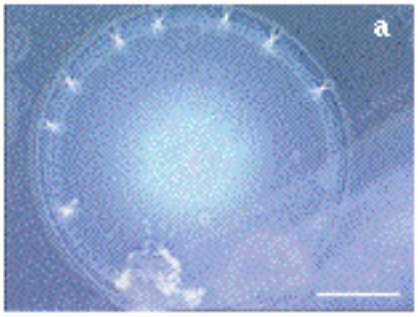


Figure 1. Single cell (a) in valve view and (b) in girdle view (LM, phase contrast, water mount). Scale bars = 20µm.



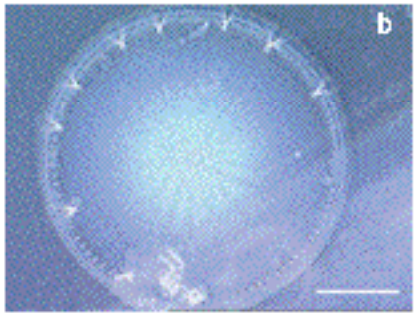


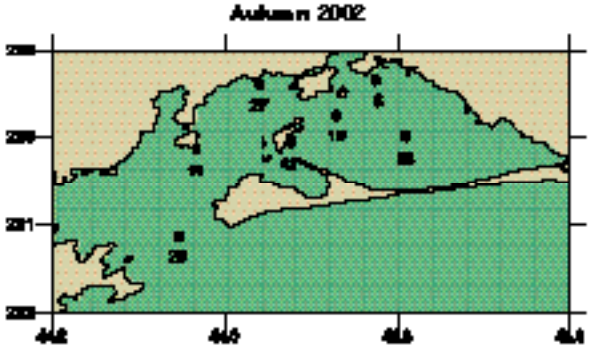
Figure 2. Single valve showing (a) areolation fasciculate and (b) densely spaced strutted processes close to valve margin and larger widely spaced occluded processes more away from valve margin (LM, bright field, Hyrax mount of cleaned material). Scale bars = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 58). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	25 - 80	58 ± 17	13
PA	20 - 45	28 ± 8	11



Division Bacillariophyta Plate 2
Class Coscinodiscophyceae
Order Thalassiosirales
Family Thalassiosiraceae

Thalassiosira rotula Meunier, 1910

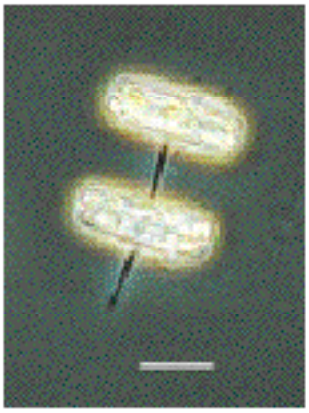


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 70).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	35	-	45	29	±	4	7
PA	10	-	15	11	±	2	7

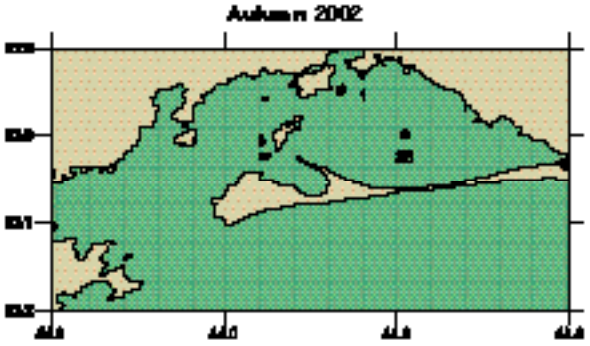


Plate 3

Division Bacillariophyta
Class Coscinodiscophyceae
Order Thalassiosirales
Family Skeletonemataceae

Detonula pumila (Castracane) Schütt, 1934

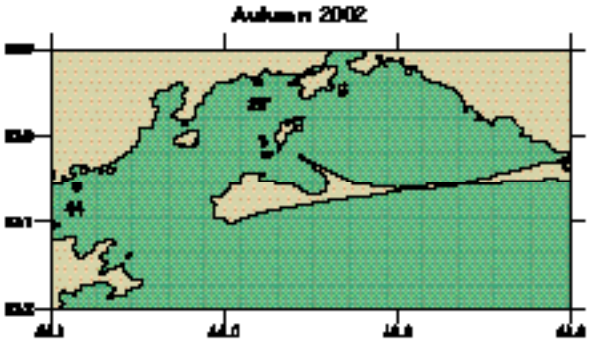
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 34). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	25 - 31	28 ± 3	14
PA	20 - 43	31 ± 8	14



Division Bacillariophyta Plate 4
Class Coscinodiscophyceae
Order Thalassiosirales
Family Skeletonemataceae

Skeletonema costatum (Greville) Cleve, 1878

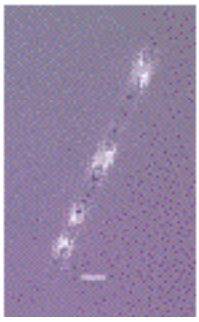


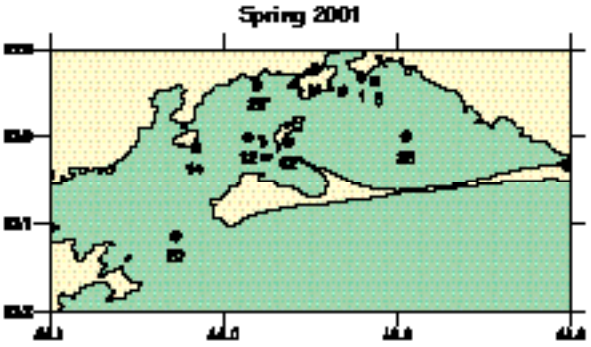
Figure 1. Chain of cells in girdle view
(LM, phase contrast, water mount). Scale bar = 10µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 44).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	4.5	-	6	5	±	0.6	8
PA	10.0	-	15	13	±	2.0	8



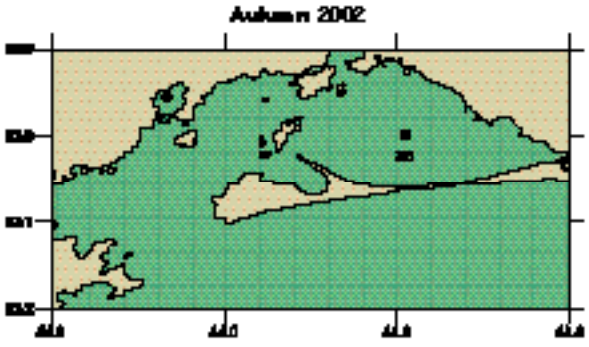


Plate 5

Division Bacillariophyta
Class Coscinodiscophyceae
Order Thalassiosirales
Family Stephanodiscaceae

Cyclotella litoralis Lange & Syvertsen, 1989

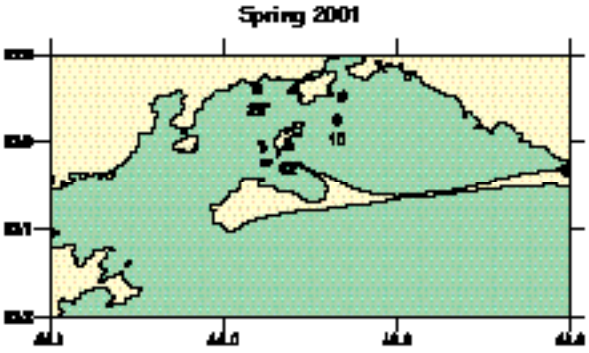
Figure 1. Single valve (LM, bright field, Hyrax mount of cleaned material). Scale bar = 20µm.

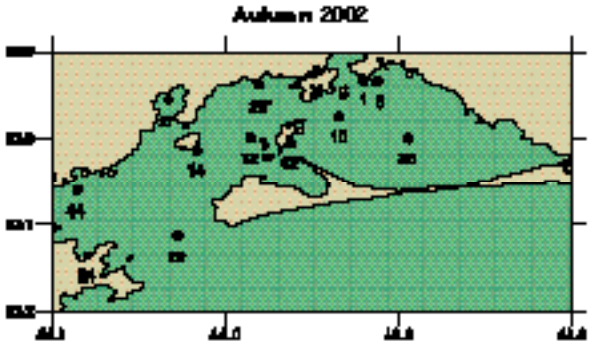
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 34). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	20 - 50	33 ± 8	3
PA	25 - 35	30 ± 7	3





Division Bacillariophyta Plate 6
Class Coscinodiscophyceae
Order Thalassiosirales
Family Stephanodiscaceae

Cyclotella stylorum Brigthwell, 1860

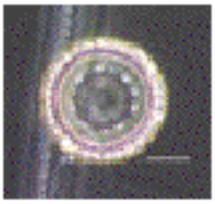


Figure 1. Single valve
(LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 34).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	20	-	50	33	±	8	3
PA	25	-	35	30	±	7	3

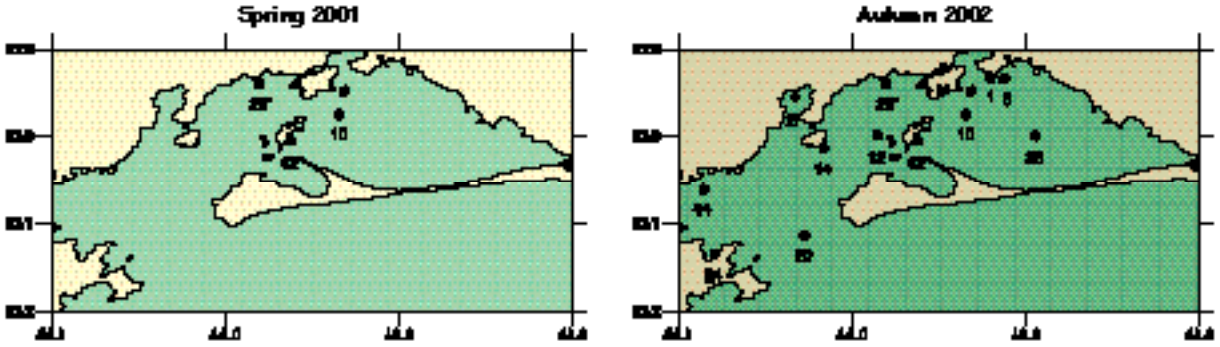


Plate 7

Division Bacillariophyta
Class Coscinodiscophyceae
Order Melosirales
Family Stephanopyxidaceae

Stephanopyxis turris (Greville & Arnott) Ralfs, 1861

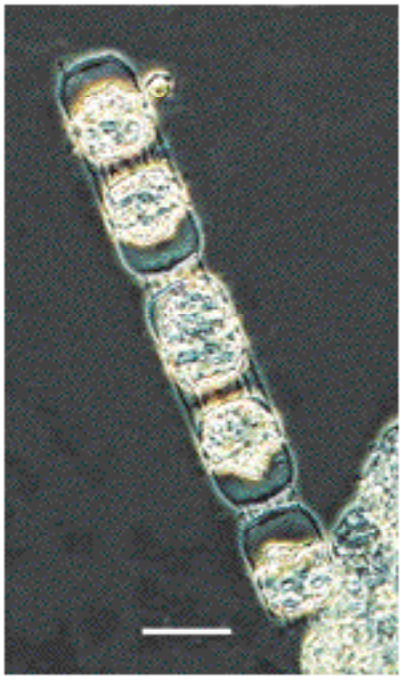


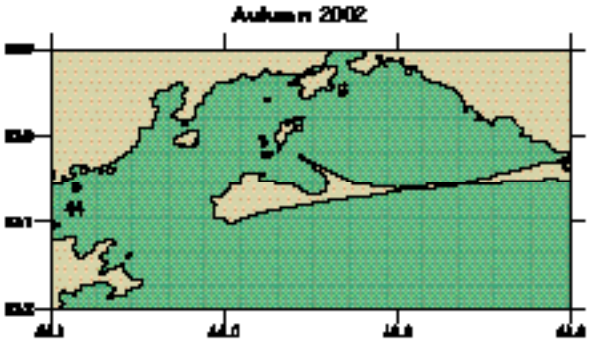
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 92). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	(µm)	n
D	40	1
PA	56	1



Division Bacillariophyta Plate 8
Class Coscinodiscophyceae
Order Paraliales
Family Paraliaceae

Paralia sulcata (Ehrenberg) Cleve, 1873

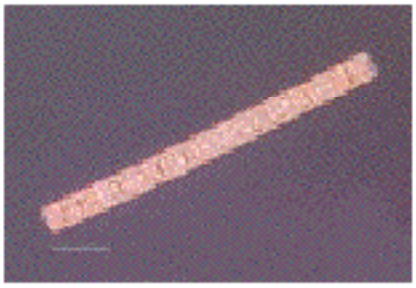


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

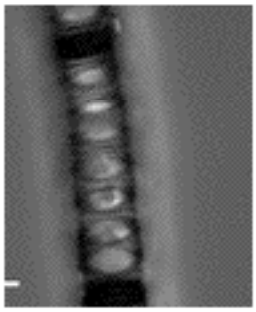


Figure 2. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 10µm.

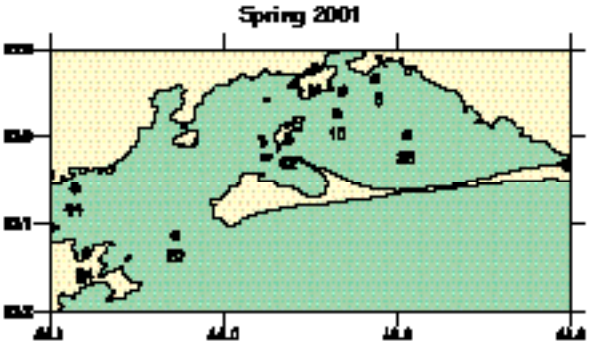
Figure 3. Single valve (LM, bright field, Hyrax mount of cleaned material). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 91).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	22 - 30	26 ± 3	15
PA	7 - 20	15 ± 4	15



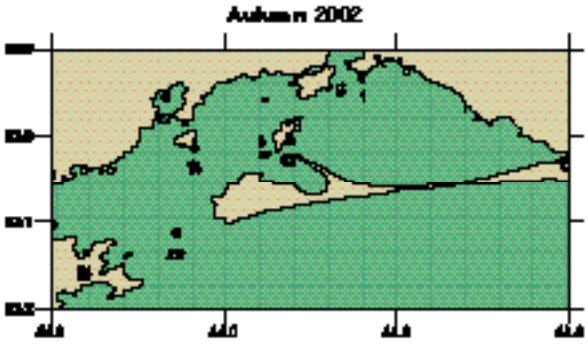


Plate 9

Division Bacillariophyta
Class Coscinodiscophyceae
Order Coscinodiscales
Family Coscinodiscaceae

Coscinodiscus gigas Ehrenberg, 1841

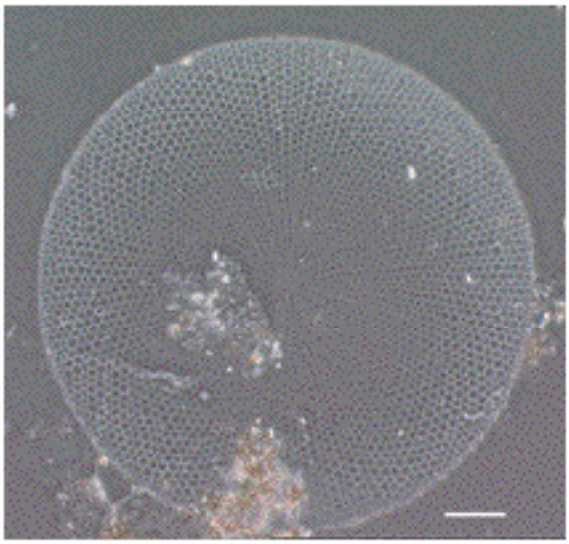


Figure 1. Single valve
(LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 20µm.

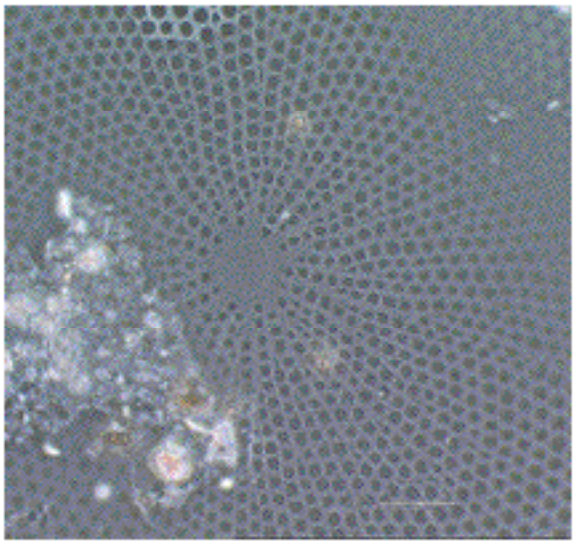


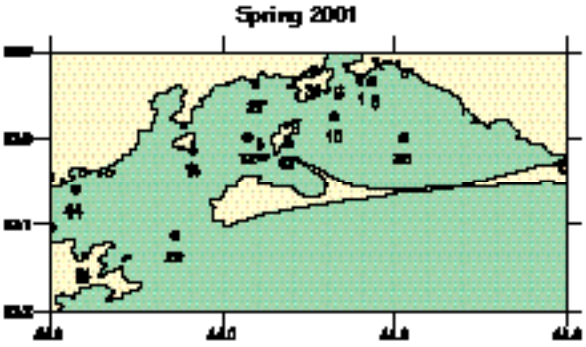
Figure 2. Detail of single valve, central area without areola.
(LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 10µm.

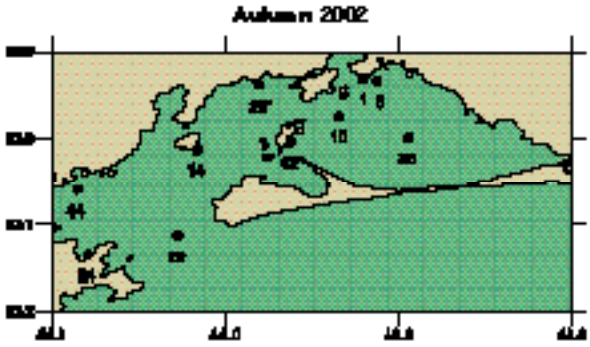
Taxonomic source

Navarro, N. (1981). The marine diatoms of Puerto Rico (p. 429). Botanica Marina. Vol. 24.

Morphometrics

	range (µm)	mean ± SD	n
D	115 - 310	177 ± 34	87





Division Bacillariophyta Plate 10
Class Coscinodiscophyceae
Order Coscinodiscales
Family Coscinodiscaceae

Coscinodiscus granii Gough, 1905

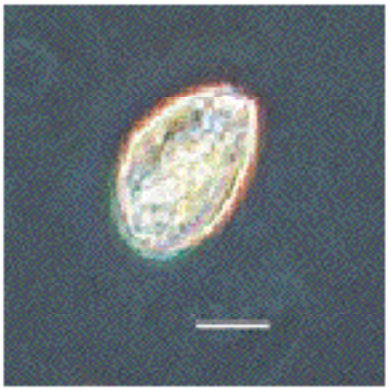


Figure 2. Cell in girdle view
(LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 109).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	(µm)	n
D	55	1
PA	30	1

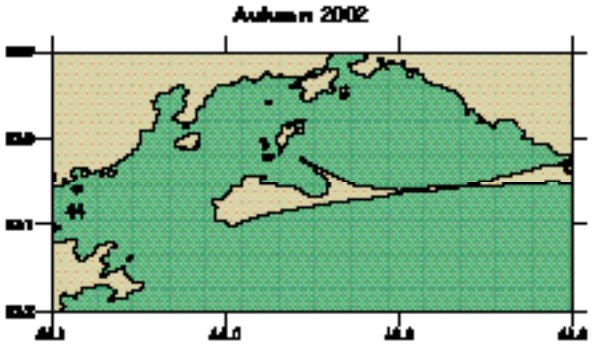


Plate 11

Division Bacillariophyta
Class Coscinodiscophyceae
Order Coscinodiscales
Family Coscinodiscaceae

Coscinodiscus cf. *asteromphalus* Ehrenberg, 1844

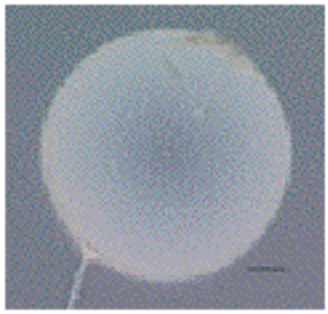


Figure 1. Single cell in valve view. (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 20µm.

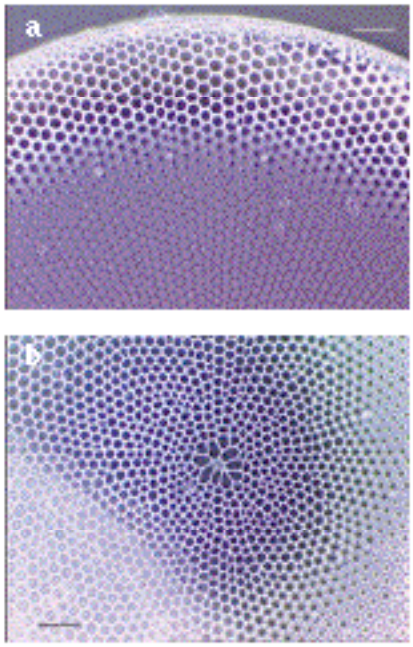


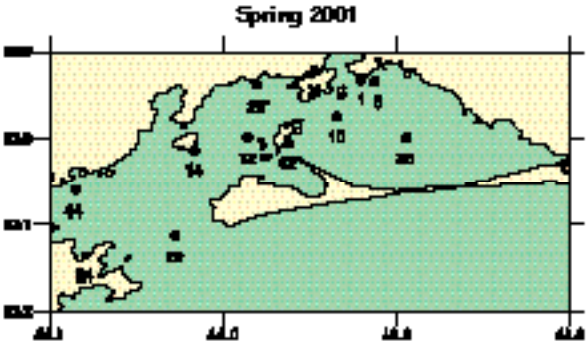
Figure 2. Detail of (a) marginal area and (b) central area (LM, phase contrast, Hyrax mount of cleaned material). Scale bars = 10µm.

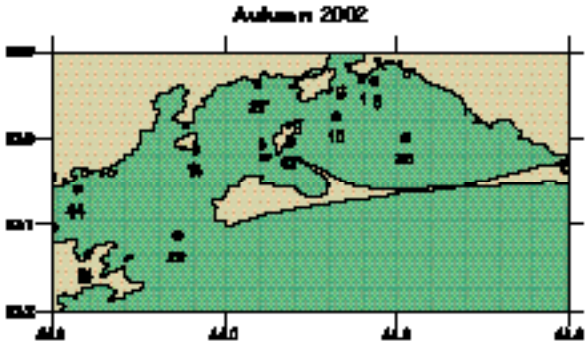
Taxonomic source

Hustedt, F. (1971). Kryptogamen - flora von Deutschland, Österreich und der Schweiz (p. 453). Johnson Reprint Corporation, New York.

Morphometrics

	range (µm)	mean ± SD	n
D	115 - 310	177 ± 34	87

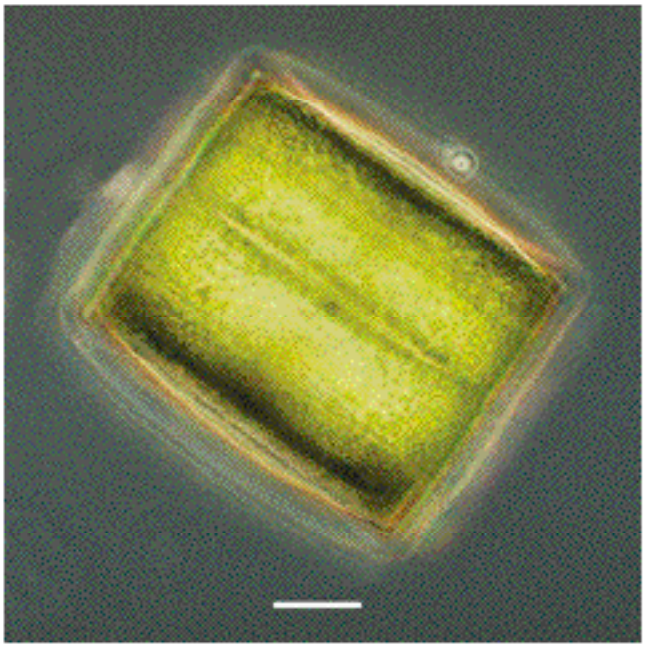




Division Bacillariophyta
Class Coscinodiscophyceae
Order Coscinodiscales
Family Coscinodiscaceae

Plate 12

Coscinodiscus wailesii Gran & Angst, 1931



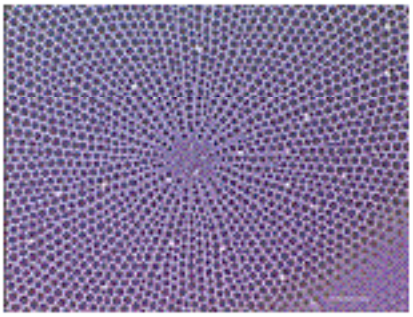


Figure 2. Detail of valve, central area without areola.
(LM, phase contrast, Hyrax mount of cleaned material).
Scale bar = 10µm.

Figure 1. Cell in girdle view
(LM, phase contrast, water mount).
Scale bar = 50µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 106).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	220 - 250	235 ± 27	4
PA	135 - 195	169 ± 12	4

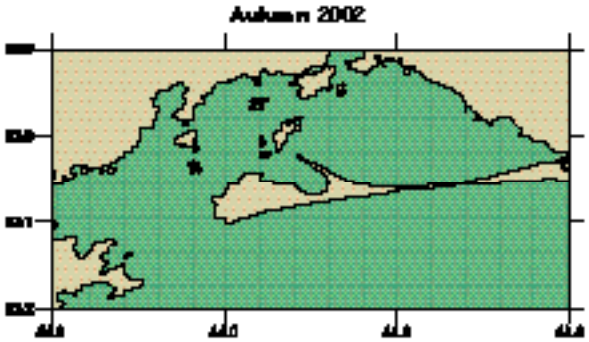


Plate 13

Division Bacillariophyta
Class Coscinodiscophyceae
Order Coscinodiscales
Family Heliopeltaceae

Actinoptychus senarius (Ehrenberg) Ehrenberg, 1843

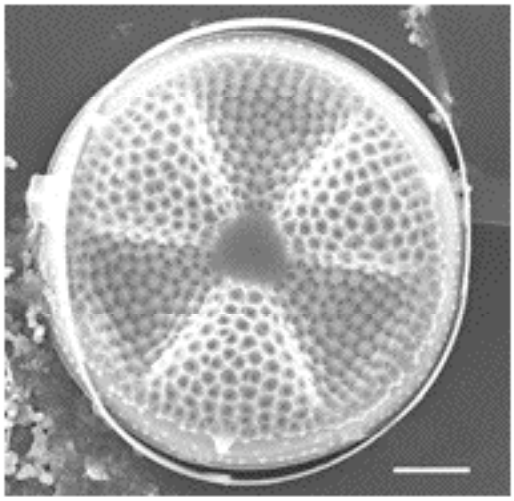


Figure 1. Single valve (SEM).
Scale bar = 10µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 93).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	(µm)	mean	n
D	55 - 65	60	2

Division Bacillariophyta Plate 14
Class Coscinodiscophyceae
Order Asterolamprales
Family Asterolampraceae

Asteromphalus cf. *flabellatus* (Brébisson) Greville, 1859

Figure 1. Single valve (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 10µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p.135).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

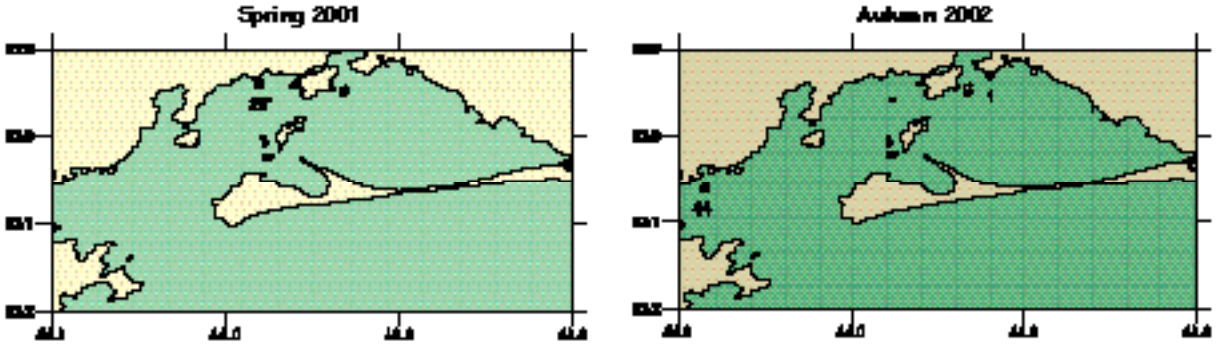


Plate 15

Division Bacillariophyta
Class Coscinodiscophyceae
Order Triceratiales
Family Triceratiaceae

Odontella mobiliensis (Bailey) Grunow, 1884

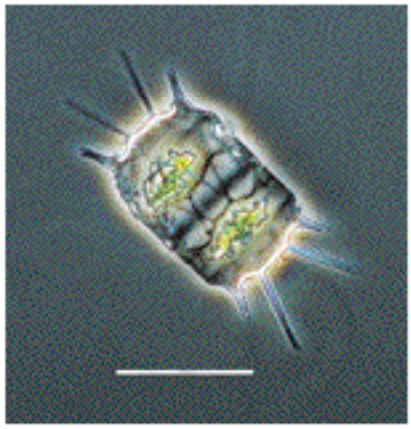


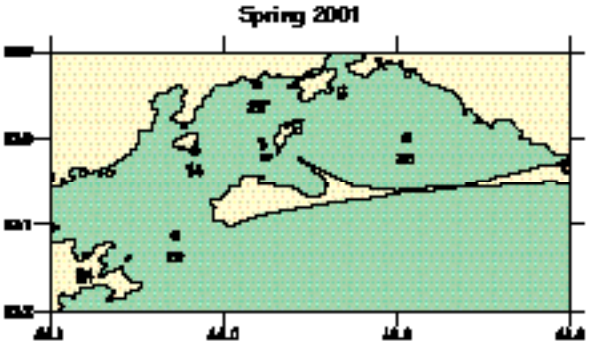
Figure 1. Dividing cell in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

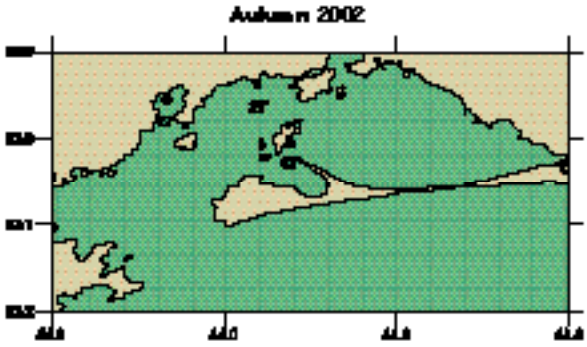
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 239). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	20.0 - 55.0	28 ± 10	13
PA	26.5 - 107.5	48 ± 19	13





Division Bacillariophyta Plate 16
Class Coscinodiscophyceae
Order Triceratiales
Family Triceratiaceae

Odontella sinensis (Greville) Grunow, 1884

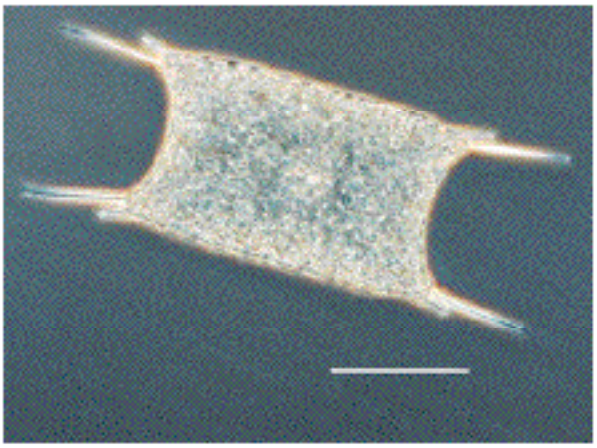


Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

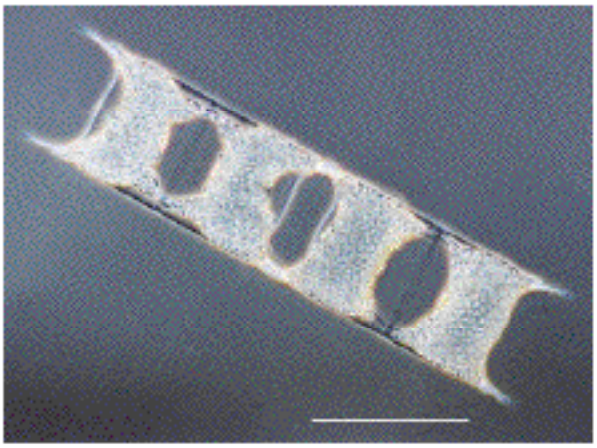


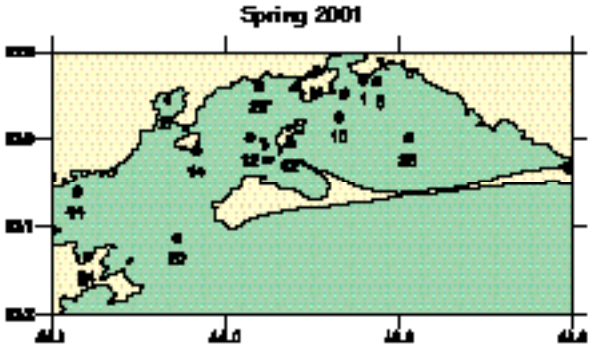
Figure 2. Dividing cells in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 239). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	90	-	265	139	±	32	62
PA	125	-	360	246	±	61	62



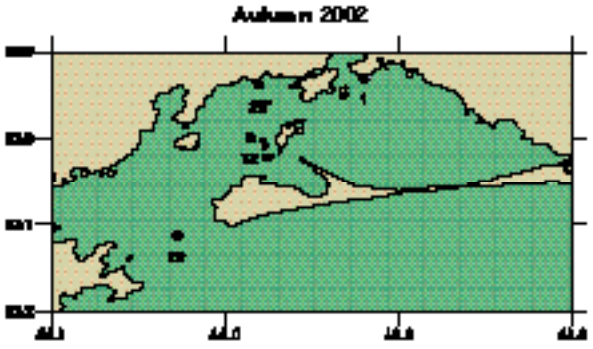


Plate 17

Division Bacillariophyta
Class Coscinodiscophyceae
Order Triceratiales
Family Triceratiaceae

Triceratium favus Ehrenberg, 1839

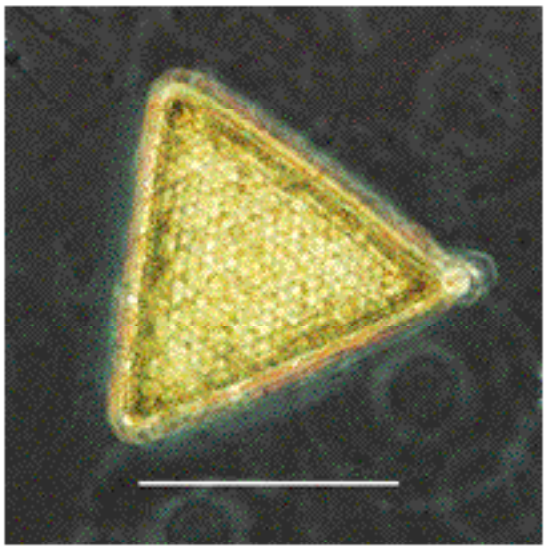


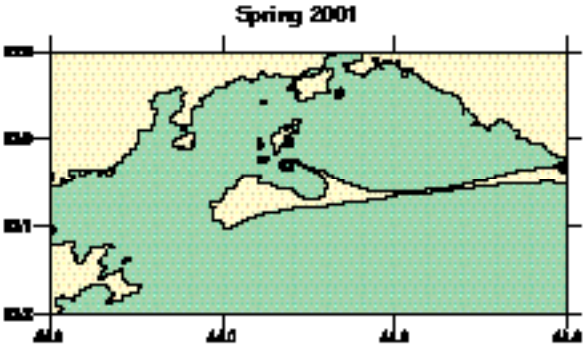
Figure 1. Single valve (LM, phase contrast, water mount). Scale bar = 100µm.

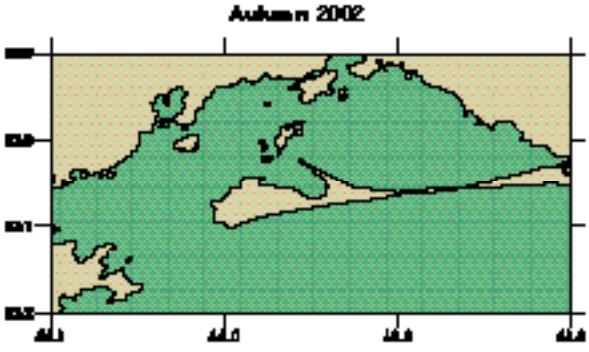
Taxonomic source

Hustedt, F. (1971). Kryptogamen - flora von Deutschland, Österreich und der Schweiz (p. 798). Johnson Reprint Corporation, New York.

Morphometrics

	range (µm)	mean ± SD	n
D	50 - 140	105 ± 48	3





Division Bacillariophyta Plate 18
Class Coscinodiscophyceae
Order Hemiaulales
Family Hemiaulaceae

Cerataulina pelagica (Cleve) Hendey, 1937

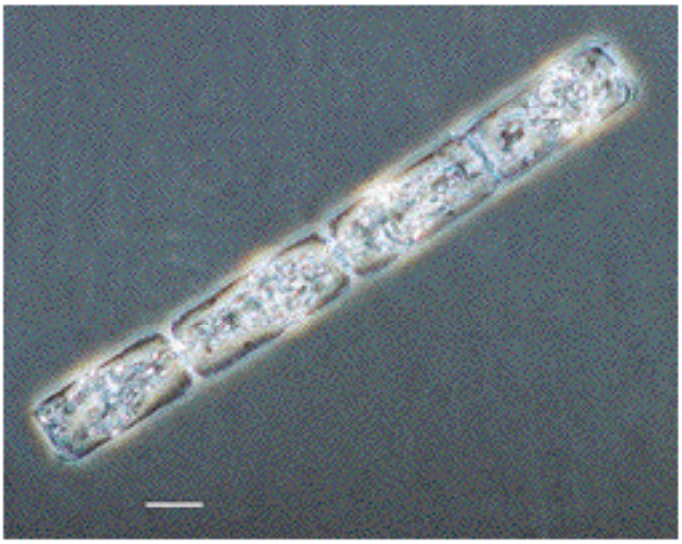


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

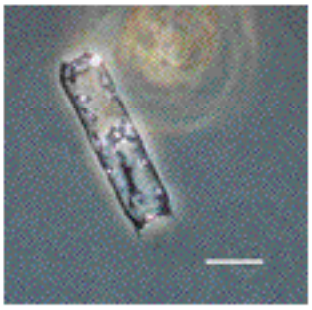


Figure 2. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p.171).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	15	-	35	24	±	5	41
PA	50	-	165	75	±	19	42

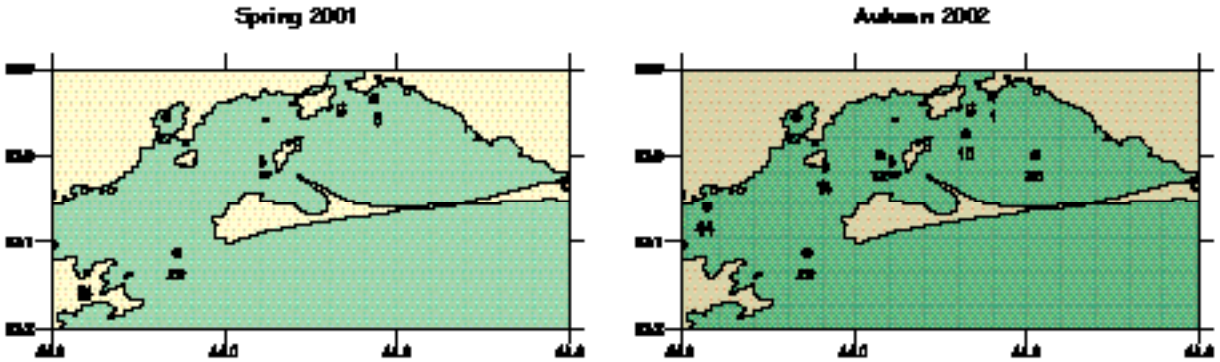


Plate 19

Division Bacillariophyta
Class Coscinodiscophyceae
Order Hemiaulales
Family Hemiaulaceae

Climacodium frauenfeldianum Grunow, 1868

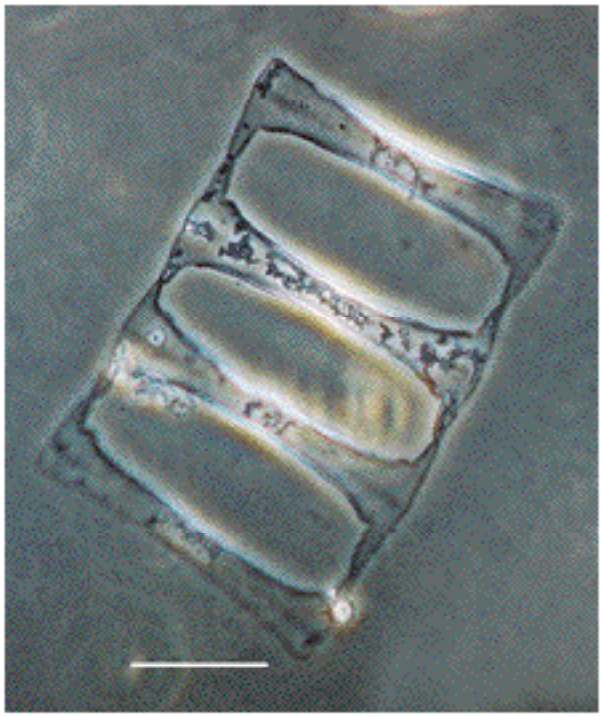


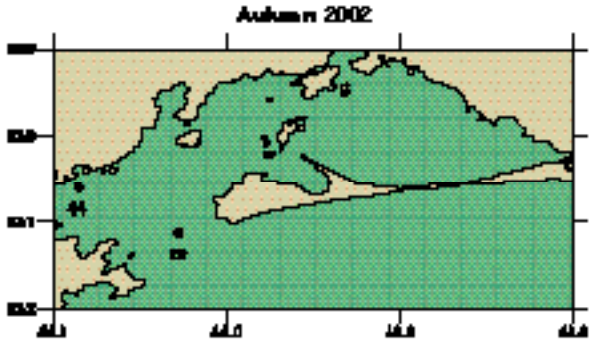
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 172). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	125 - 130	128 ± 3	3
PA	40 - 60	53 ± 12	3



Division Bacillariophyta Plate 20
Class Coscinodiscophyceae
Order Hemiaulales
Family Hemiaulaceae

Eucampia cornuta (Cleve) Grunow, 1882

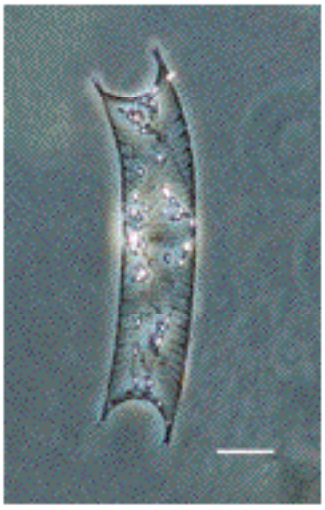


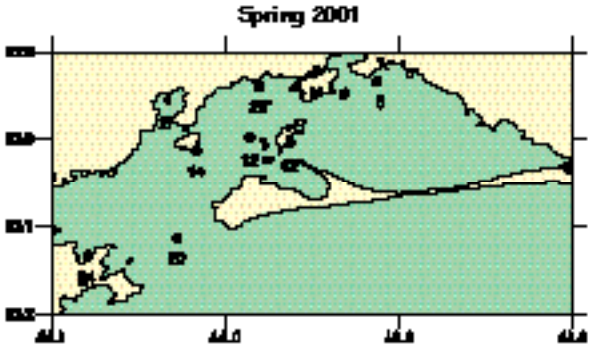
Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 175). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	10 - 30	21 ± 8	10
PA	35 - 155	75 ± 44	10



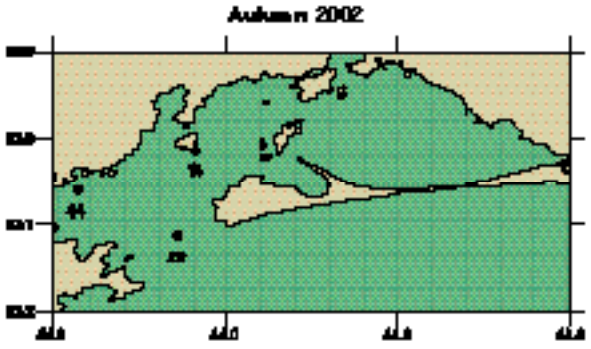


Plate 21

Division Bacillariophyta
Class Coscinodiscophyceae
Order Hemiaulales
Family Hemiaulaceae

Eucampia zodiacus Ehrenberg, 1839

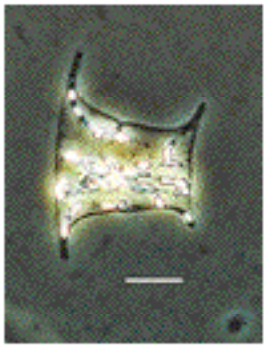


Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

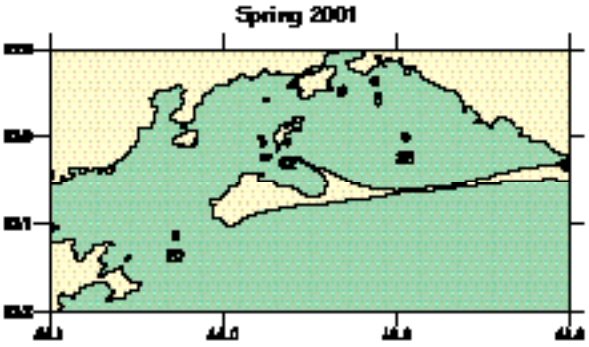
Figure 2. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 176). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	10 - 50	26 ± 22	5
PA	30 - 50	36 ± 9	5



Division Bacillariophyta Plate 22
Class Coscinodiscophyceae
Order Hemiaulales
Family Hemiaulaceae

Hemiaulus hauckii Grunow, 1881

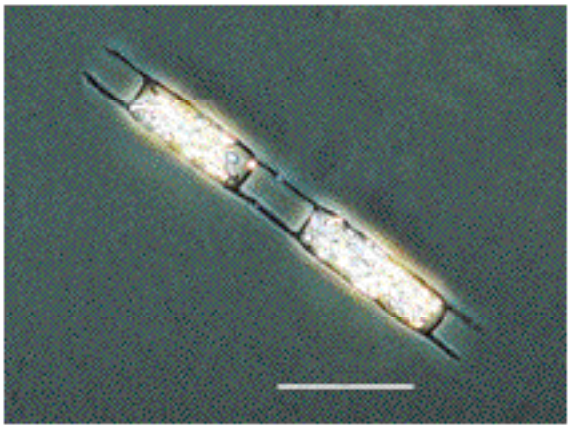


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 177). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	10 - 29	16 ± 4	23
PA	30 - 100	56 ± 18	23

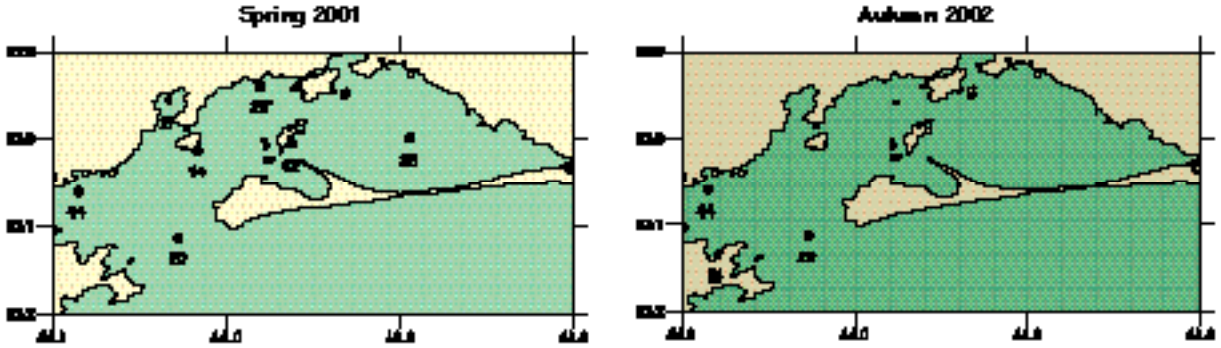


Plate 23

Division Bacillariophyta
Class Coscinodiscophyceae
Order Hemiaulales
Family Hemiaulaceae

Hemiaulus membranaceus Cleve, 1873

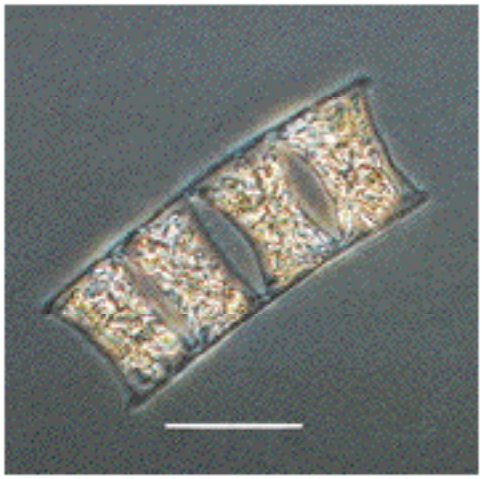


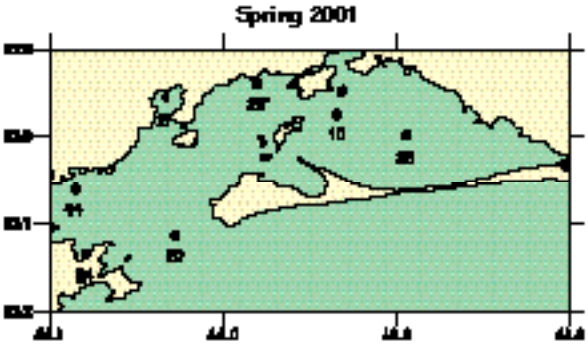
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

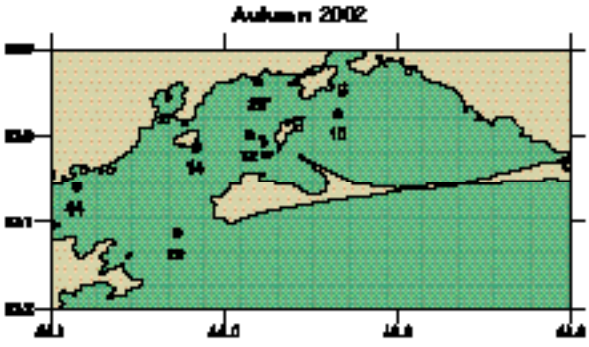
Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p 177). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	20 - 82	46 ± 14	72
PA	30 - 175	55 ± 23	72





Division Bacillariophyta Plate 24
Class Coscinodiscophyceae
Order Hemiaulales
Family Hemiaulaceae

Hemiaulus sinensis Greville, 1865

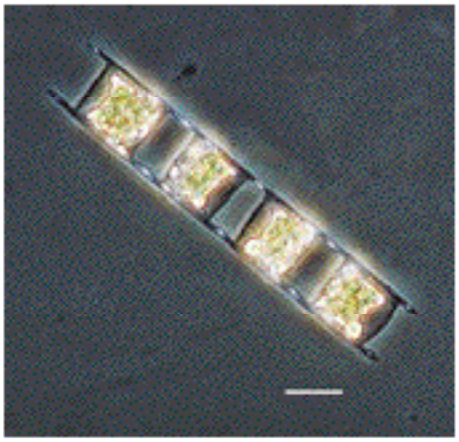


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p 177).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	18	-	35	26	±	4	30
PA	30	-	80	49	±	12	30

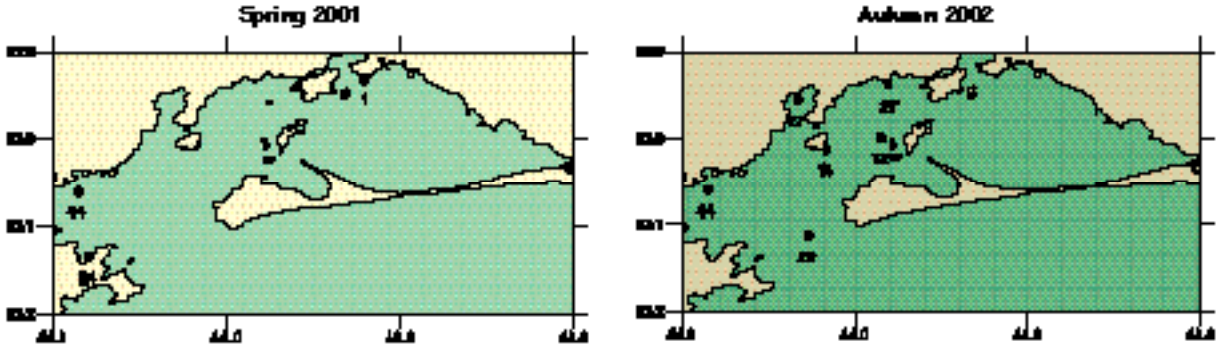


Plate 25

Division Bacillariophyta
Class Coscinodiscophyceae
Order Lithodesmiales
Family Lithodesmiaceae

Ditylum brightwellii (T. West) Grunow,1883

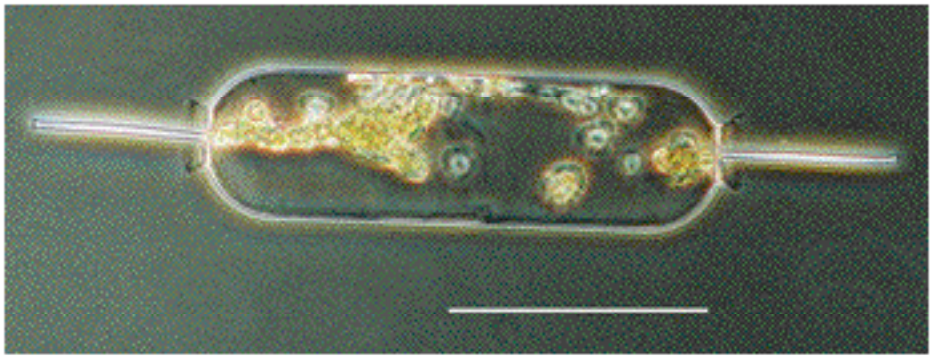


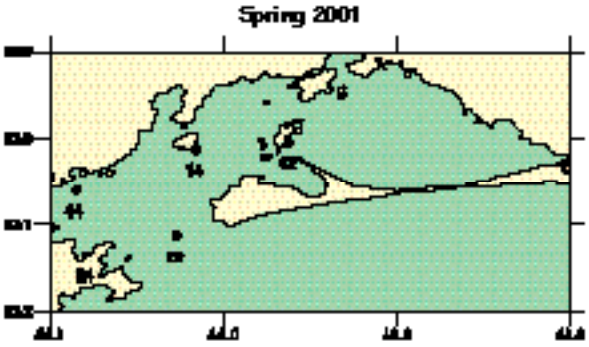
Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 230).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	63		1
PA	136 - 138	137 ± 1	2



Division Bacillariophyta Plate 26
Class Coscinodiscophyceae
Order Lithodesmiales
Family Lithodesmiaceae

Helicotheca tamesis (Shrubsole) Ricard, 1987

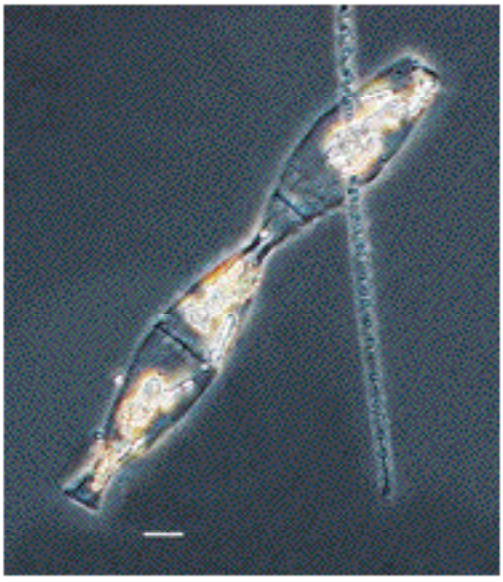


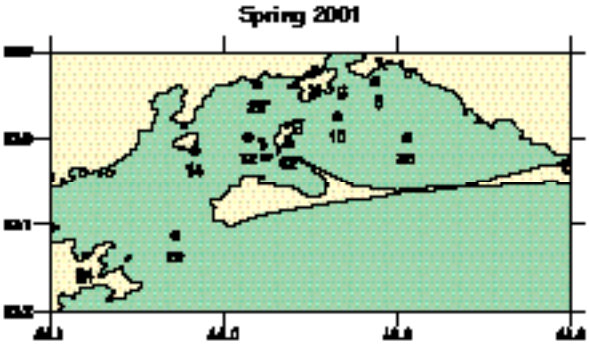
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 234). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	9.5 - 60	47 ± 12	24
PA	32.5 - 150	104 ± 31	24



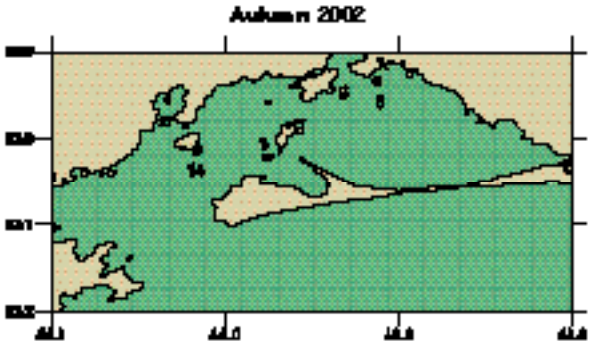


Plate 27

Division Bacillariophyta
Class Coscinodiscophyceae
Order Lithodesmiales
Family Lithodesmiaceae

Lithodesmium undulatum Ehrenberg, 1839

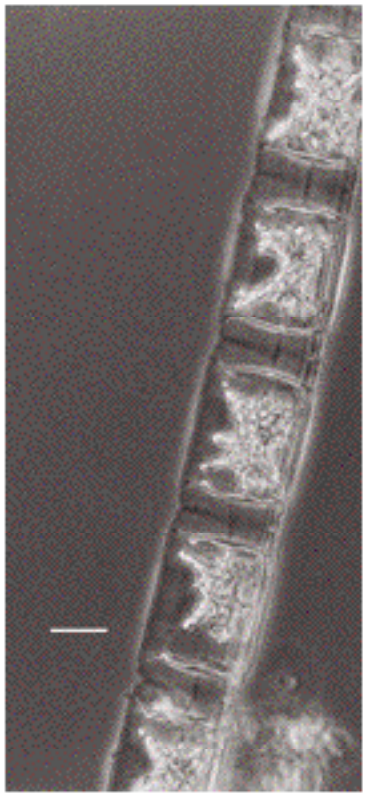


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

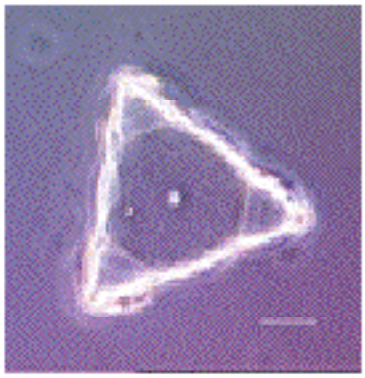


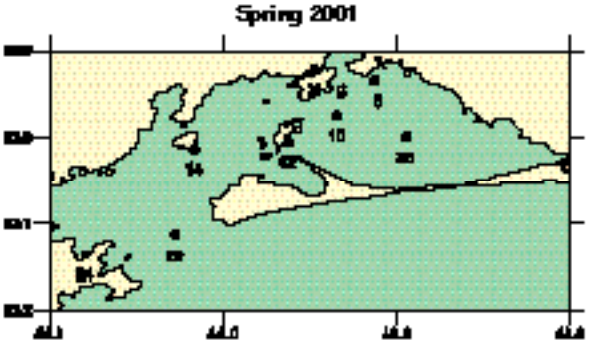
Figure 2. Single valve (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 10µm

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 234). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	30 - 60	46 ± 6	28
PA	30 - 60	48 ± 9	28



Division Bacillariophyta **Plate 28**
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Dactyliosolen fragilissimus (Bergon) Hasle, 1996

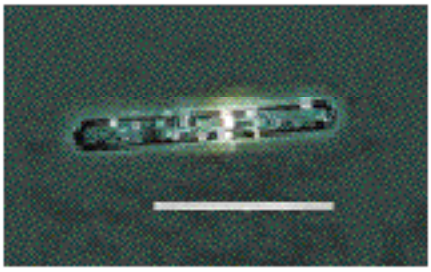


Figure 1. Single cell in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 167).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	5 - 10	7 ± 2	7
PA	50 - 95	65 ± 15	7

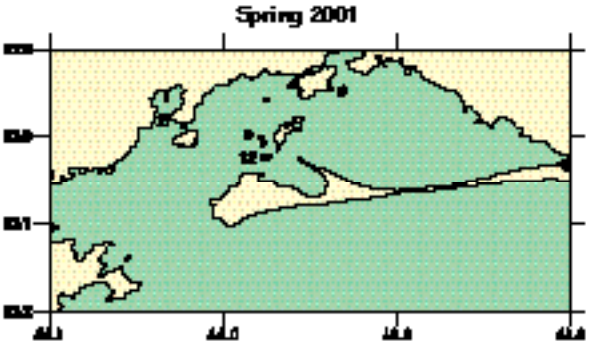


Plate 29

Division Bacillariophyta
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Guinardia delicatula (Cleve) Hasle, 1996

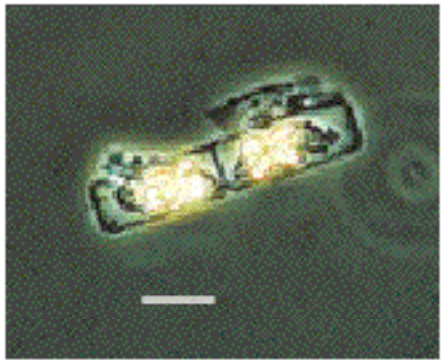


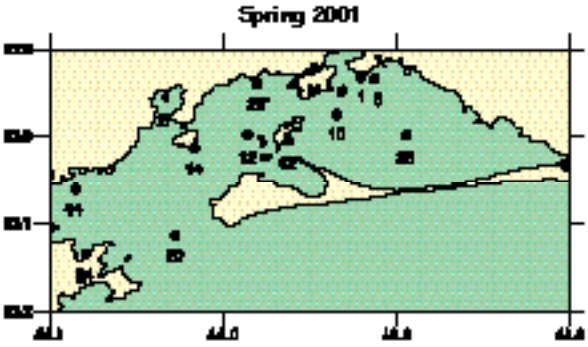
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

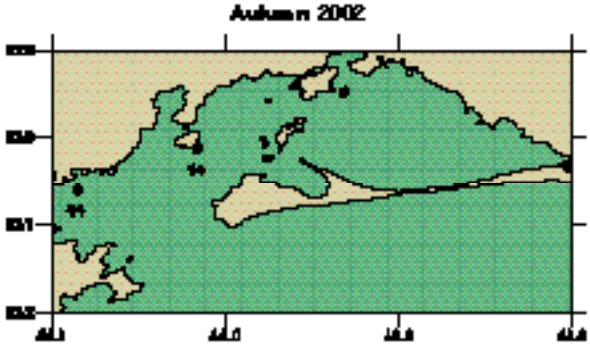
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 161). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	15 - 26	17 ± 2	43
PA	25 - 60	43 ± 8	43





Division Bacillariophyta Plate 30
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Guinardia flaccida (Castracane) H. Peragallo, 1892

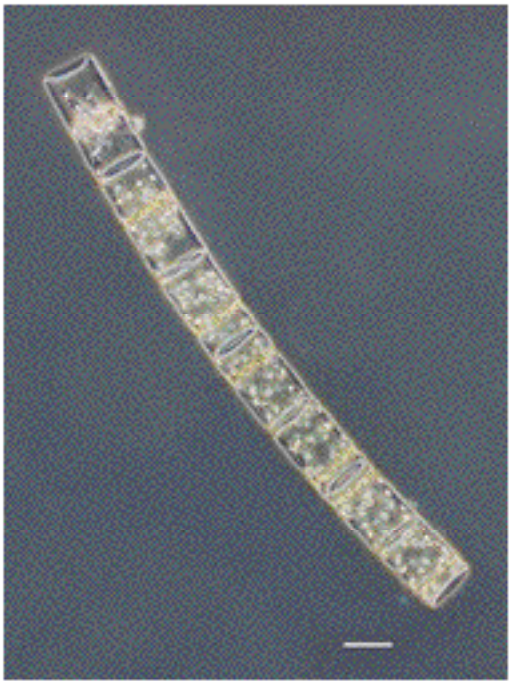


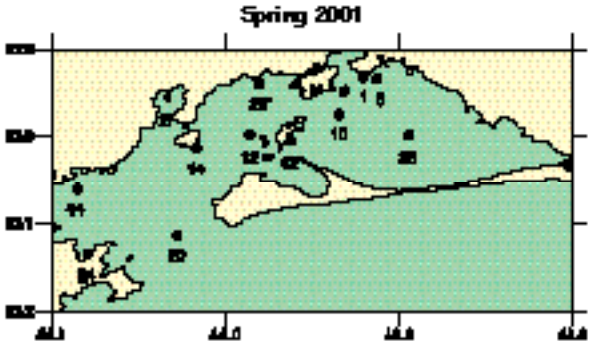
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 163).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	12.5 - 92	53 ± 12	76
PA	50.0 - 200	112 ± 298	76



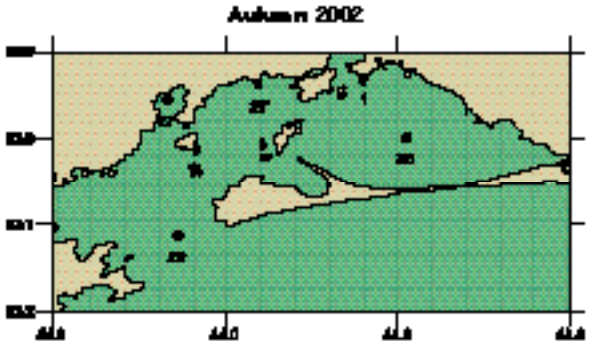


Plate 31

Division Bacillariophyta
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Guinardia striata (Stolterfoth) Hasle, 1996

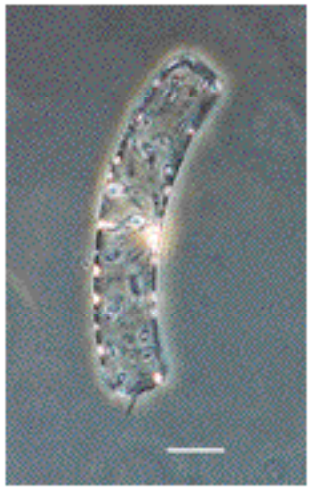


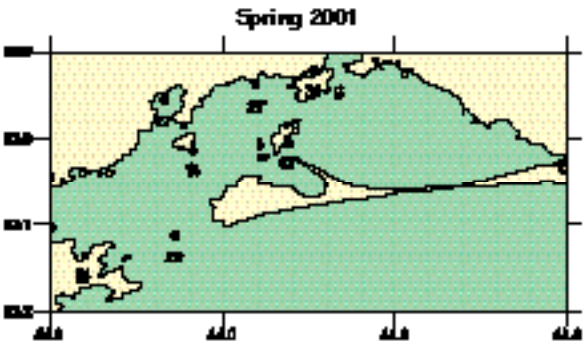
Figure 1. Cell in girdle view. Cells can often form curved, spiraling chains (LM, phase contrast, water mount). Scale bar = 20µm.

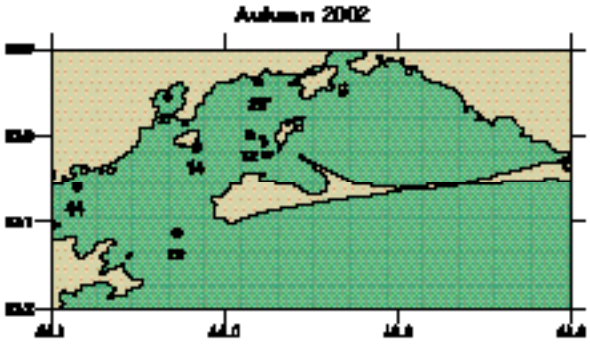
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 163). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	5 - 40	19 ± 29	30
PA	30 - 135	85 ± 9	30





Division Bacillariophyta Plate 32
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Proboscia alata (Brightwell) Sundström, 1986

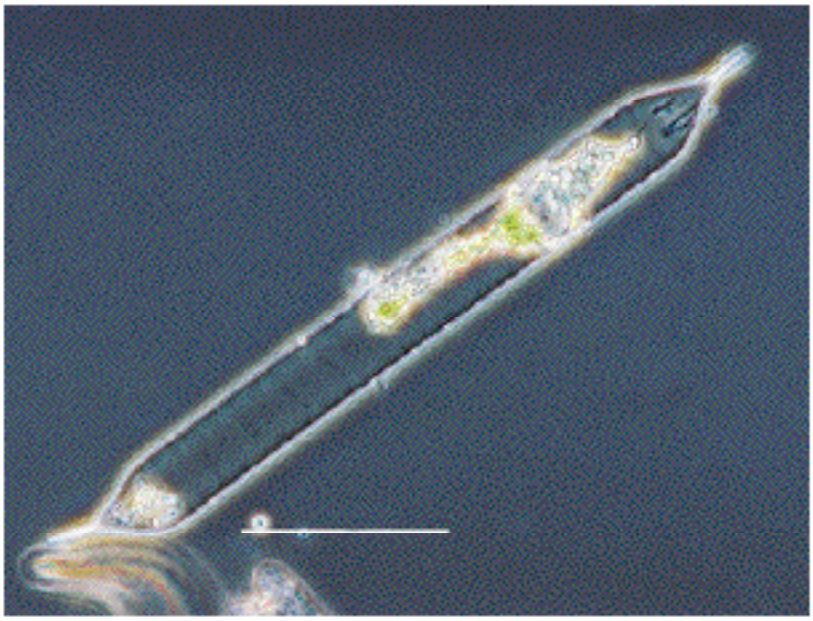


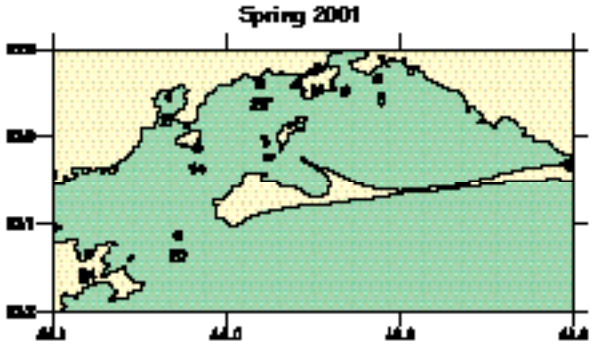
Figure 1. Cell in girdle view
(LM, phase contrast, water mount).
Scale bar = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 159).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	6	-	50	18	±	14	20
PA	295	-	650	442	±	120	20



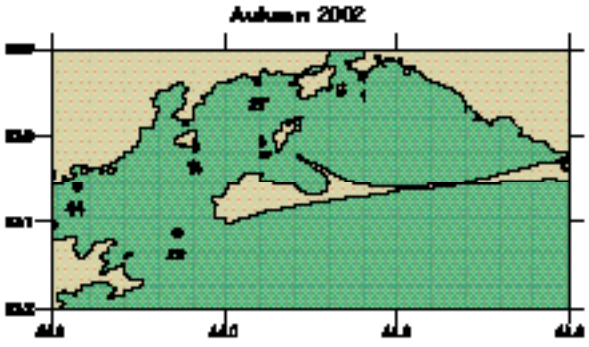


Plate 33

Division Bacillariophyta
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Pseudosolenia calcar-avis (Schultze) Sundström, 1986

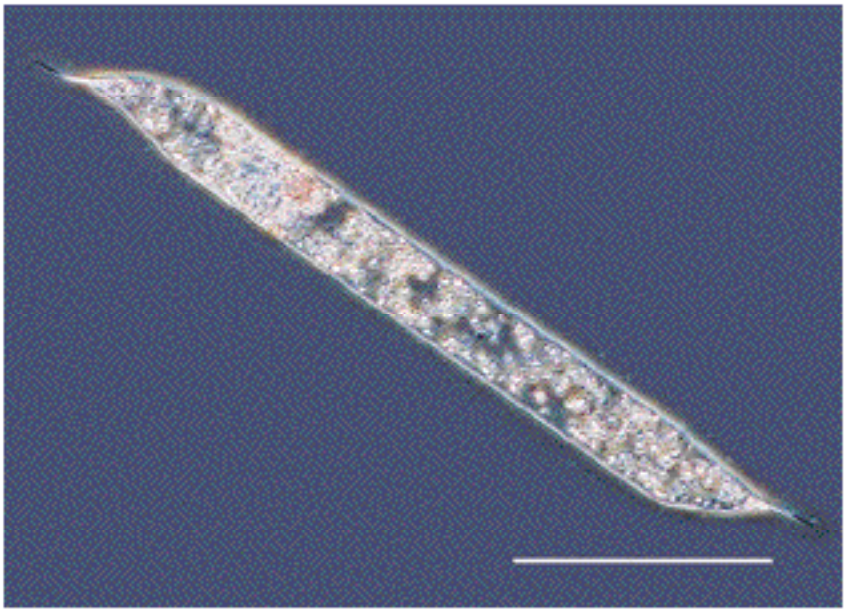


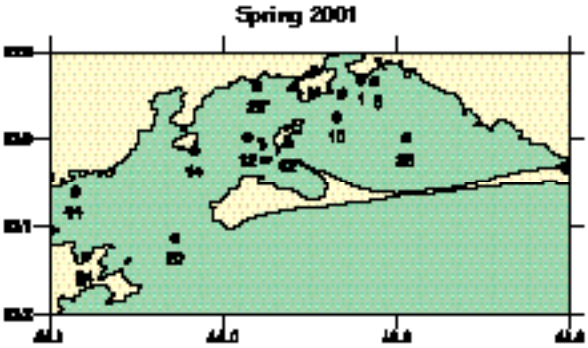
Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

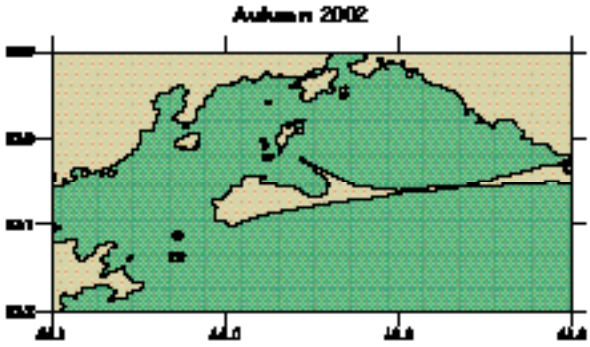
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 160). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	15 - 55	43 ± 7	33
PA	315 - 750	470 ± 95	33





Division Bacillariophyta Plate 34
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Rhizosolenia cf. *castracanei* H. Peragallo, 1888

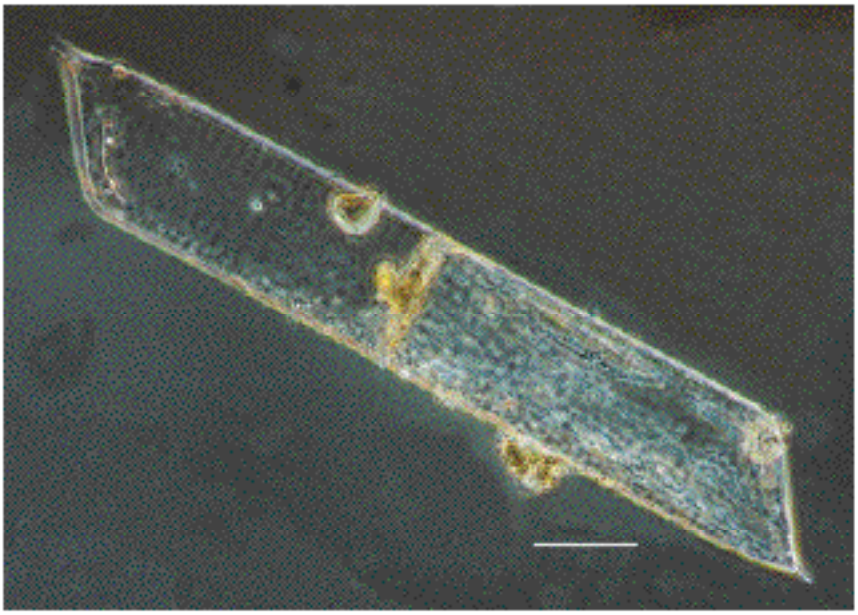


Figure 1. Cell in girdle view
(LM, phase contrast, water mount).
Scale bar = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 151).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	105 - 130	118 ± 18	2
PA	360 - 550	455 ± 134	2

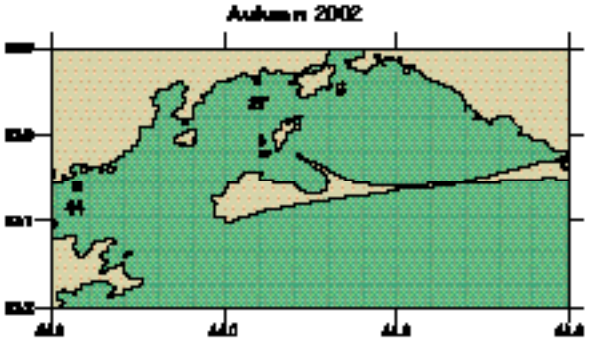


Plate 35

Division Bacillariophyta
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Rhizosolenia hebetata Bailey, 1856

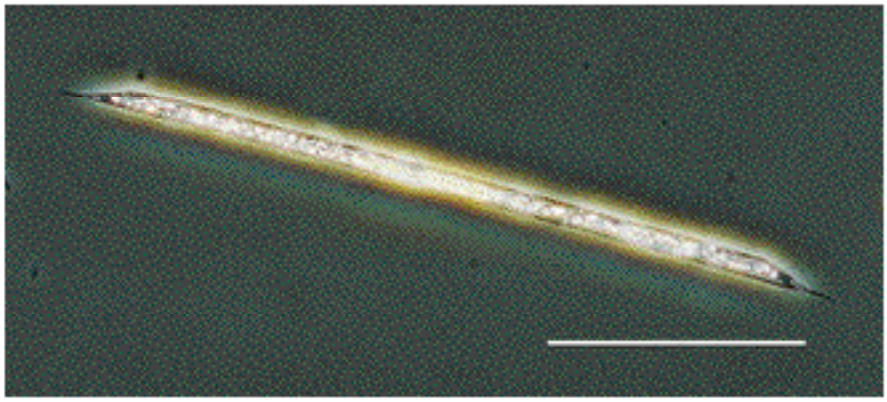


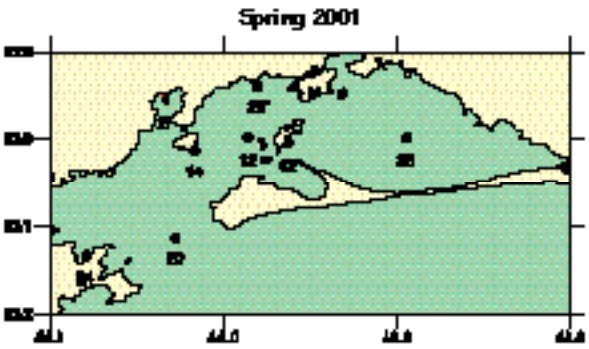
Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

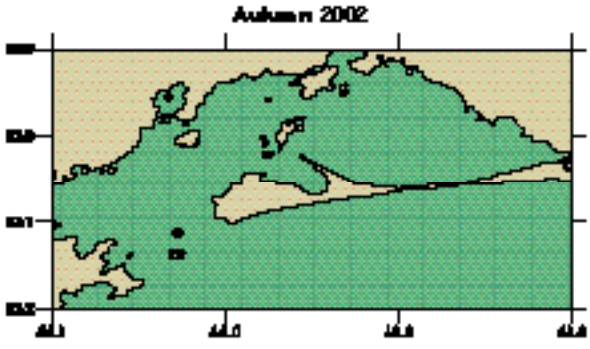
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 149). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	7 - 30	18 ± 9	15
PA	245 - 640	342 ± 113	15





Division Bacillariophyta Plate 36
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Rhizosolenia hyalina Ostenfeld, 1901

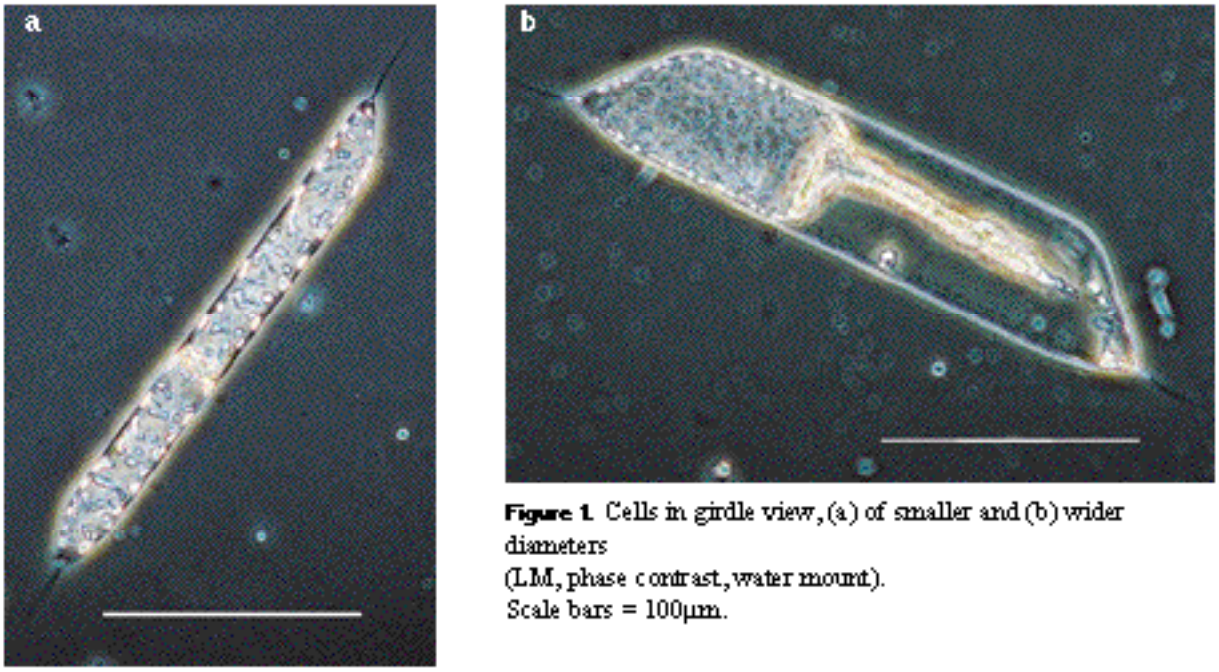


Figure 1. Cells in girdle view, (a) of smaller and (b) wider diameters
(LM, phase contrast, water mount).
Scale bars = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 151).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	13 - 50	24 ± 7	26
PA	125 - 340	222 ± 59	26

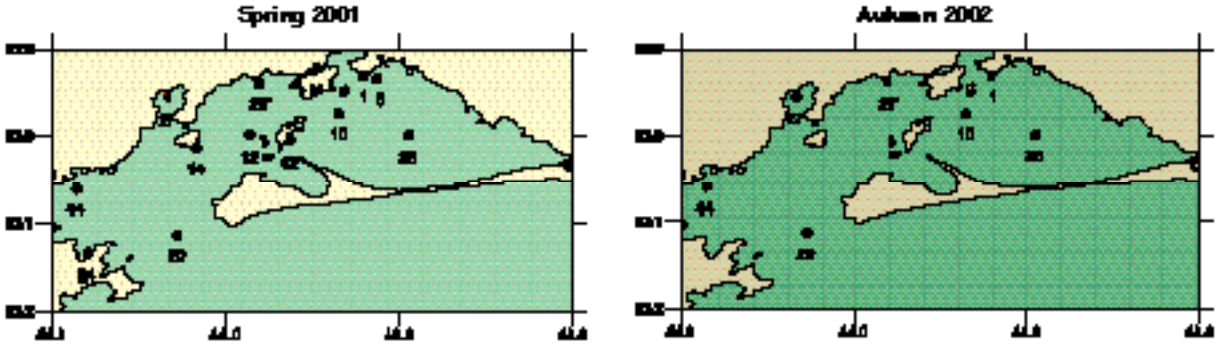


Plate 37

Division Bacillariophyta
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Rhizosolenia pungens Cleve-Euler,1937

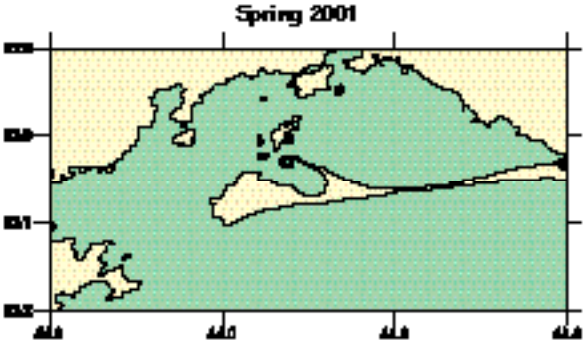
Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 157). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	8.0 - 15	11 ± 32	9
PA	182.5 - 280	233 ± 3	9



Division Bacillariophyta Plate 38
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Rhizosolenia robusta Norman ex Ralfs, 1861

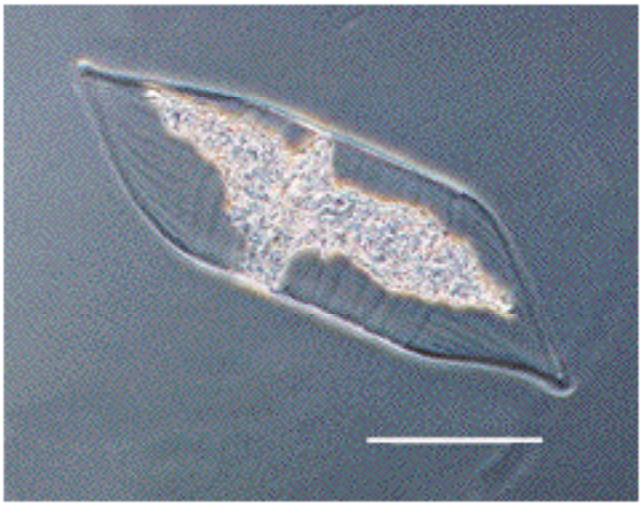


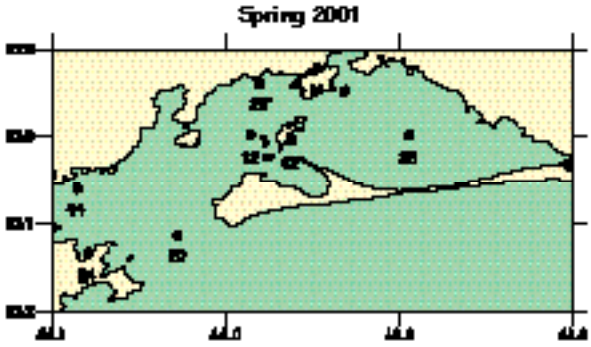
Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 159).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	16.0	-	170	74	±	33	29
PA	67.5	-	475	327	±	70	29



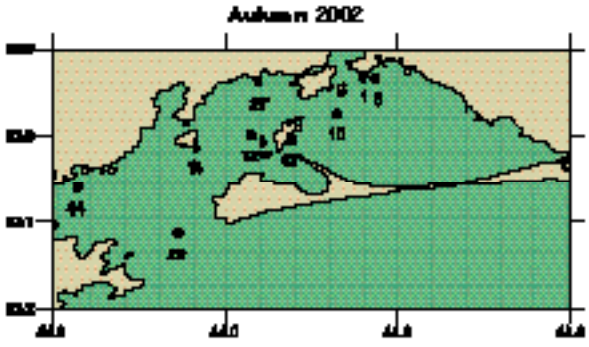


Plate 39

Division Bacillariophyta
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Rhizosolenia setigera Brightwell, 1858

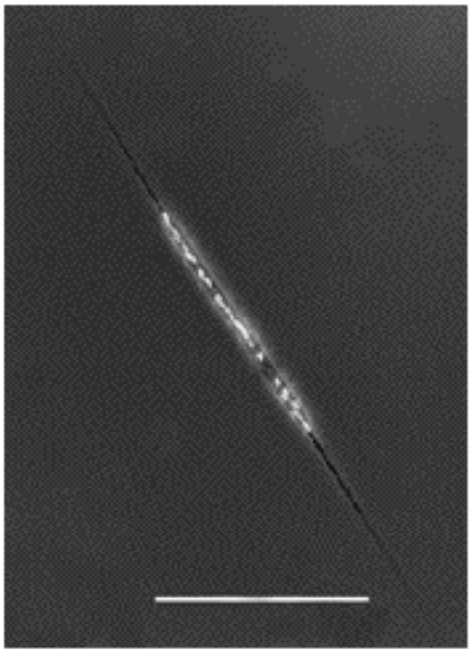


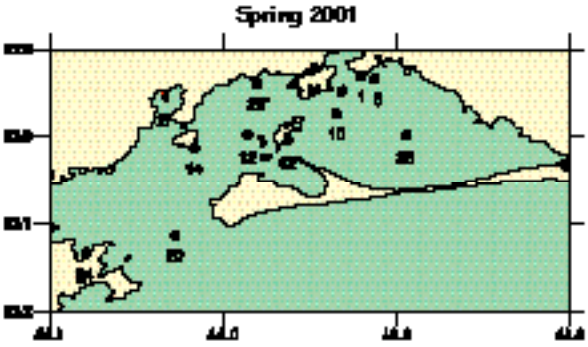
Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

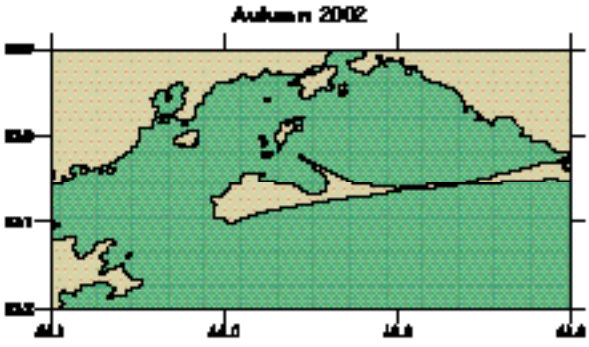
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 157). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	3 - 24.5	6 ± 5	30
PA	25 - 475.5	238 ± 100	30





Division Bacillariophyta Plate 40
Class Coscinodiscophyceae
Order Rhizosoleniales
Family Rhizosoleniaceae

Rhizosolenia cf. *styliformis* Brightwell, 1858

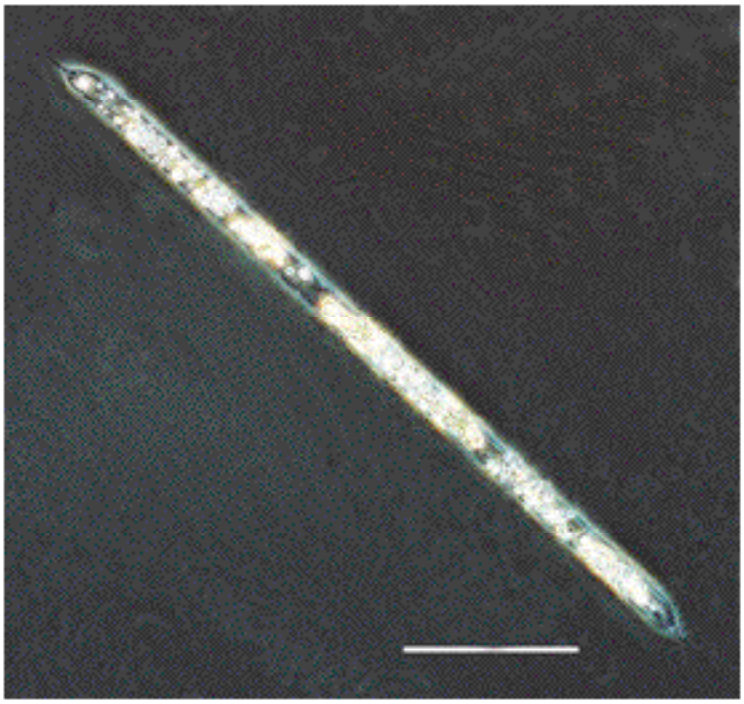


Figure 1. Cell in girdle view (LM, phase contrast, water mount). Scale bar = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 146).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	13 - 20	15 ± 3	5
PA	225 - 575	404 ± 124	5

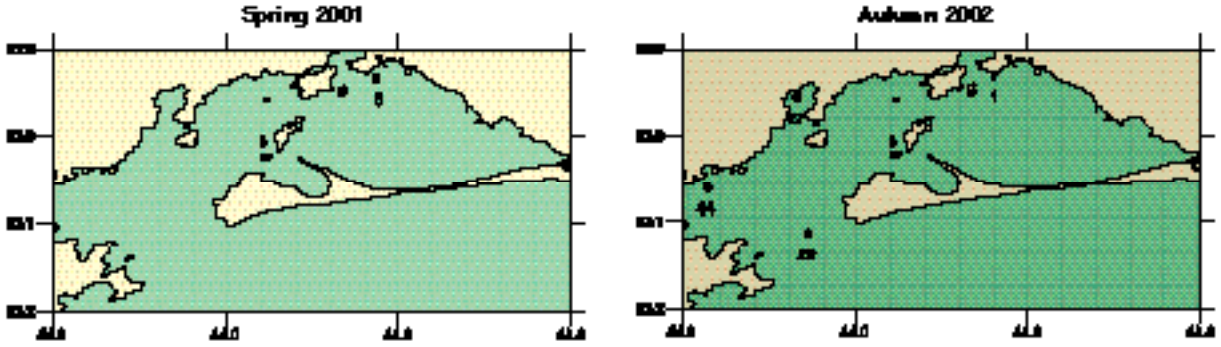


Plate 41

Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Bacteriastrum cf. *delicatulum* Cleve, 1897

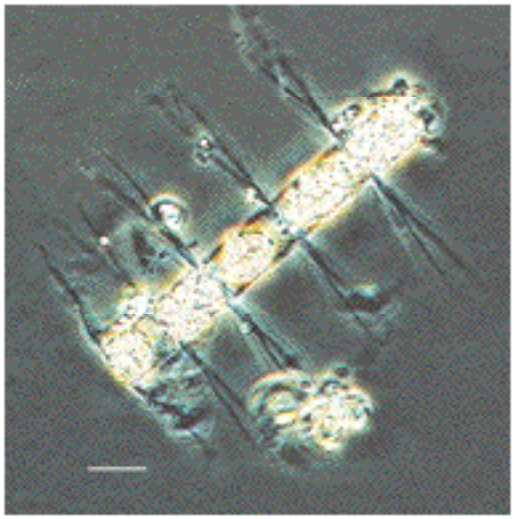


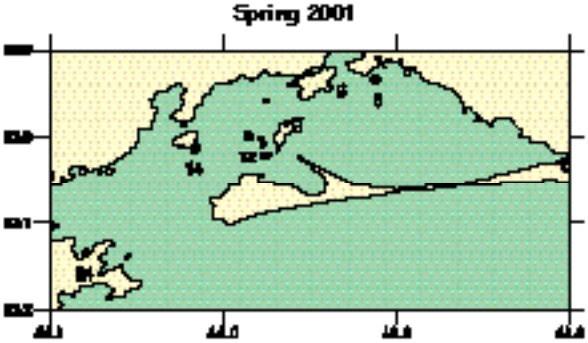
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

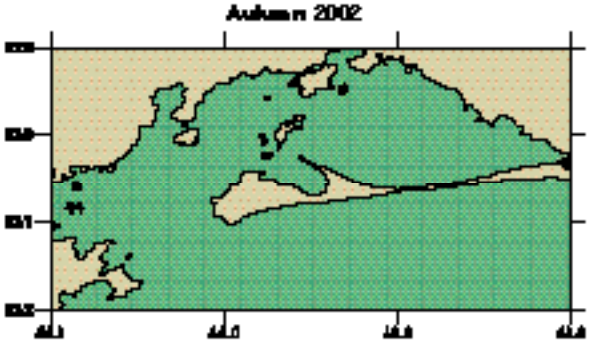
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p.188). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	20 - 32.5	22 ± 4	21
PA	20 - 35.0	28 ± 4	21





Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Plate 42

Bacteriastrum hyalinum Lauder, 1864

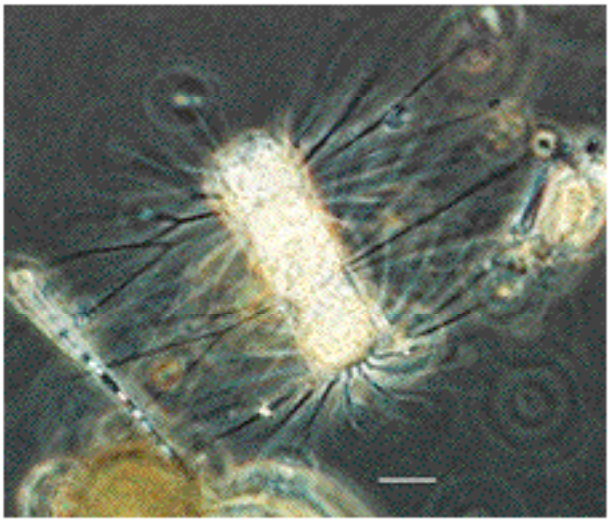


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 188).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	20	-	33	27	±	4	15
PA	20	-	40	27	±	6	13

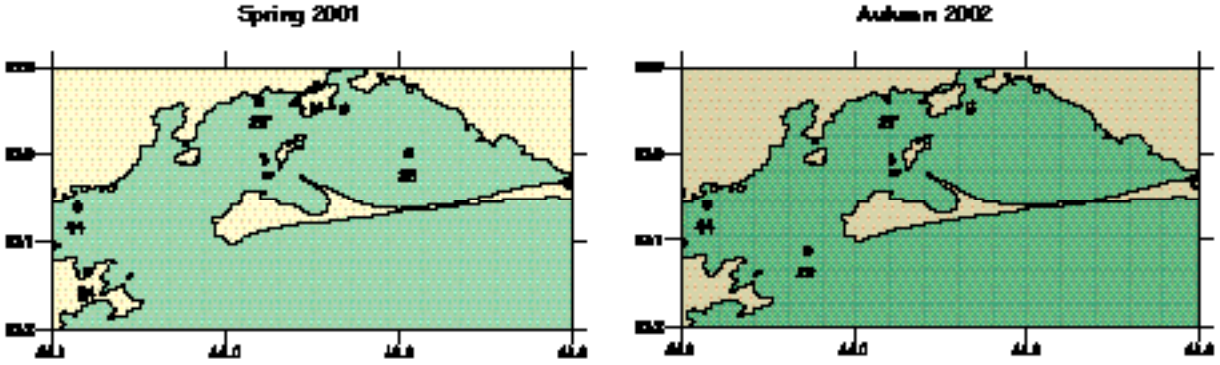


Plate 43

Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros aequatorialis Cleve, 1873

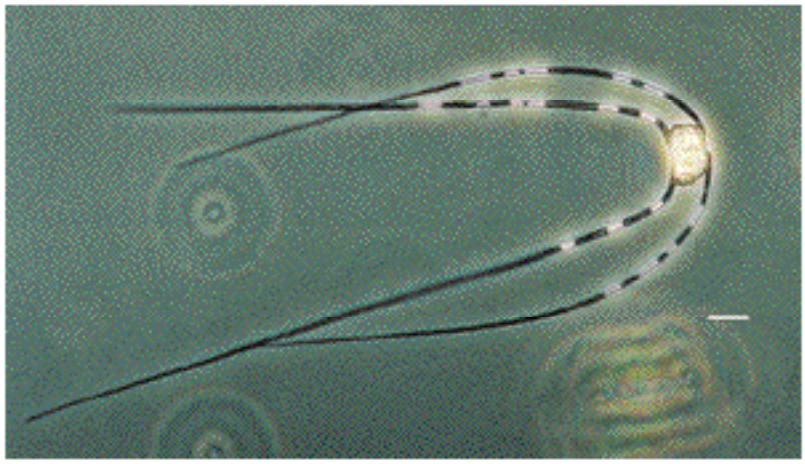
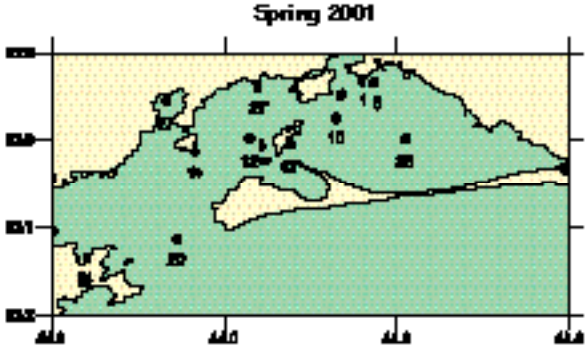


Figure 1. Solitary cell in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 193). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.



Division Bacillariophyta Plate 44
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros affinis Lauder, 1864

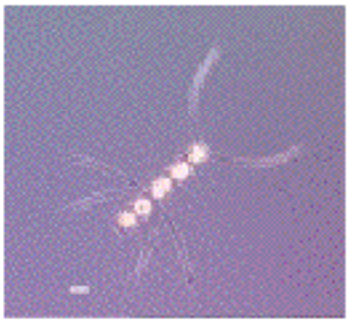


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 216).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego

Morphometrics

	range (µm)			mean ± SD			n
D	15	-	26	19	±	5	5
PA	15	-	20	18	±	3	5

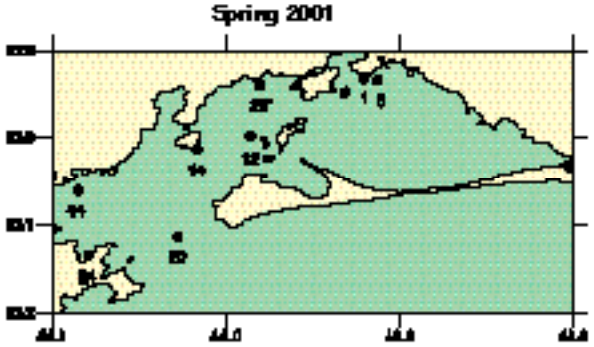


Plate 45

Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros cf. *brevis* Schütt, 1895

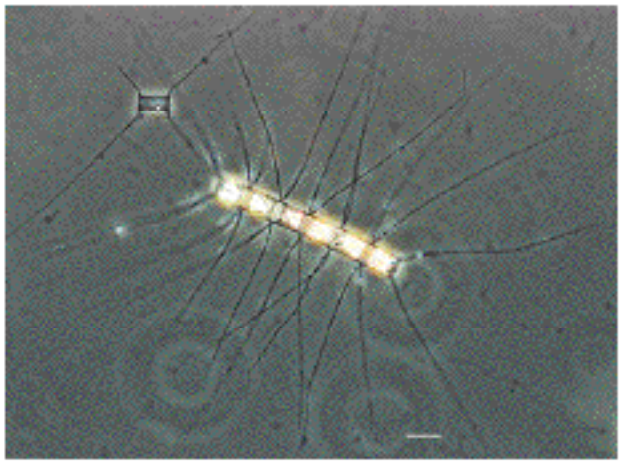


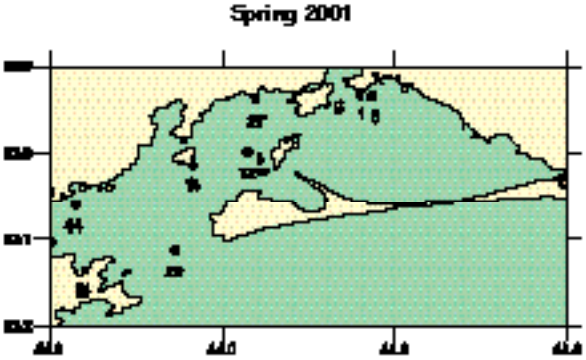
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Cupp (1943). Marine plankton diatoms of the west coast of North America (p. 129). University of California Press, Berkeley and Los Angeles.

Morphometrics

	range (µm)	mean ± SD	n
D	15 - 23.0	18 ± 3	5
PA	12 - 22.5	19 ± 3	5



Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Plate 46

Chaetoceros coarctatus Lauder,1864

Figure 1. Chain of cells in girdle view. Arrow indicates a *Vorticella* sp., an epiphytic flagellate usually found attached on to the cells (LM, phase contrast, water mount). Scale bar = 50µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 199).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	30 - 37	33 ± 4	6
PA	40 - 50	42 ± 5	5

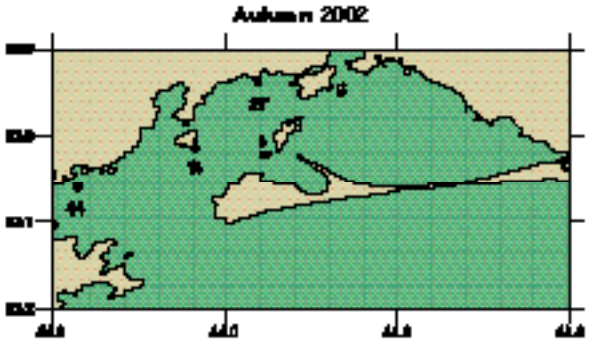


Plate 47

Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros cf. *concavicornis* Mangin, 1917

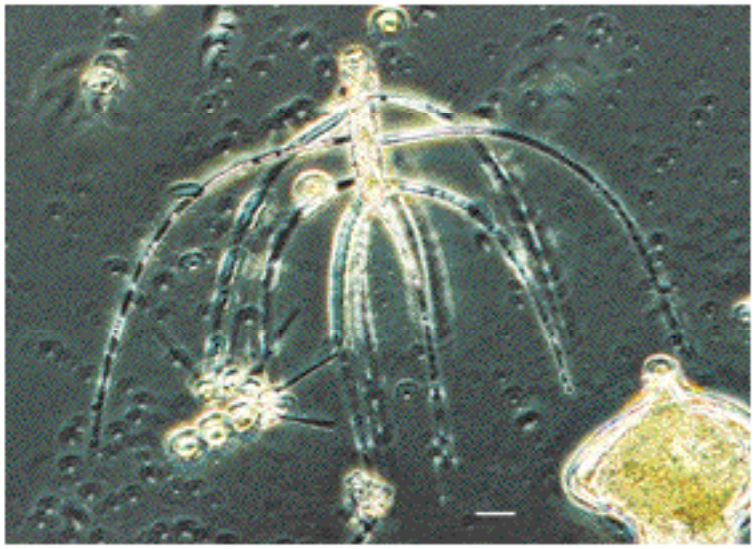


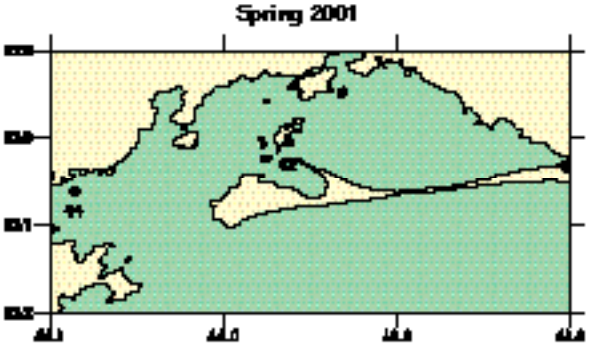
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 199). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	20 - 56	32 ± 16	10
PA	25 - 70	36 ± 13	10



Division Bacillariophyta Plate 48
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros curvisetus Cleve,1889

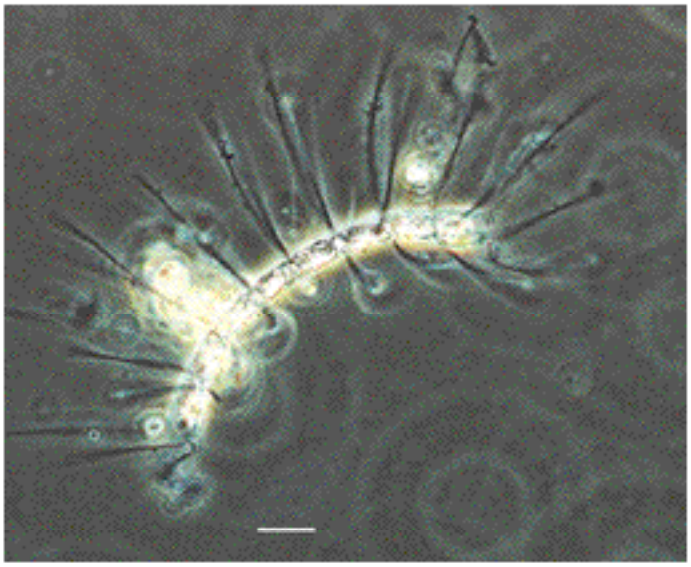


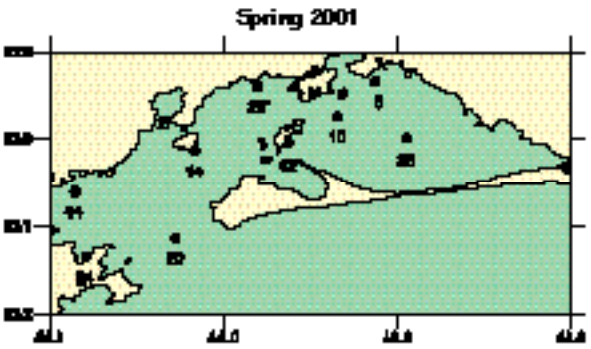
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 211).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	8 - 15	12 ± 2	33
PA	15 - 35	21 ± 6	33



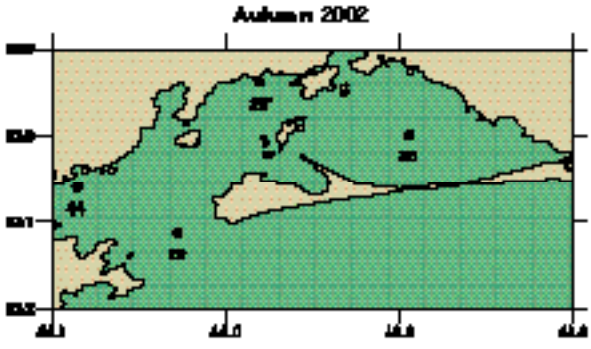


Plate 49

Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros danicus Cleve,1889

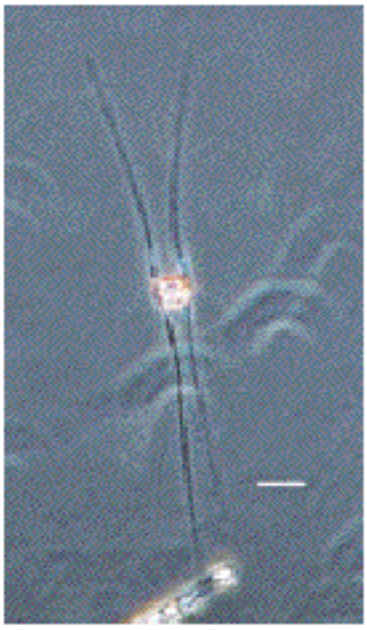


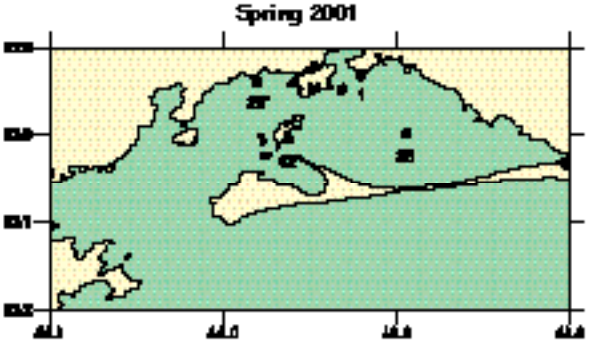
Figure 1. Solitary cell in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 195). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	10 - 22.5	14 ± 3	12
PA	9 - 20.0	13 ± 3	12



Division Bacillariophyta Plate 50
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros decipiens Cleve, 1873

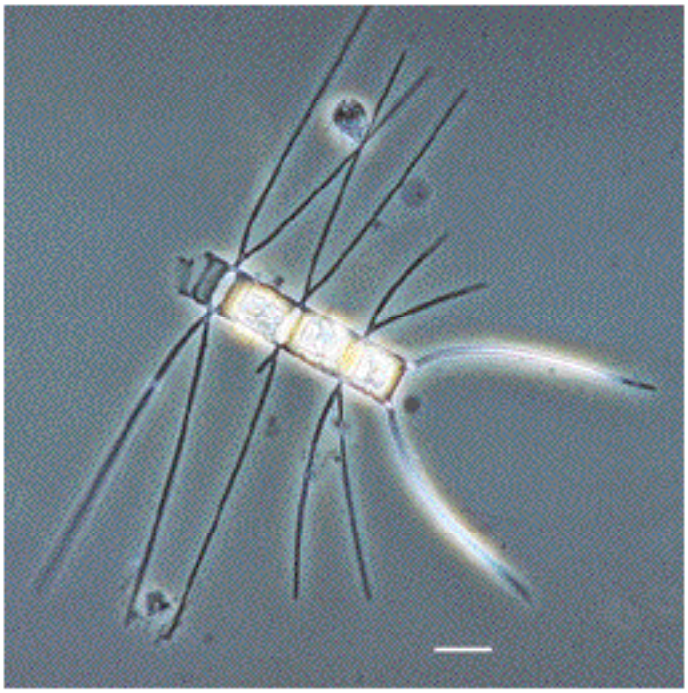


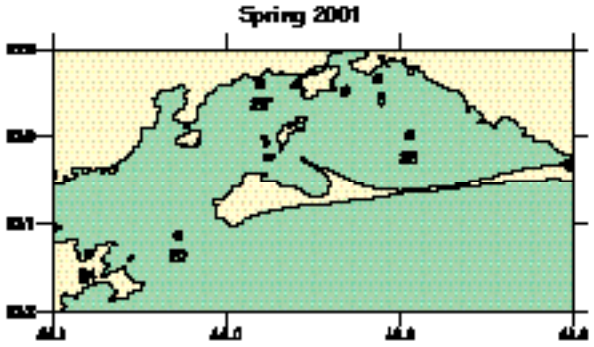
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 204).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	17.5 - 45	30 ± 9	21
PA	20.0 - 42	28 ± 6	21



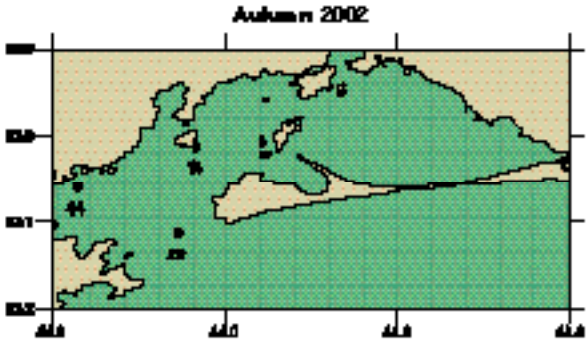


Plate 51

Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros didymus Ehrenberg, 1845

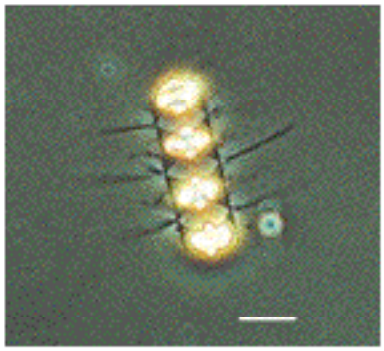


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

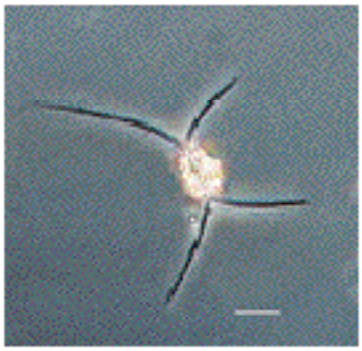


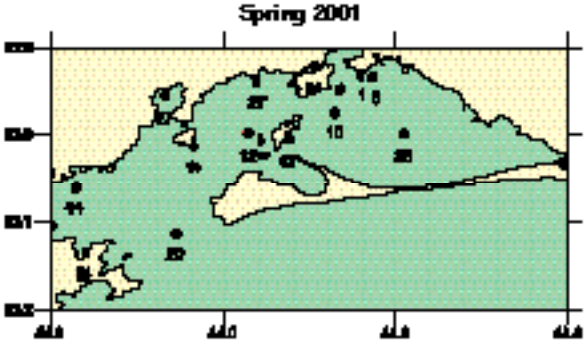
Figure 2. Typical resting spores free from parent cells (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 207). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	14.0 - 35	23 ± 6	18
PA	12.5 - 20	16 ± 2	18



Division Bacillariophyta Plate 52
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros peruvianus Brightwell, 1856

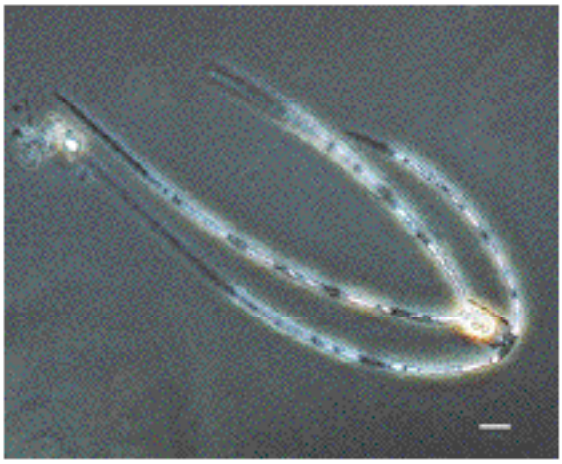


Figure 1. Solitary cell in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 195).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

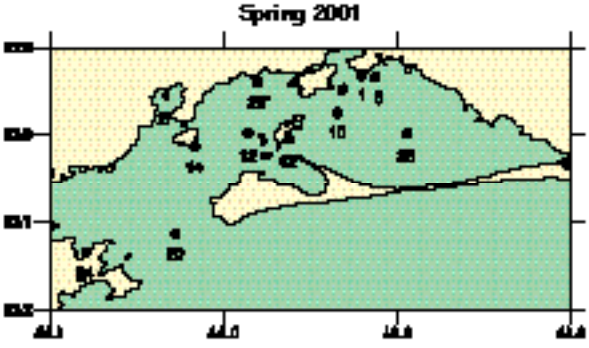


Plate 53

Division Bacillariophyta
Class Coscinodiscophyceae
Order Chaetocerotales
Family Chaetocerotaceae

Chaetoceros sp. 1

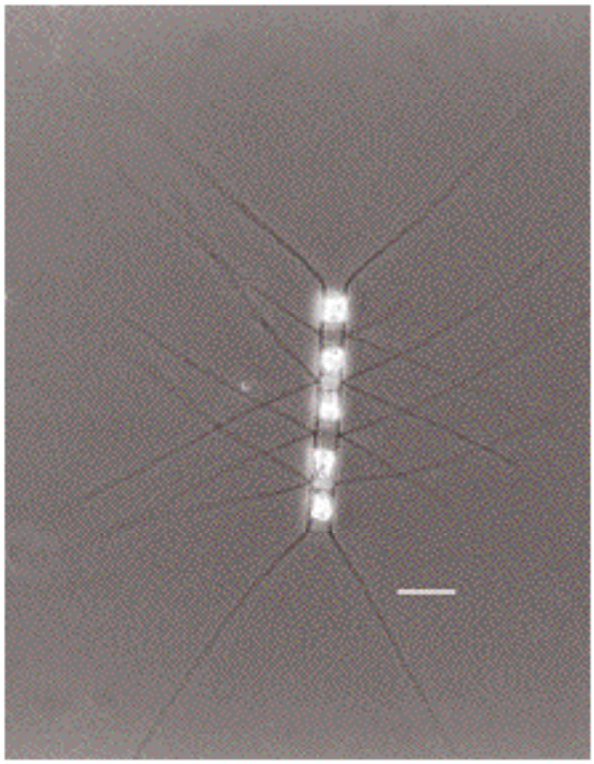
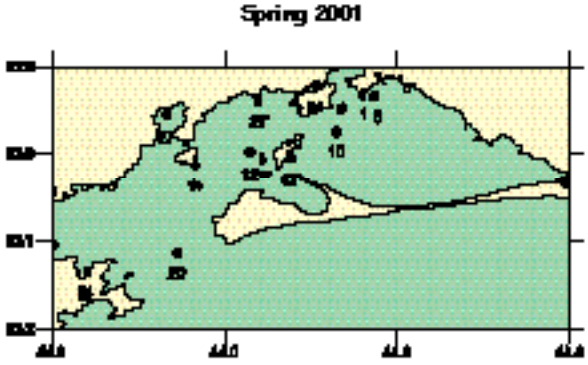


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bars = 20µm.

Morphometrics

	range (µm)			mean ± SD			n
D	10	-	20	12	±	3	18
PA	17	-	25	20	±	2	18



Division Bacillariophyta Plate 54
Class Coscinodiscophyceae
Order Leptocylindrales
Family Leptocylindraceae

Leptocylindrus danicus Cleve, 1889

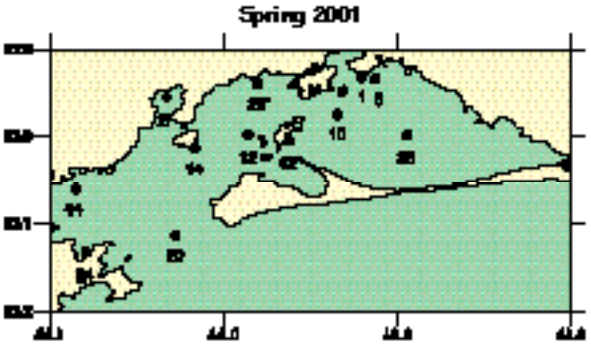
Figure 1. Chain of cells in girdle view; cells with numerous chloroplasts (LM, phase contrast, water mount). Scale bar = 50µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 93).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	5.0	-	20	10	±	3	110
PA	22.5	-	80	46	±	10	111



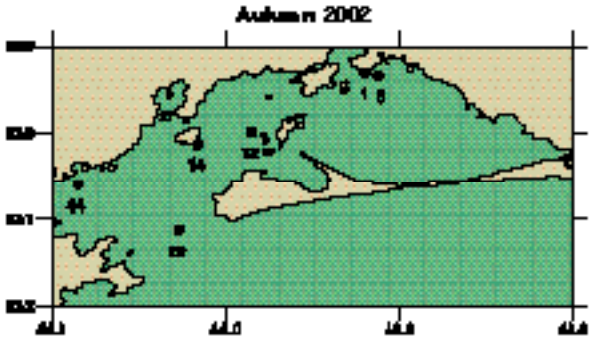


Plate 55

Division Bacillariophyta
Class Coscinodiscophyceae
Order Leptocylindrales
Family Leptocylindraceae

Leptocylindrus minimus Gran, 1915

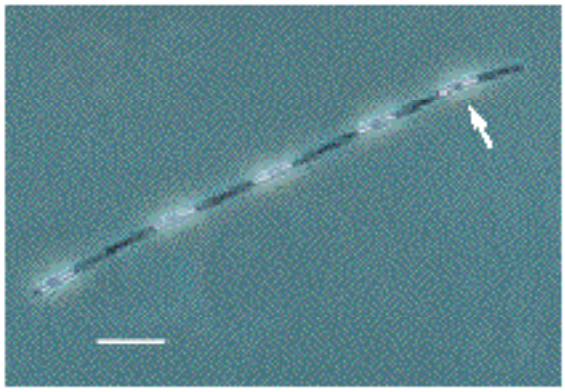


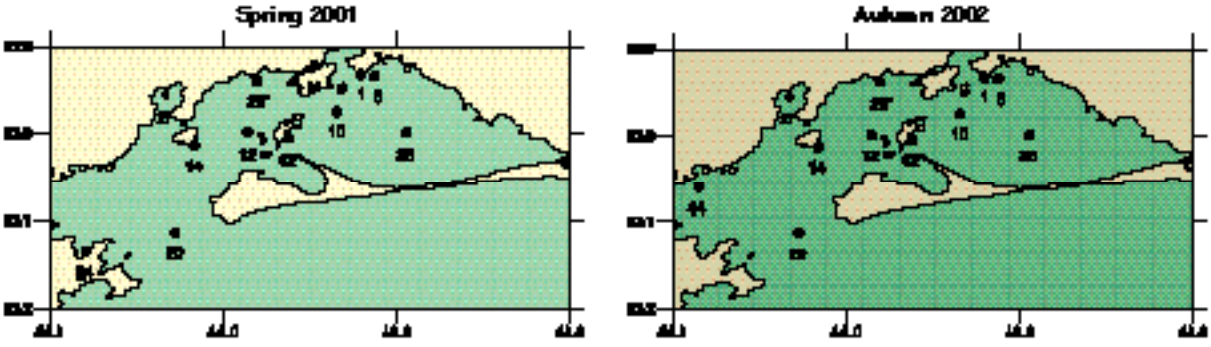
Figure 1. Chain of cells in girdle view; arrow indicates presence of 2 chloroplasts per cell (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 94). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
D	2 - 5	3 ± 0.8	100
PA	4 - 65	42 ± 10.0	101



Division Bacillariophyta
Class Fragilariophyceae
Order Fragilariales
Family Fragilariaceae

Plate 56

Asterionellopsis glacialis (Castracane) Round, 1990

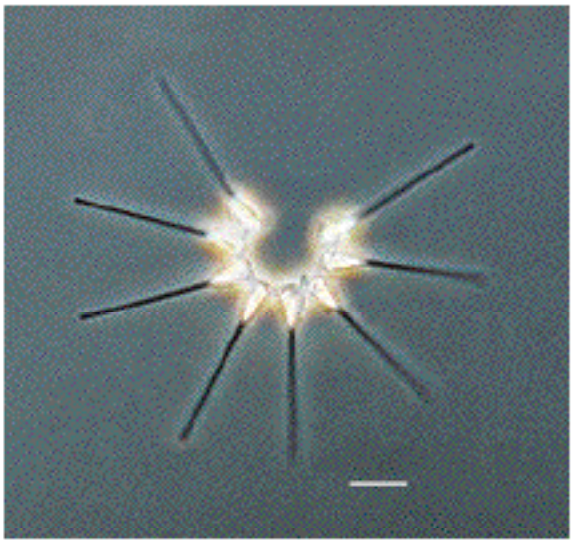


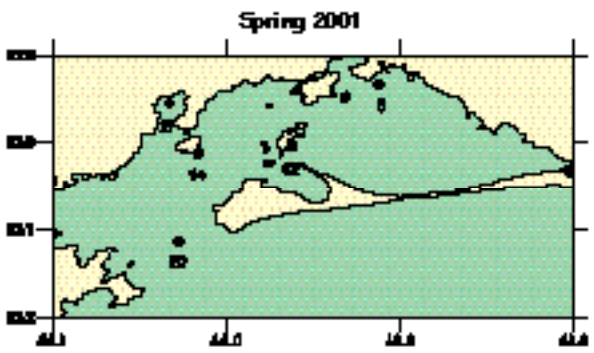
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 243).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
D	50	-	60	55	±	4	7
PA	7	-	15	10	±	2	7



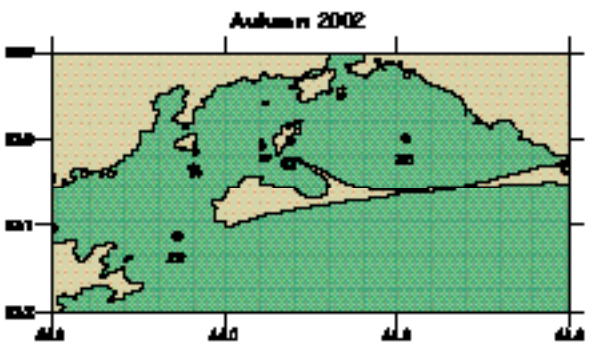


Plate 57

Division Bacillariophyta
Class Fragilariophyceae
Order Rhaphoneidales
Family Rhaphoneidaceae

Delphineis sp.

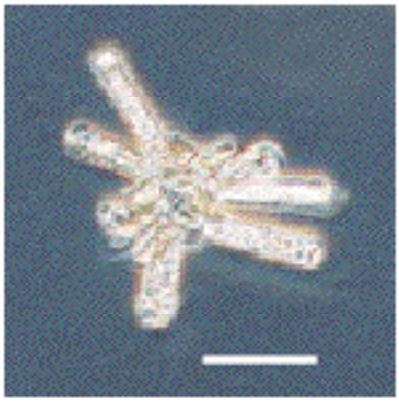


Figure 1. Chain of cells in girdle view, attached to a particle (LM, phase contrast, water mount). Scale bar = 50µm.

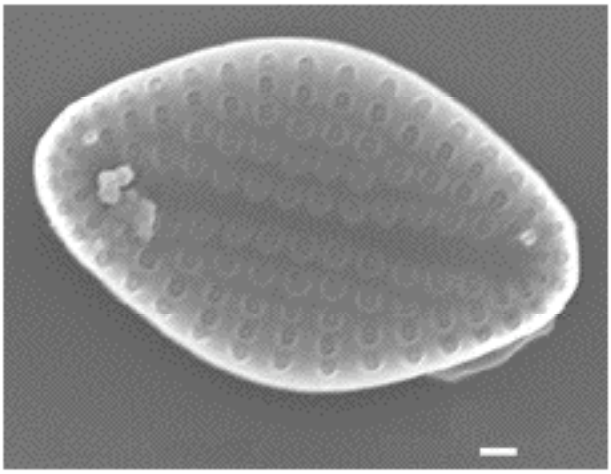
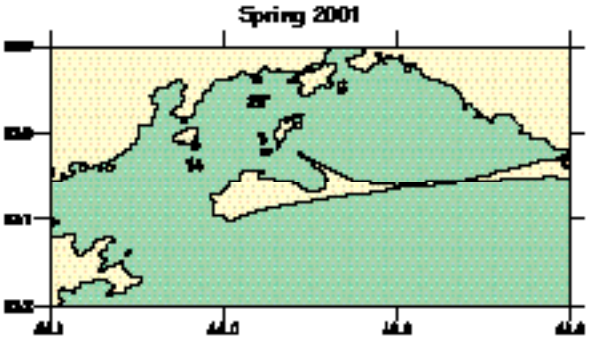
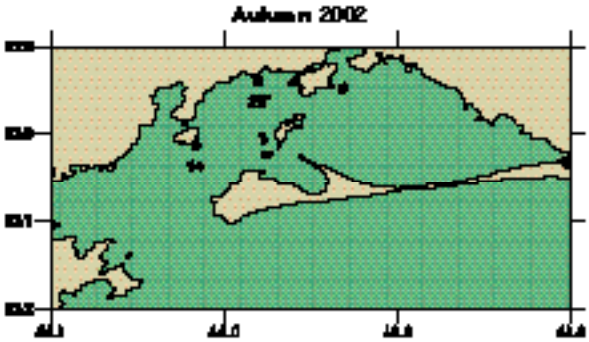


Figure 2. Single valve (SEM). Scale bar = 1µm.





Division Bacillariophyta
Class Fragilariophyceae
Order Thalassionematales
Family Thalassionemataceae

Plate 58

Lioloma pacificum (Cupp) Hasle, 1996

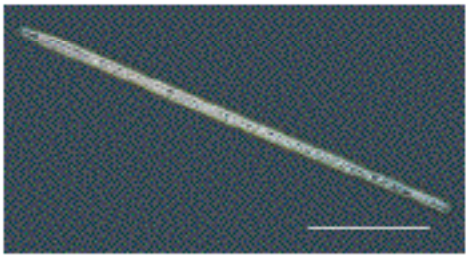


Figure 1. Cell in valve view (LM, phase contrast, water mount). Scale bar = 100µm.

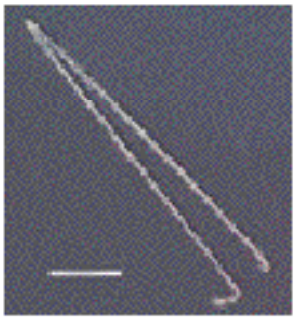


Figure 2. Colony of cells in valve view (LM, phase contrast, water mount). Scale bar = 20µm.

Figure 3. Detail of a head pole with two apical spines (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 10µm.

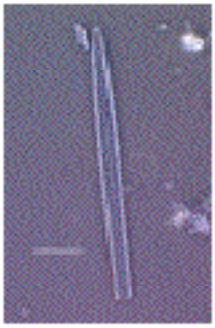


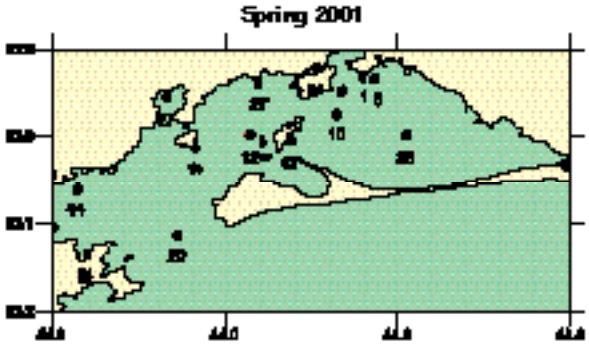
Figure 4. Detail of the foot pole of a valve (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 10µm.

Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 257).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
AA	257.50	-	990	530	±	2	62
TA	1.25	-	11	8	±	171	62



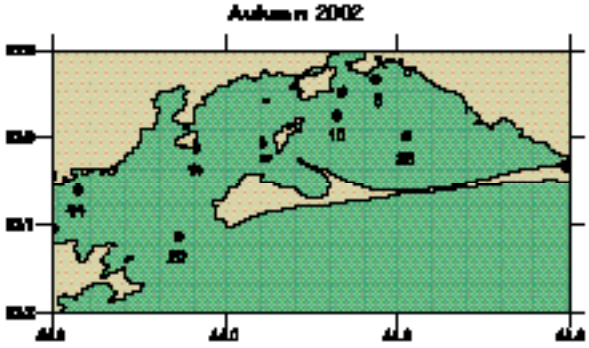


Plate 59

Division Bacillariophyta
Class Fragilariophyceae
Order Thalassionematales
Family Thalassionemataceae

Thalassionema nitzschioides (Grunow) Grunow ex Hustedt, 1932

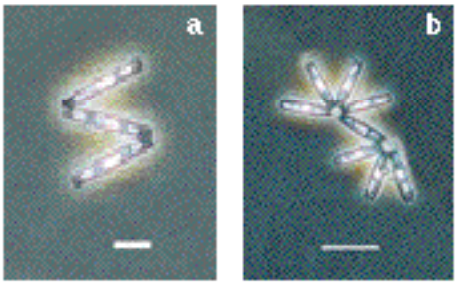


Figure 1. Chains of cells in girdle view, showing (a) zig-zag formation and (b) fan-like formation (LM, phase contrast, water mount). Scale bars = (a) 10µm; (b) 20µm.

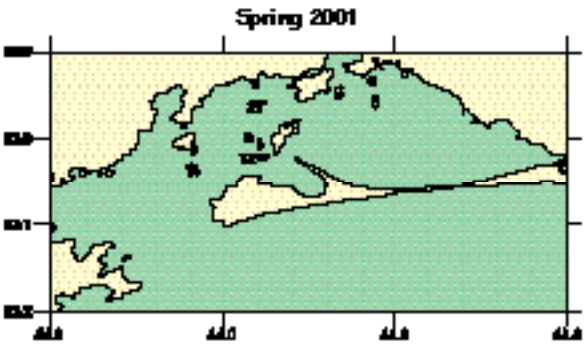
Figure 2. Single valve (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 10µm.

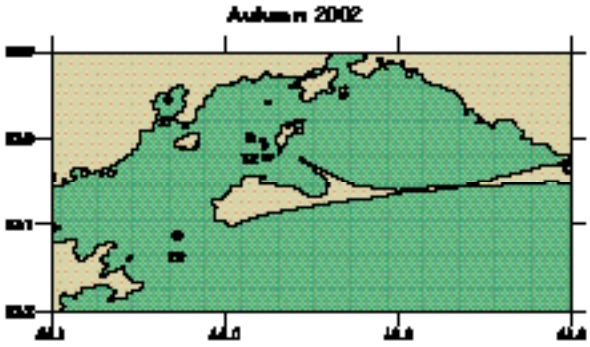
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 261). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
AA	20	-	75	49	±	20	28
TA	4	-	10	6	±	2	28





Division Bacillariophyta
Class Fragilariophyceae
Order Thalassionematales
Family Thalassionemataceae

Plate 60

Thalassionemataceae 1

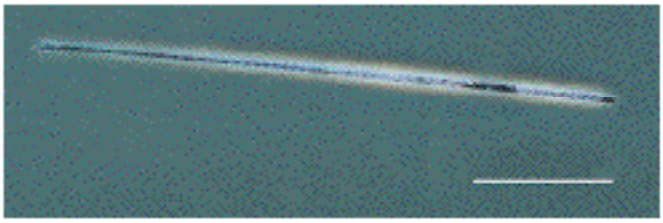


Figure 1. Cell in valve view
(LM, phase contrast, water mount).
Scale bar = 100µm.

Morphometrics

	range (µm)			mean ± SD			n
AA	270.0	-	550	423	±	76	36
TA	1.5	-	6	4	±	1	36

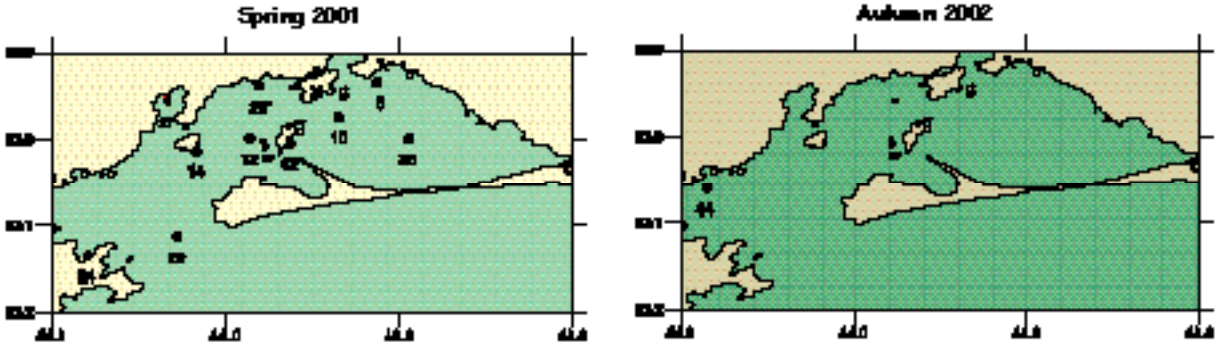


Plate 61

Division Bacillariophyta
Class Bacillariophyceae
Order Naviculales
Family Naviculaceae

Haslea wawrikae (Hustedt) Simonsen, 1974

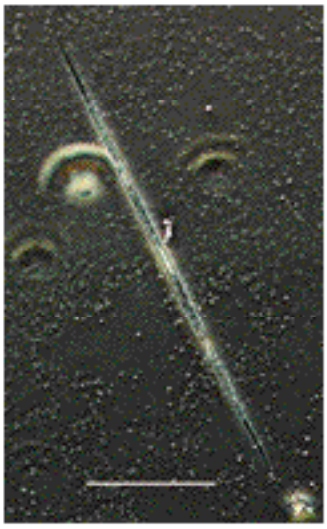


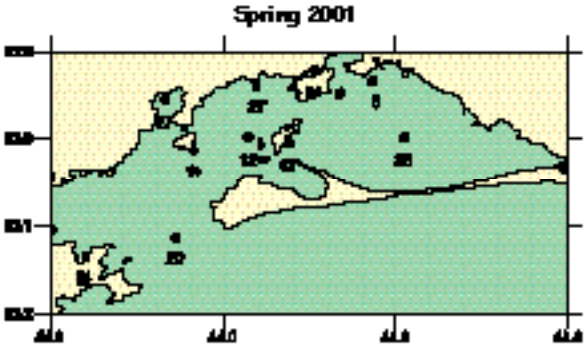
Figure 1. Cell in valve view (LM, phase contrast, water mount). Scale bar = 100µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 278). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
AA	295 - 625	392 ± 52	31
TA	4 - 8	6 ± 1	31



Division Bacillariophyta
Class Bacillariophyceae
Order Naviculales
Family Naviculaceae

Plate 62

Haslea sp.1

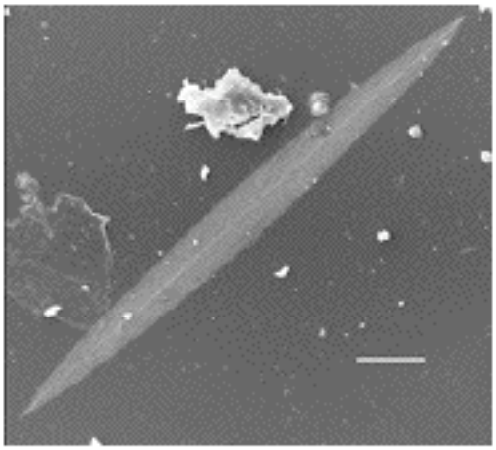


Figure 1. Single valve, internal view (SEM). Scale bar = 20µm.

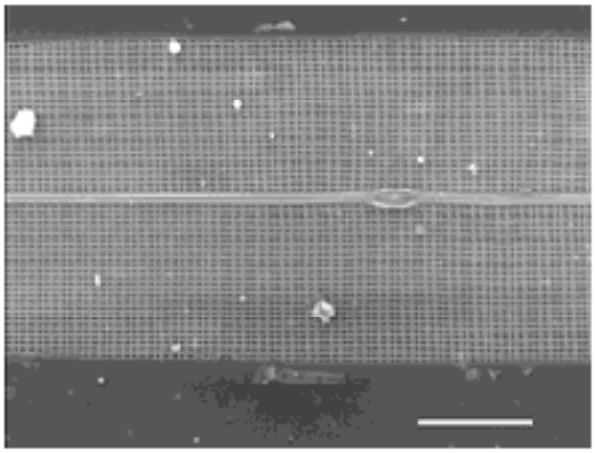


Figure 2. Detail of the central part of the valve (SEM). Scale bar = 5µm.

Plate 63

Division Bacillariophyta
Class Bacillariophyceae
Order Naviculales
Family Naviculaceae

Meuniera membranacea (Cleve) P.C. Silva, 1996

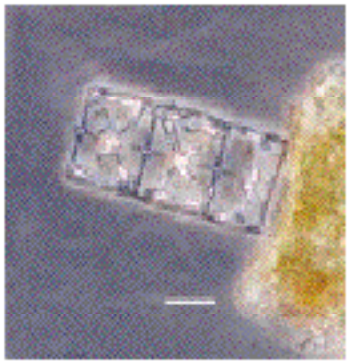


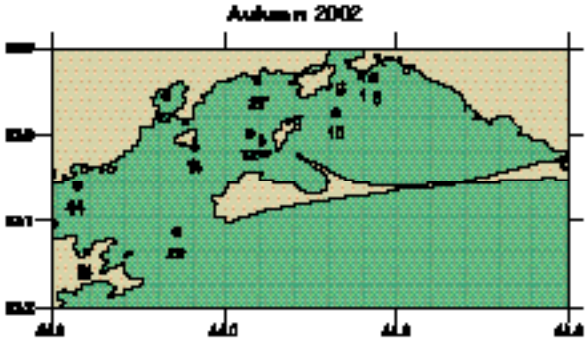
Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 273). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

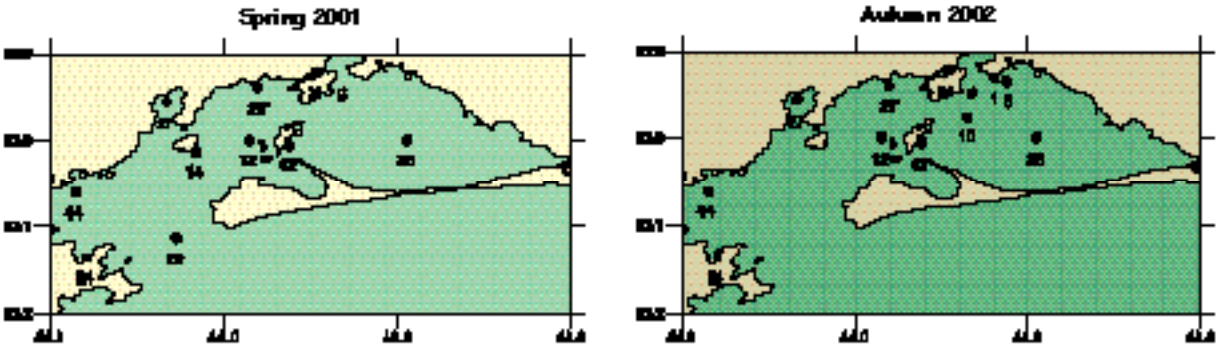
	range (µm)			mean ± SD			n
AA	20	-	60	33	±	6	53
TA	30	-	67	48	±	11	53



Division Bacillariophyta
Class Bacillariophyceae
Order Naviculales
Family Pleurosigmataceae

Plate 64

Pleurosigmataceae



The distribution of the Pleurosigmataceae includes taxa of the genera *Pleurosigma*, *Gyrosigma* and *Donkinia*, as presented in Plates 64-67.

Pleurosigma angulatum
(Quekett) Wm. Smith, 1852

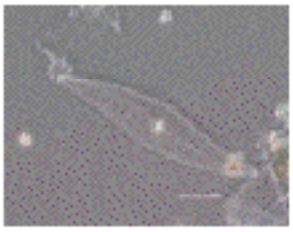


Figure 1. Single valve (LM, phase contrast, Hyrax mount of cleaned material).
Scale bar = 20µm.

Pleurosigma diverse-striatum
Meister, 1935

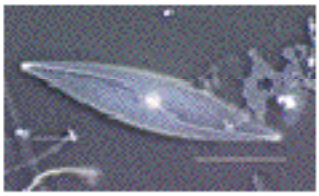


Figure 2. Single valve (LM, phase contrast, Hyrax mount of cleaned material).
Scale bar = 20µm.

Taxonomic source

Navarro, N. (1982). A survey of the Marine diatoms of Puerto Rico V. Suborder Raphidineae: Families Achnanthaceae and Naviculaceae (excluding Navicula and Mastogloia) (p. 325). Botanica Marina, 25.

Pleurosigma cf. *strigosum*
Wm. Smith, 1852

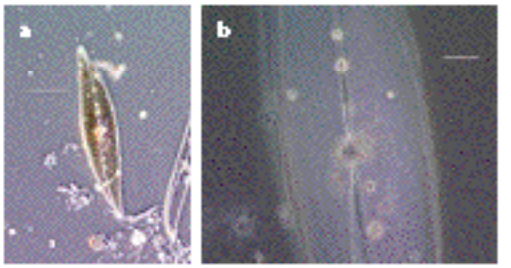


Figure 3. (a) Whole valve and (b) detail of the central raphe endings.
(LM, phase contrast, Hyrax mount of cleaned material).
Scale bars = (a) 50µm; (b) 10µm.

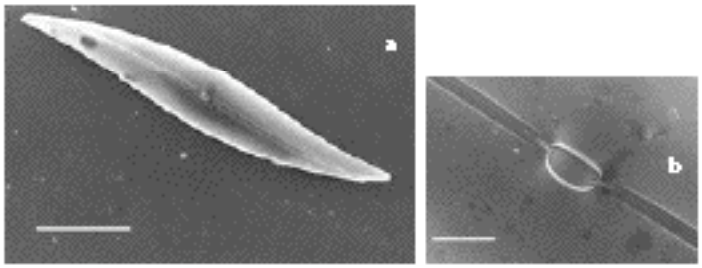


Figure 4. (a) Inside view of whole valve and (b) of a detail of the central raphe endings (SEM).
Scale bars = (a) 50µm; (b) 5µm.

Taxonomic source

Peragallo, H. & Peragallo, M. (1965). Diatomées marines de France et des districts maritimes voisins (p.163). A. Ascher & Co. Amsterdam.
Cardinal, A.; Poulin, M. & Bérard-Therriault (1989). New criteria for species characterization in the genera *Donkinia*, *Gyrosigma* and *Pleurosigma* (Naviculaceae, Bacillariophyceae) (p.15-27). Phycologia. Vol 28 (1).

Plate 65

Division Bacillariophyta
Class Bacillariophyceae
Order Naviculales
Family Pleurosigmataceae

Pleurosigmataceae

Pleurosigma sp. 1

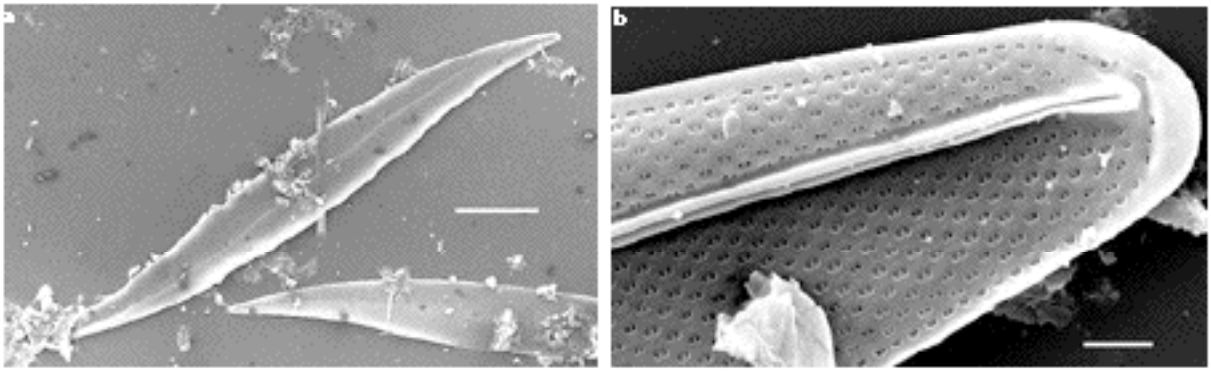


Figure 1. (a) Inside view of whole valve and (b) of a detail of the valve apex with terminal raphe ending (SEM). Scale bars = (a) 50µm; (b) 2µm.

Pleurosigma sp. 2

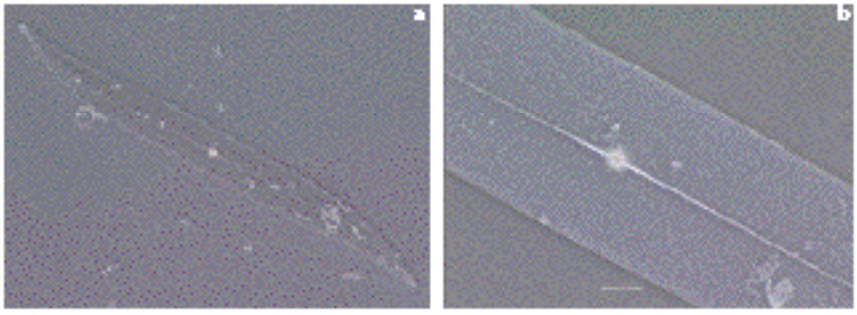


Figure 2. (a) Whole valve and (b) detail of the central raphe endings. (LM, phase contrast, Hyrax mount of cleaned material). Scale bars = (a) 20µm; (b) 10µm.

Pleurosigma sp. 3

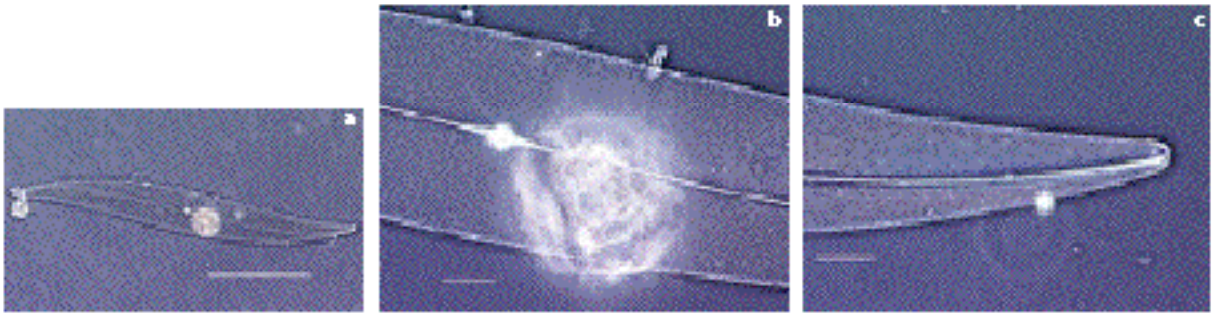


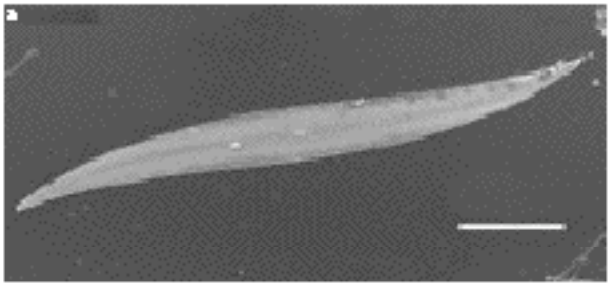
Figure 3. (a) Whole valve, (b) detail of the central raphe endings, and (c) of the valve apex with terminal raphe ending (LM, phase contrast, Hyrax mount of cleaned material). Scale bars = (a) 100µm; (b) 10µm; (c) 10µm.

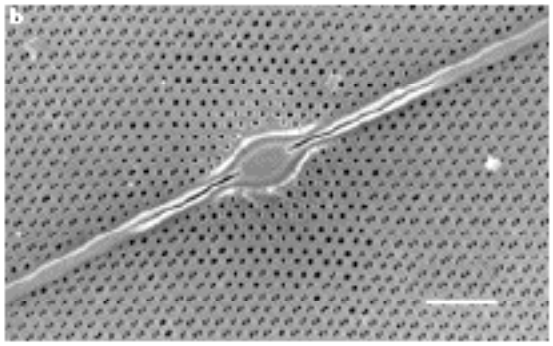
Division Bacillariophyta
Class Bacillariophyceae
Order Naviculales
Family Pleurosigmataceae

Plate 66

Pleurosigmataceae

Pleurosigma sp. 4





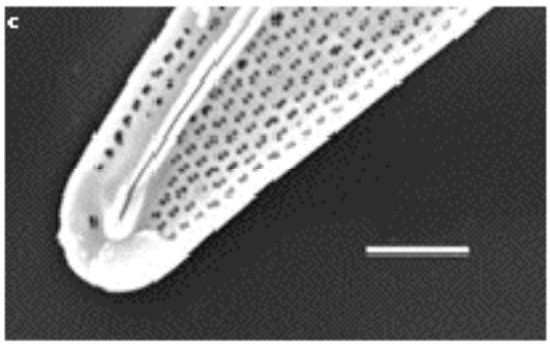


Figure 1.
(a) Inside view of whole valve, (b) of a detail of the central raphe endings, and (c) of the valve apex with terminal raphe ending (SEM).
Scale bars = (a) 50µm; (b) 2µm; (c) 2µm.

Pleurosigmataceae 1

Figure 3. Single valve
(LM, phase contrast, Hyrax mount of cleaned material).
Scale bar = 10µm.

Pleurosigmataceae 2

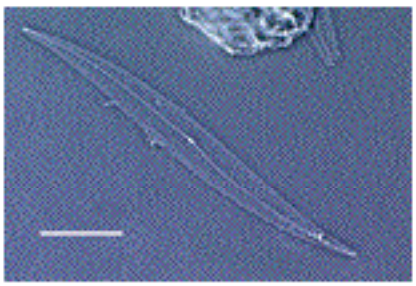


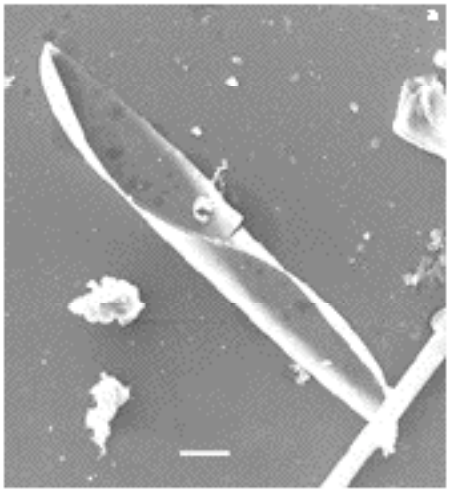
Figure 2. Single valve
(LM, phase contrast, Hyrax mount of cleaned material).
Scale bar = 50µm.

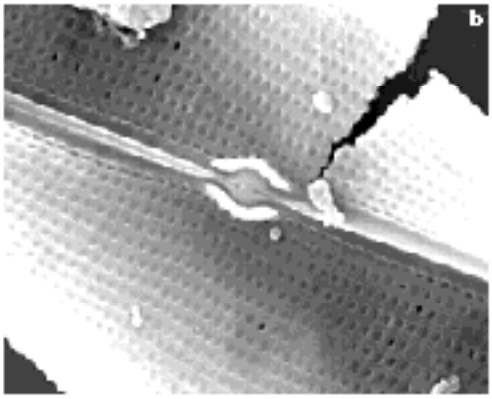
Plate 67

Division Bacillariophyta
Class Bacillariophyceae
Order Naviculales
Family Pleurosigmataceae

Pleurosigmataceae

Donkinia sp.





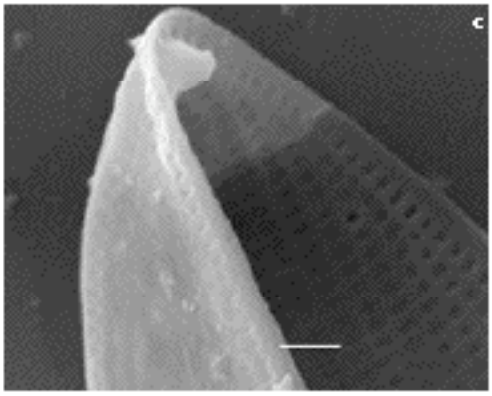
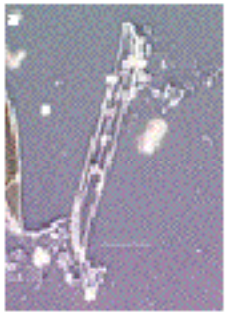


Figure 1. (a) Inside view of whole valve, (b) of a detail of the central raphe endings and (c) of the valve apex with terminal raphe ending (SEM).
Scale bars = (a) 10µm; (b) 1µm; (c) 1µm.

Gyrosigma balticum (Ehrenberg) Rabenhorst, 1853



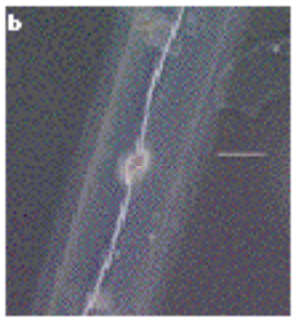


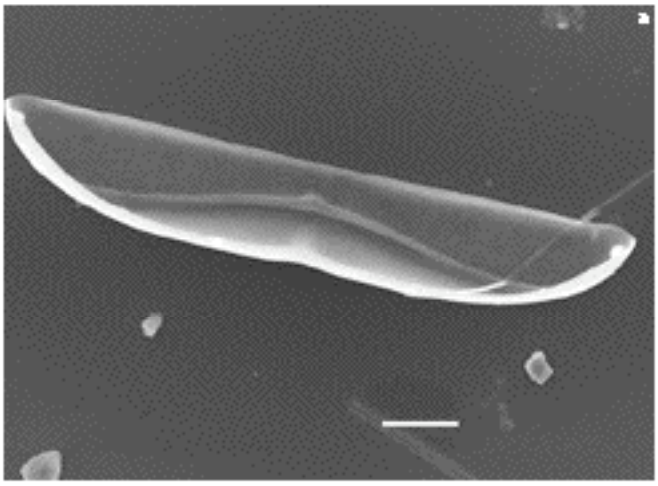
Figure 2. (a) Whole valve and (b) detail of the central raphe endings
(LM, phase contrast, Hyrax mount of cleaned material).
Scale bars = (a) 50µm; (b) 10µm.

Taxonomic source
Hustedt, F. (1985). The pennate diatoms.
A translation of Hustedt´s "Die Kieselalgen, 2. Teil" (p. 783). Koenigstein, Germany.

Division Bacillariophyta
Class Bacillariophyceae
Order Thalassiophysales
Family Catenulaceae

Plate 68

Amphora cf. *arenaria* Donkin, 1853



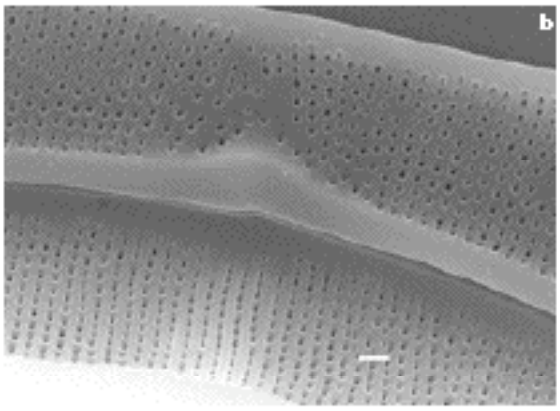


Figure 1 (a) Inside view of whole valve and (b) detail of the central part of the valve (SEM).
Scale bar = (a) 10µm; (b) 1µm.

Taxonomic source
Peragallo, H. & Peragallo, M. (1965). Diatomées marines de France et des districts maritimes voisins (p.217). A. Ascher & Co. Amsterdam.

Plate 69

Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

Bacillaria paxillifera (O. F. Müller) Hendey, 1951

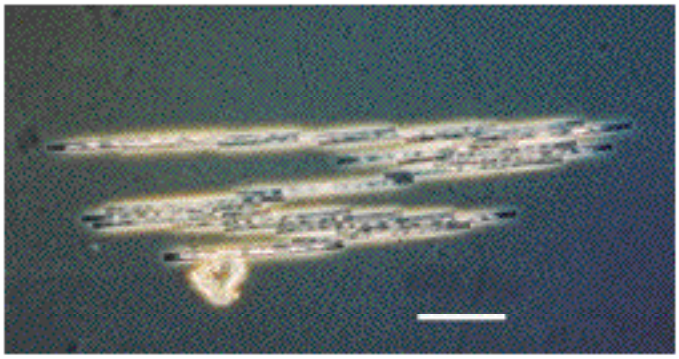


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 50µm.

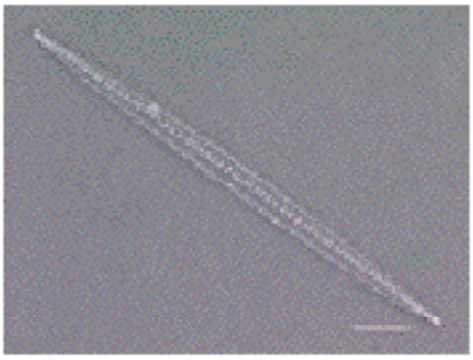


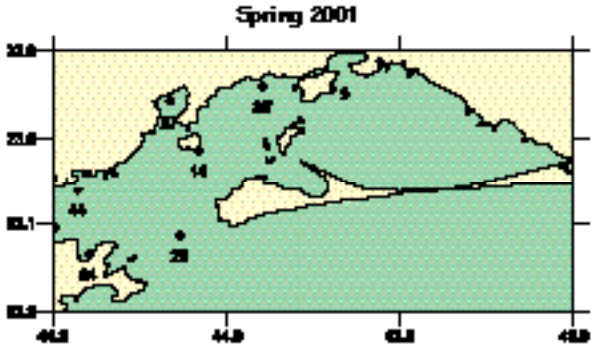
Figure 2. Single valve, showing raphe along its main axis (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 10µm.

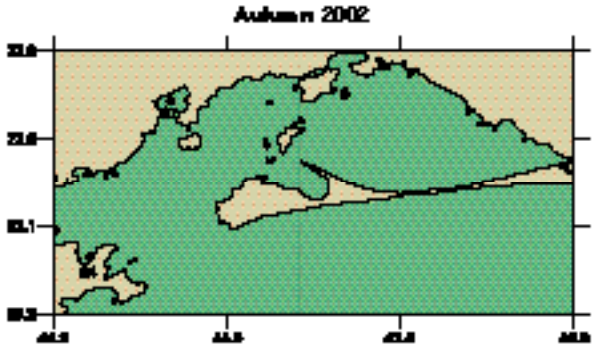
Taxonomic source

Hasle, G. R. & Syvertsen, E. E. (1997). Diatoms (p. 293). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
AA	95.0 - 115	106 ± 9.0	5
TA	4.5 - 5	5 ± 0.2	5





Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

Plate 70

Cylindrotheca closterium (Ehrenberg) Lewin & Reimann, 1964

Figure 1. Two solitary cells in valve view (LM, phase contrast, water mount). Scale bar = 20µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 294).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
AA	40 - 165.5	83 ± 23.0	159
TA	2 - 5.5	4 ± 0.8	159

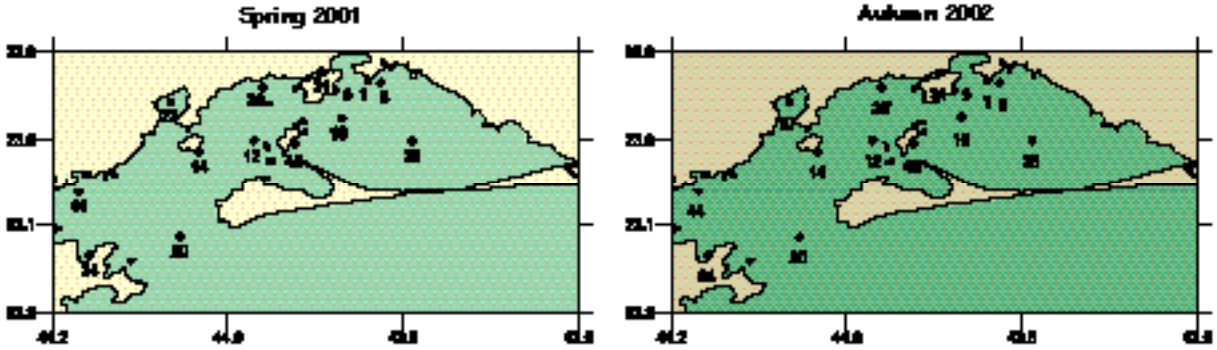


Plate 71

Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

Fragilariopsis doliolus (Wallich) Medlin & Sims, 1993

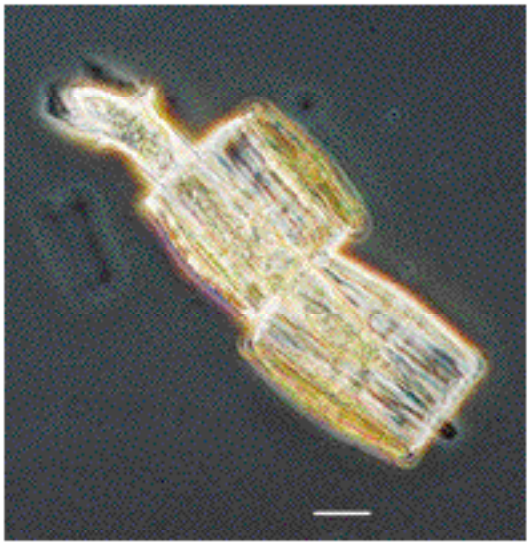


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

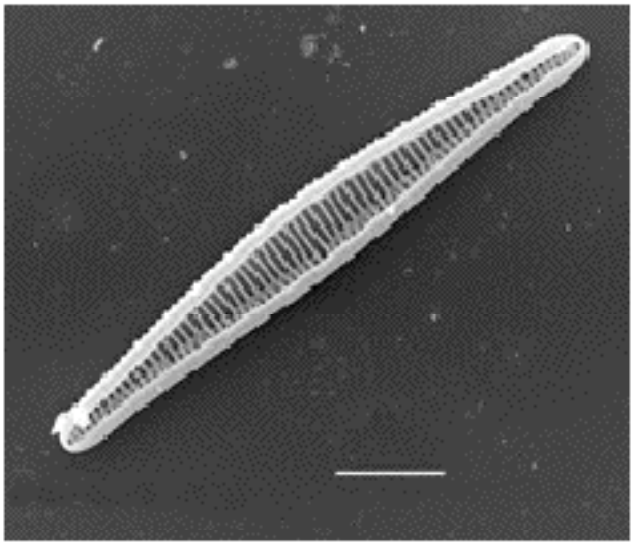


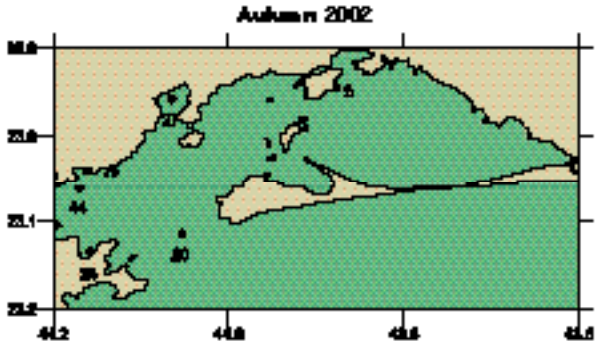
Figure 2. Inside view of single valve (SEM). Scale bar = 10µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 303). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
AA	47 - 80	65 ± 101	22
TA	5 - 12	10 ± 1	22



Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

Plate 72

Nitzschia constricta (Gregory) Grunow, 1880

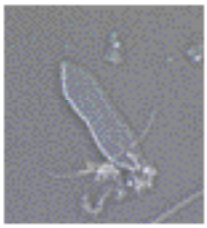


Figure 1. Single valve
(LM, phase contrast, Hyrax mount of cleaned material).
Scale bar = 10µm.

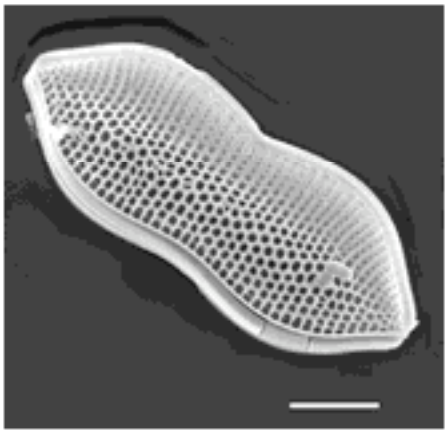


Figure 2. External view of single valve (SEM).
Scale bar = 5µm.

Taxonomic source

Navarro, N. (1982). A survey of the Marine diatoms of Puerto Rico VII. Suborder Raphidineae: Families Auriculaceae, Epithemiaceae, Nitzschiaceae and Surirellaceae (p. 394). Botanica Marina, 25.

Plate 73

Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

Nitzschia lorenziana var. *subtilis* Grunow, 1880

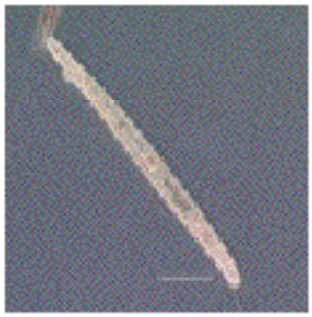


Figure 1. Single cell (LM, phase contrast, water mount). Scale bar = 50µm.

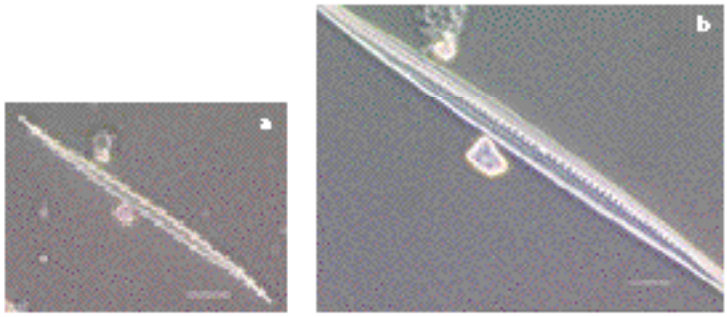


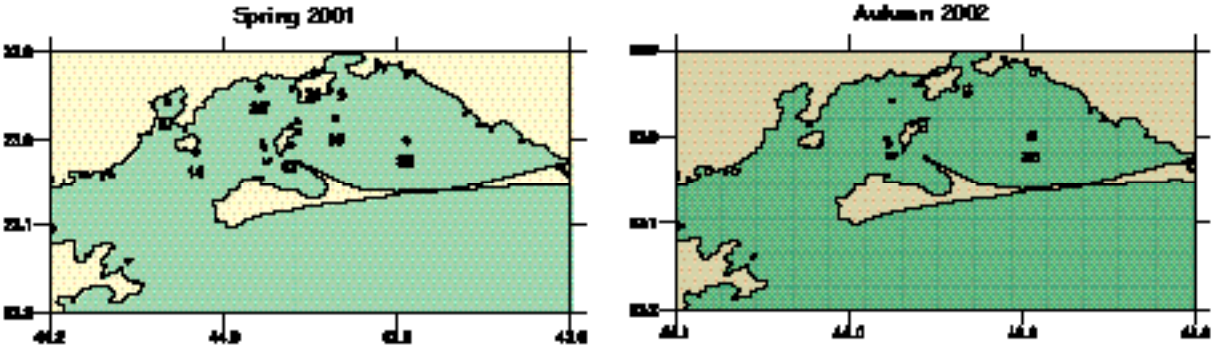
Figure 2. (a) Whole valve and (b) detail of the central part of the valve (LM, phase contrast, Hyrax mount of cleaned material). Scale bars = (a) 20µm; (b) 10µm.

Taxonomic source

Peragallo, H. & Peragallo, M. (1965). Diatomées marines de France et des districts maritimes voisins (p. 294). A. Ascher & Co. Amsterdan.

Morphometrics

	(µm)	mean	n
AA	170	170	2
TA	15	15	2



Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

Plate 74

Nitzschia morphotype *Nitzschiella* (*sensu* H. Peragallo, 1965)

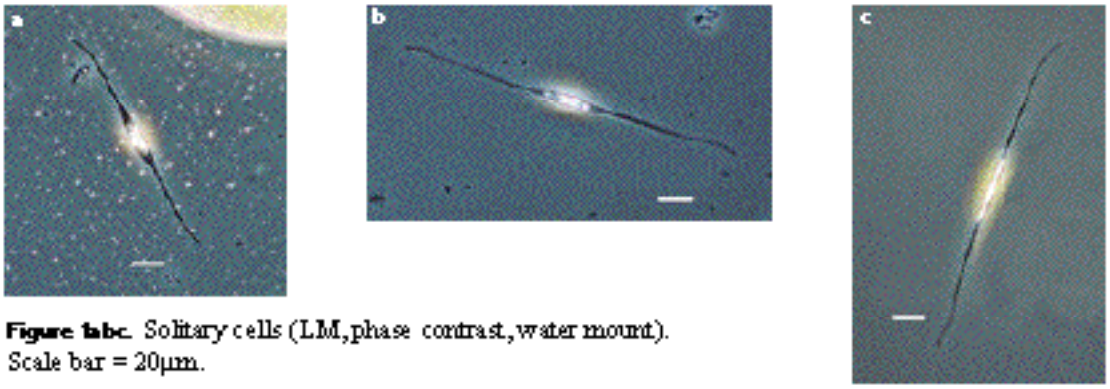


Figure 1a-c. Solitary cells (LM, phase contrast, water mount). Scale bar = 20µm.

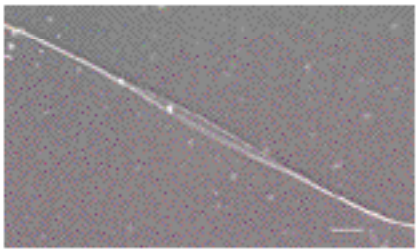


Figure 2. Single valve (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 20µm.

Taxonomic source

Peragallo, H. & Peragallo, M. (1965). Diatomées marines de France et des districts maritimes voisins (p. 292). A. Ascher & Co. Amsterdan.

Morphometrics

	range (µm)	mean ± SD	n
AA	50 - 160.0	95 ± 35	12
TA	6 - 7.5	6 ± 1	12

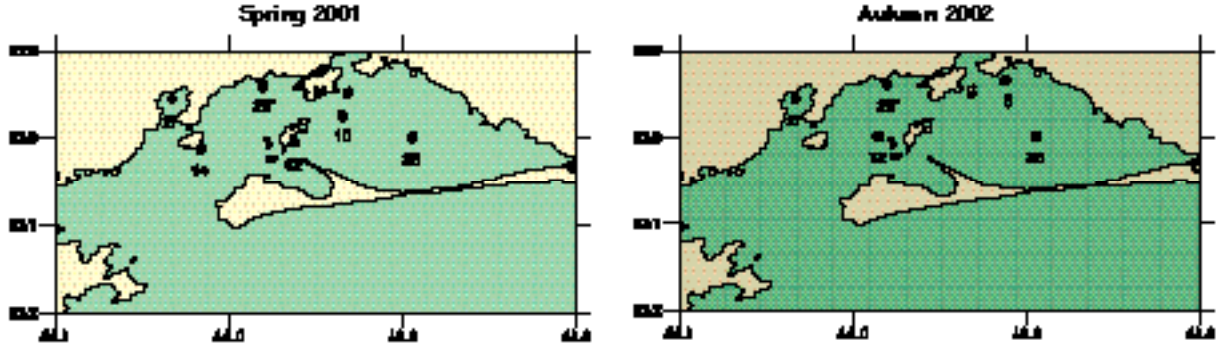


Plate 75

Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

"Nitzschia delicatissima complex" (*sensu* Hasle, 1965)

Pseudo-nitzschia "delicatissima" sp.1

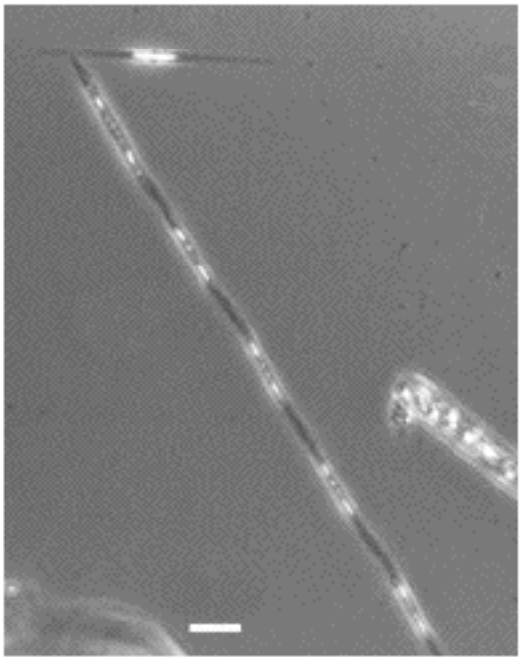


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Pseudo-nitzschia "delicatissima" sp.2

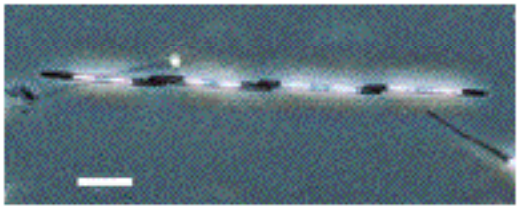


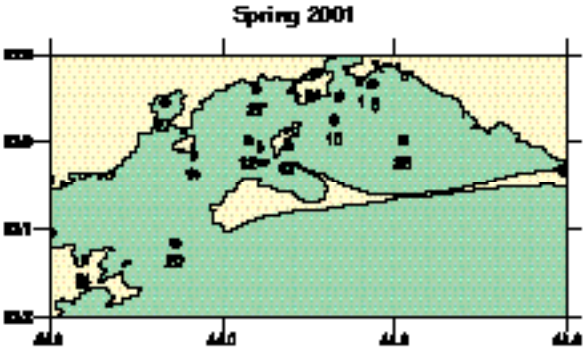
Figure 2. Chain of cells in girdle view, showing truncated cell ends (LM, phase contrast, water mount). Scale bar = 20µm.

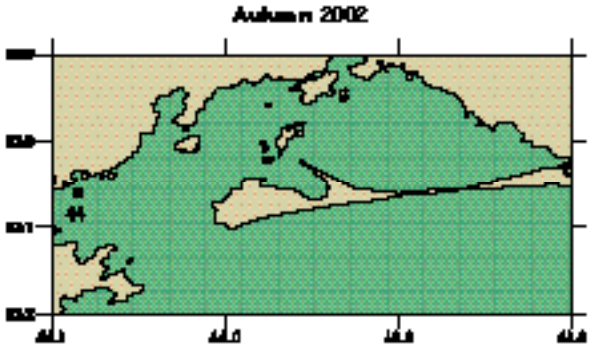
Morphometrics

	range (µm)	mean ± SD	n
PA	30 - 65.0	49 ± 12	13
AA	2 - 2.5	2 ± 12	13

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 307). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.





Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

Plate 76

"Nitzschia seriata complex" (*sensu* Hasle, 1965)

Pseudo-nitzschia "seriata" sp.1

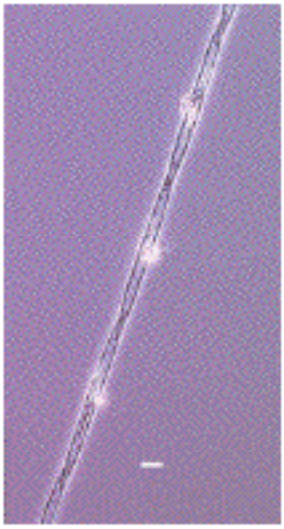


Figure 3. Chain of cells in girdle and valve views (LM, phase contrast, water mount). Scale bar = 10µm.

Pseudo-nitzschia "seriata" sp.2

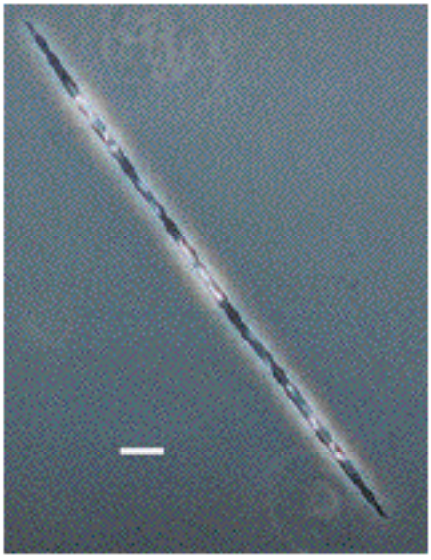


Figure 1. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 20µm.

Pseudo-nitzschia "seriata" sp.3

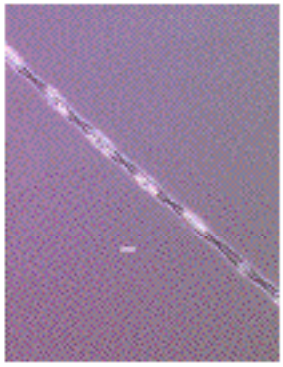
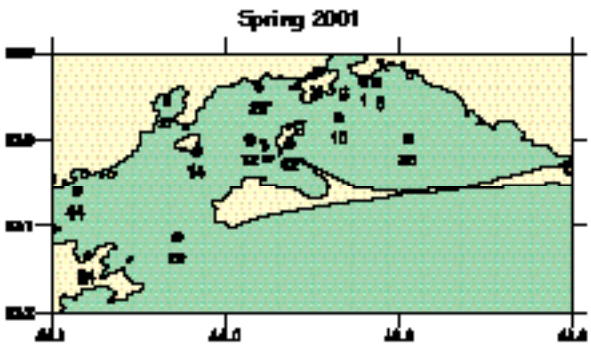


Figure 2. Chain of cells in girdle view (LM, phase contrast, water mount). Scale bar = 10µm.

Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 307).
In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.



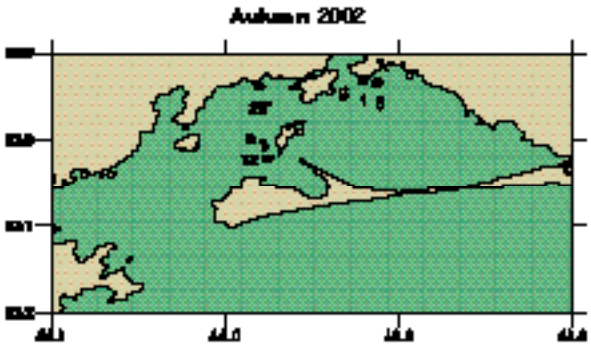


Plate 77

Division Bacillariophyta
Class Bacillariophyceae
Order Bacillariales
Family Bacillariaceae

Pseudo-nitzschia pungens (Grunow ex Cleve) Hasle, 1993

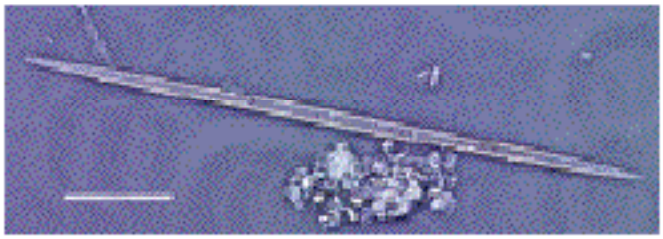


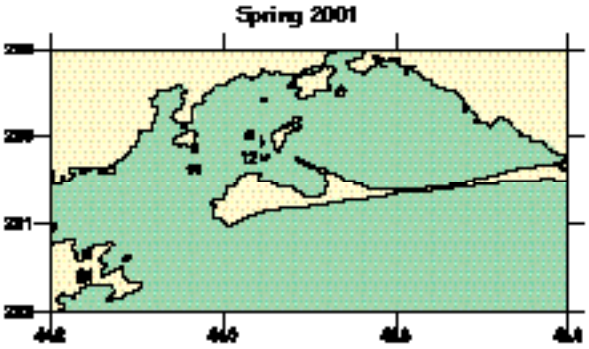
Figure 1. Single valve; 2 rows of poroids (not shown here) were seen during analysis (LM, phase contrast, Hyrax mount of cleaned material). Scale bar = 20µm.

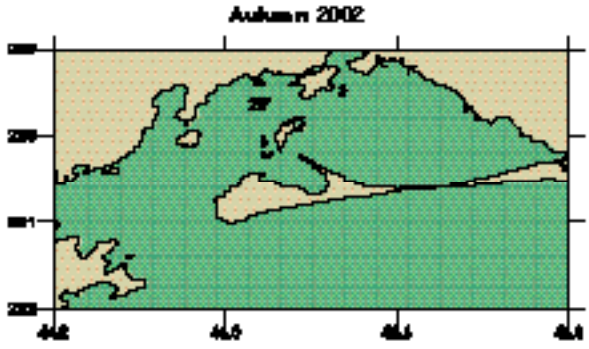
Taxonomic source

Hasle & Syvertsen (1997). Diatoms (p. 312). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
AA	76 - 123.75	96 ± 18.0	10
TA	4 - 4.50	4 ± 0.2	10





DINOFLAGELLATES

Plate 78

Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Gonyaulacaceae

Gonyaulax spinifera (Claparède & Lachmann) Diesing, 1866

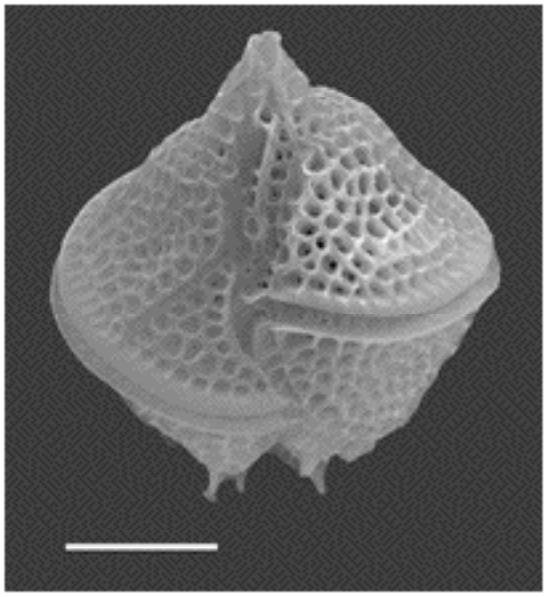


Figure 1. Cell in ventral view (SEM). Scale bar = 20µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 214). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)		
TL	39	-	45
D	32	-	37

Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Ceratiaceae

Plate 79

Ceratium breve (Ostenfeld & Schmidt) Schröder, 1906

Figure 1. Chain of cells in dorsal view (LM, phase contrast). Scale bar = 100µm.

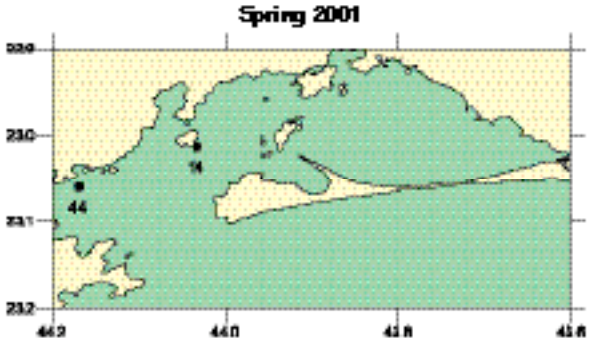
Figure 2. Chain of cells in dorsal view (LM, phase contrast). Scale bar = 100µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 232). Her Majesty's Stationary Office, London.

Morphometrics

	range (µm)			mean ± SD			n
TL	210	-	275	244	±	22	6
D	65	-	120	82	±	20	6
EL	35	-	50	42	±	5	6
AHL	130	-	185	157	±	20	6
RANHL	105	-	140	116	±	13	6
LANHL	110	-	130	123	±	7	6



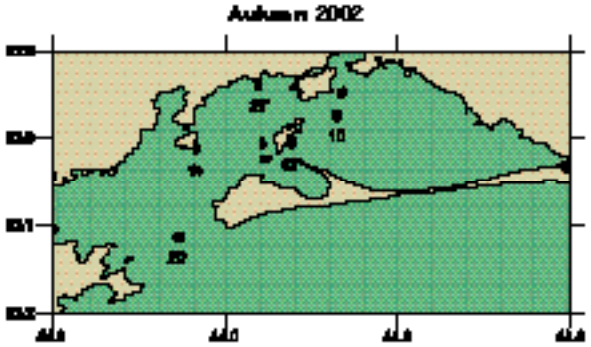


Plate 80

Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Ceratiaceae

Ceratium furca (Ehrenberg) Claparède & Lachmann, 1858

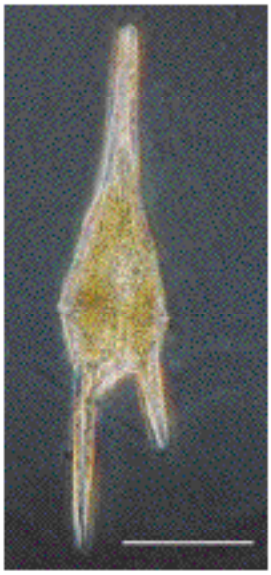


Figure 1.
Cell in dorsal view (LM, phase contrast). Scale bar = 50µm.

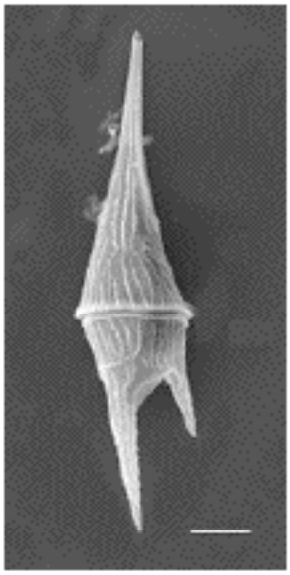


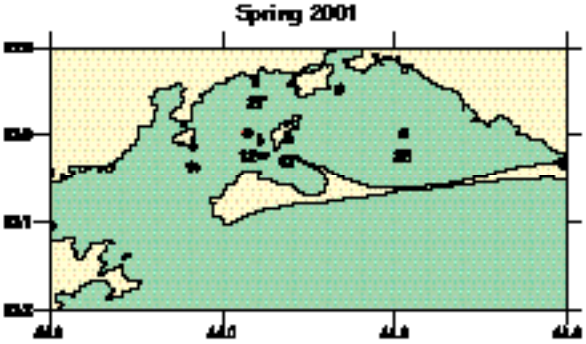
Figure 2.
Cell in dorsal view (SEM). Scale bar = 20µm.

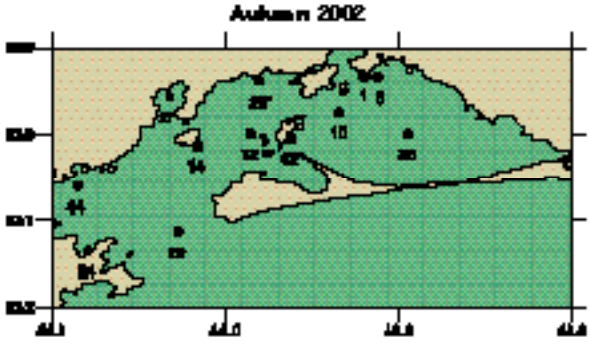
Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 228). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	mean ± SD	n
TL	210 - 275	244 ± 22	6
D	65 - 120	82 ± 20	6
EL	35 - 50	42 ± 5	6
AHL	130 - 185	157 ± 20	6
RANHL	105 - 140	116 ± 13	6
LANHL	110 - 130	123 ± 7	6





Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Ceratiaceae

Plate 81

Ceratium fusus (Ehrenberg) Dujardin, 1841

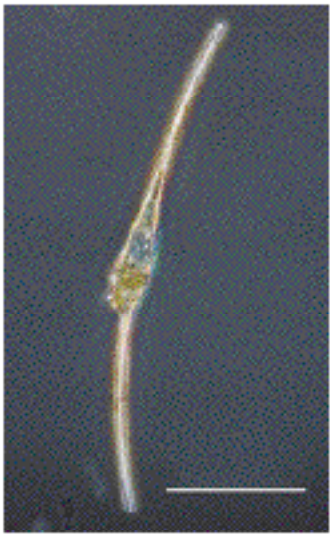


Figure 1. Cell in ventral view (LM, phase contrast). Scale bar = 100µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 231). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)			mean ± SD			n
TL	155	-	475	349	±	47	86
D	22	-	40	29	±	4	86
EL	25	-	115	66	±	17	75
AHL	85	-	200	121	±	21	85
RANHL	1	-	10	3	±	1	71
LANHL	102	-	245	147	±	20	71

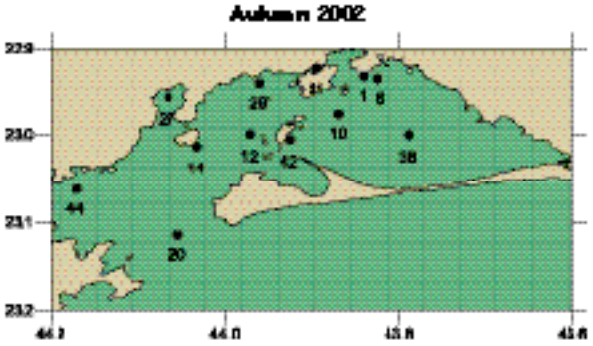


Plate 82

Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Ceratiaceae

Ceratium gibberum Gourret, 1883

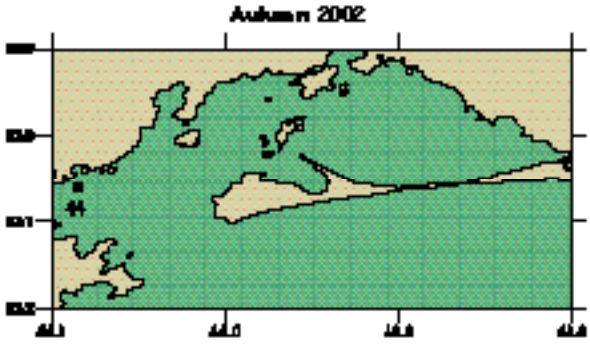
Figure 1. Cell in dorsal view (LM, phase contrast). Scale bar = 100µm.

Taxonomic source

Graham, H. W. & Bronikowsky, N. (1944). The genus *Ceratium* in the Pacific and North Atlantic Oceans. Carnegie Institute of Washington Publ., Ser. Biol., (p.33).

Morphometrics

	range (µm)	mean ± SD	n
TL	105 - 340	245 ± 134	2
D	95 - 95	95 ± 0	2
EL	50 - 50	50 ± 0	2
AHL	40 - 230	135 ± 134	2
LANHL	110 - 120	115 ± 7	2



Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Ceratiaceae

Plate 83

Ceratium horridum (Cleve) Gran, 1902

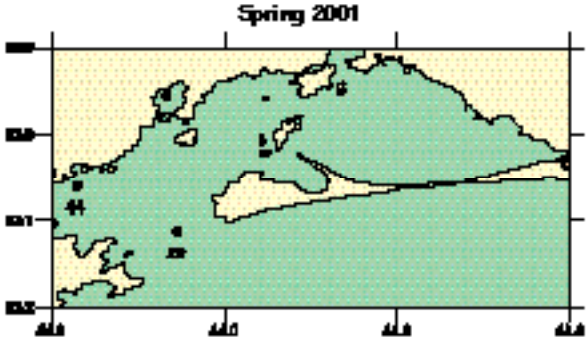
Figure 1. Cell in dorsal view (LM, phase contrast). Scale bar = 100µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 240). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	mean ± SD	n
TL	145 - 375	214 ± 53	38
D	32 - 70	44 ± 6	38
EL	15 - 40	26 ± 5	38
AHL	85 - 310	151 ± 50	38
RANHL	60 - 195	110 ± 35	38
LANHL	15 - 195	110 ± 37	38



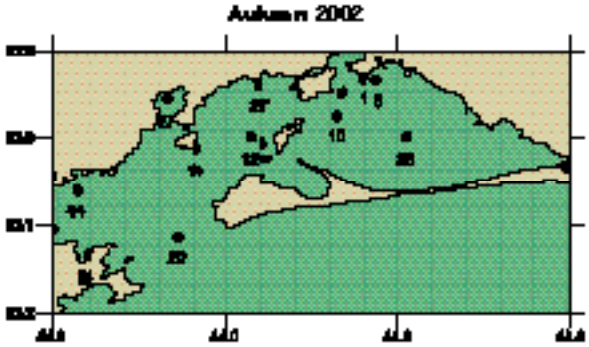


Plate 84

Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Ceratiaceae

Ceratium massiliense (Gourret) Jörgensen, 1911

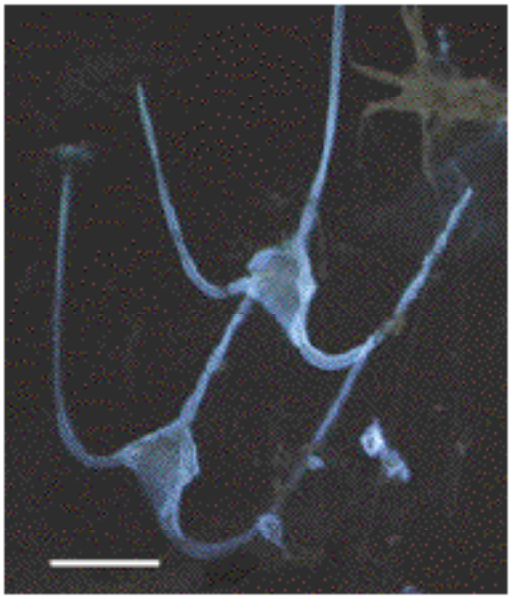


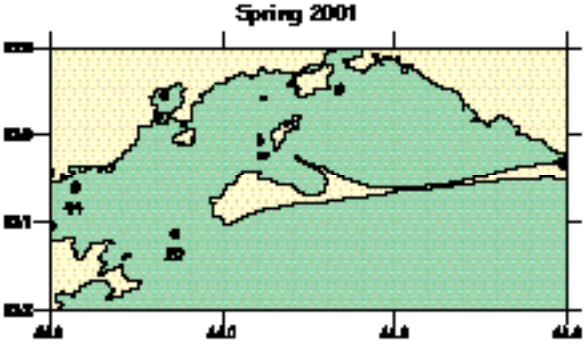
Figure 1. Chain of cells in dorsal view (FM, calcofluor stained cells). Scale bar = 50µm.

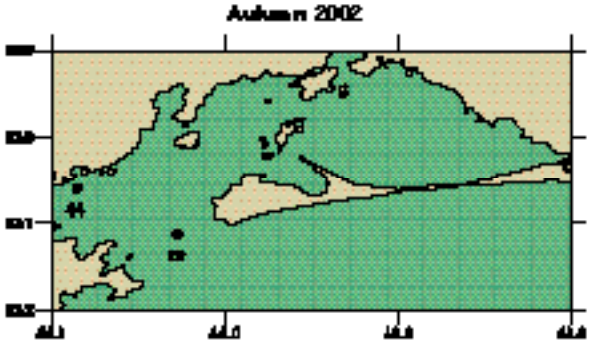
Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 236). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	mean ± SD	n
TL	35 - 385	272 ± 94	12
D	35 - 80	46 ± 12	11
EL	20 - 55	31 ± 12	12
AHL	125 - 330	237 ± 57	12
RANHL	100 - 310	186 ± 67	12
LANHL	120 - 305	190 ± 63	11





Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Ceratiaceae

Plate 85

Ceratium cf. *trichoceros* (Ehrenberg) Kofoid, 1908

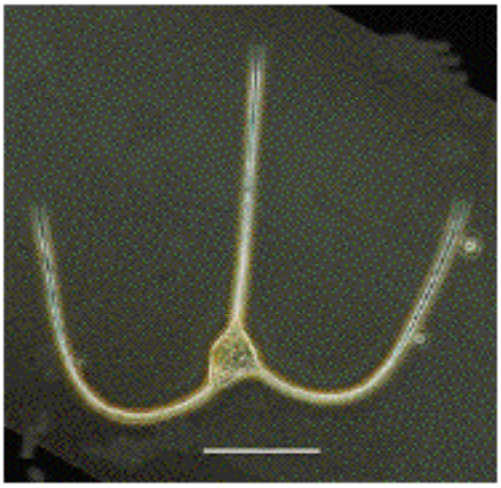


Figure 1. Cell in dorsal view (LM, phase contrast). Scale bar = 100µm.

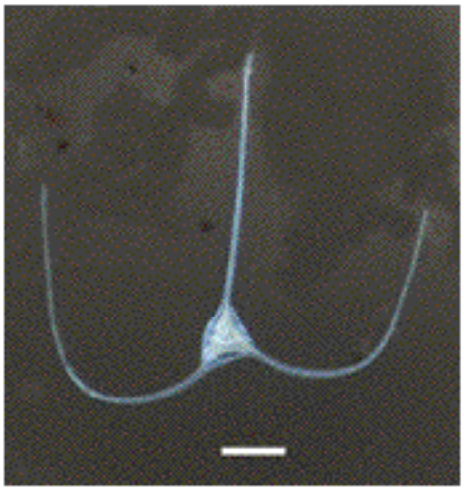


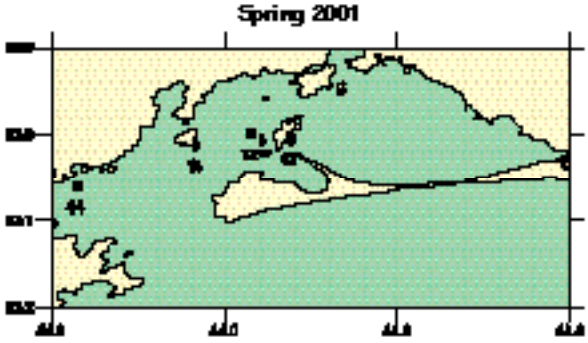
Figure 2. Cell in dorsal view (FM, calcofluor stained cell). Scale bar = 20µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 236). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	n
TL	265	1
D	45	1
EL	25	1
AHL	190	1
RANHL	195	1
LANHL	215	1



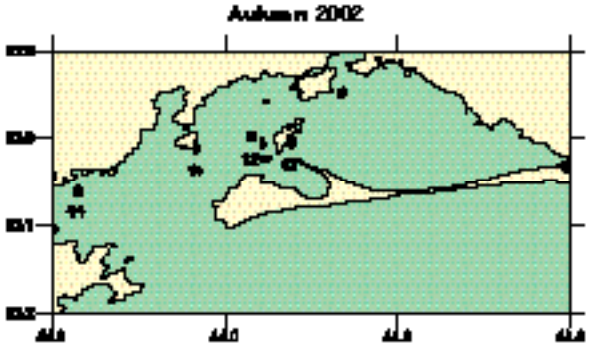


Plate 86

Division Dinophyta
Class Dinophyceae
Order Gonyaulacales
Family Goniodomaceae

Alexandrium cf. *fraterculus* (Balech) Balech, 1985

Figure 1. Chain of cells in ventral view (LM, phase contrast). Scale bar = 20µm.

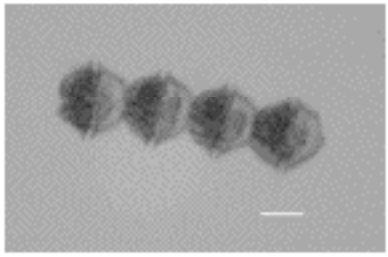


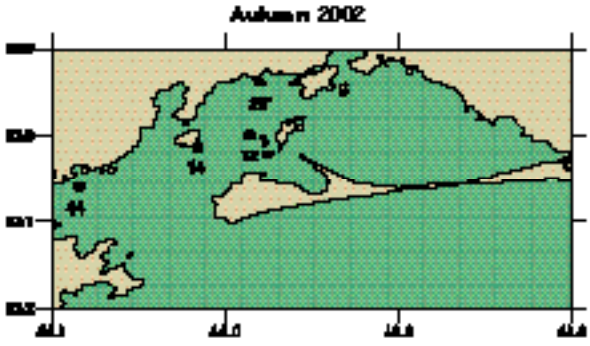
Figure 2. Chain of cells in ventral view (LM, bright field). Scale bar = 25µm.

Taxonomic source

Steindinger, K. A. & Tangen, K. (1997). Dinoflagellates (p. 494). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
TL	32	-	50	40	±	7	16
D	35	-	55	41	±	8	16



Division Dinophyta
Class Dinophyceae
Order Peridiniales
Family Peridiniaceae

Plate 87

Scrippsiella spinifera Honsell & Cabrini, 1991

Figure 1. Cell in ventral view (LM, phase contrast). Scale bar = 10µm.

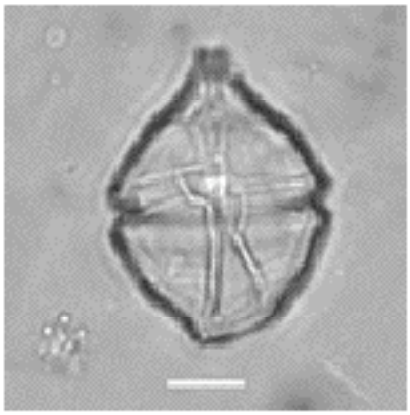


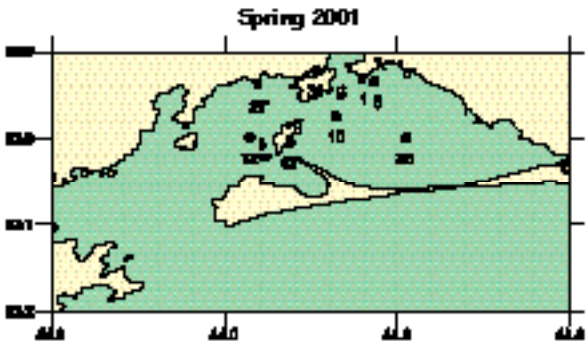
Figure 2. Cell in ventral view (LM, bright field). Scale bar = 10µm.

Taxonomic source

Honsell, G. & Cabrinni, M. (1991). *Scrippsiella spinifera* sp. nov. (Pyrrhophyta): A New Dinoflagellate from the Northern Adriatic Sea. Botanica Marina Vol. 34, pp. 167 - 175.

Morphometrics

	range (µm)	mean ± SD	n
TL	30 - 80	41 ± 6	294
D	21 - 55	29 ± 4	294
EL	15 - 42	24 ± 4	273
HL	10 - 40	17 ± 3	273



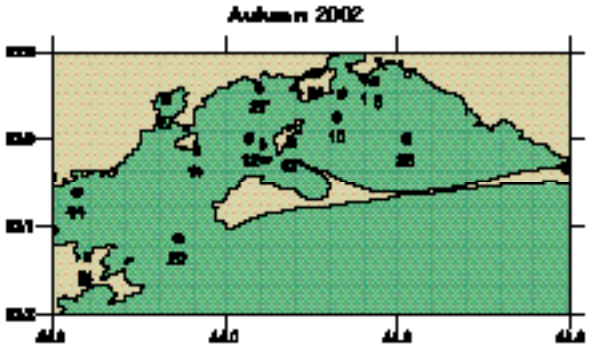


Plate 88

Division Dinophyta
Class Dinophyceae
Order Peridiniales
Family Peridiniaceae

Scrippsiella trochoidea (Stein) Loeblich III, 1976

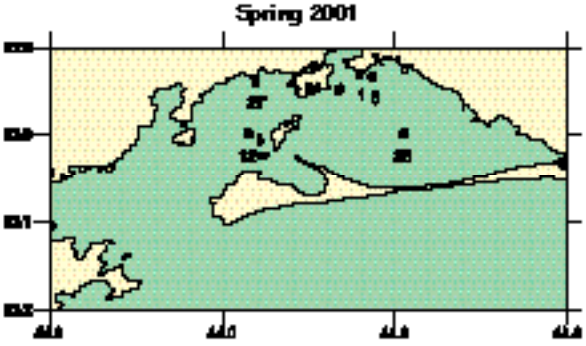
Figure 1. Cell in dorsal view (LM, phase contrast). Scale bar = 20µm.

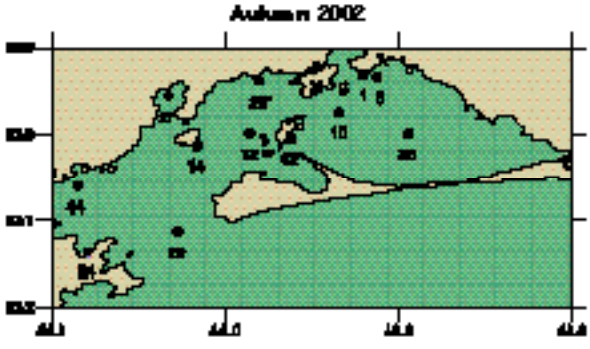
Taxonomic source

Dodge, J. D. 1982. Marine Dinoflagellates of the British Isles (p. 163). Her Majesty's Stationery Office, London

Morphometrics

	range (µm)	mean ± SD	n
TL	23 - 45	34 ± 3	81
D	20 - 35	27 ± 3	81
EL	8 - 25	20 ± 3	81
HL	10 - 25	14 ± 3	81





Division Dinophyta Plate 89
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Protoperidinium divergens (Ehrenberg) Balech, 1974

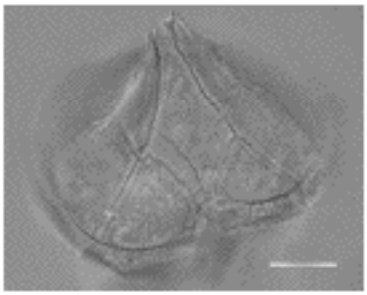


Figure 1. Epitheca in ventral view (LM, bright field, cleared material). Scale bar = 20µm.

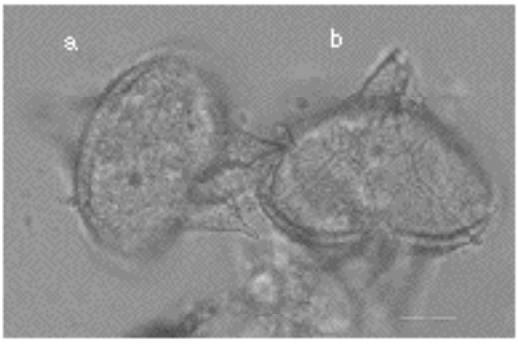


Figure 3. Hypotheca in (a) dorsal view and (b) epitheca in ventral view (LM, bright field, cleared material). Scale bar = 20µm.

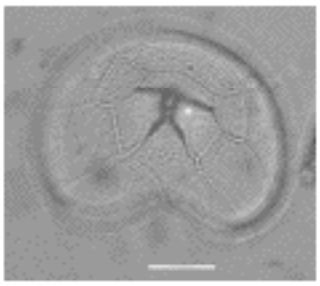


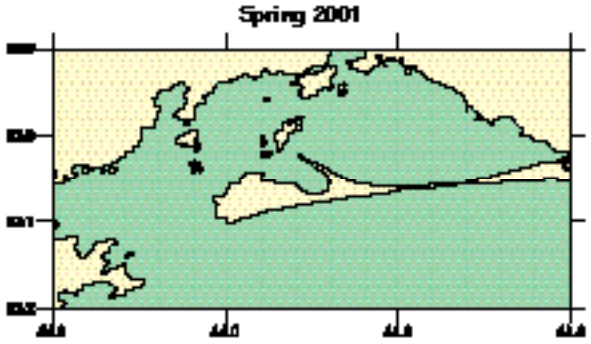
Figure 2. Apical view (LM, bright field, cleared material). Scale bar = 20µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 193). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	mean ± SD	n
TL	40		1
D	40 - 48	43 ± 5	3
EL	28		1
HL	12		1



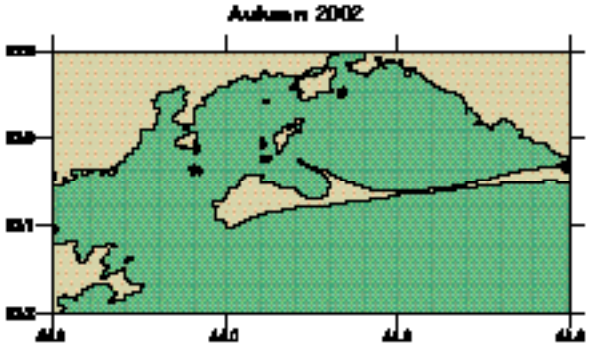


Plate 90

Division Dinophyta
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Protoperidinium leonis (Pavillard) Balech, 1974

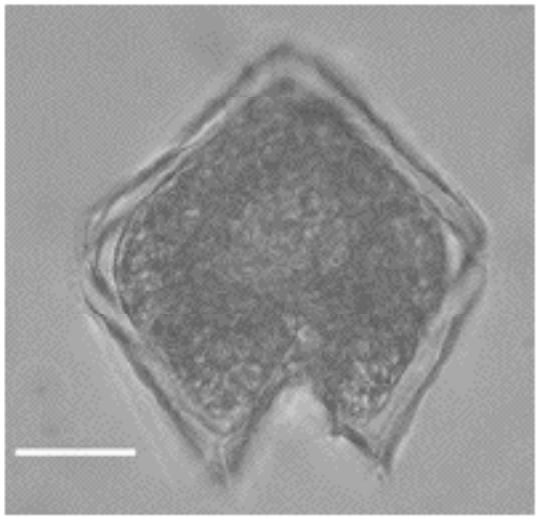


Figure 1. Cell in ventral view, cyst formation (LM, bright field). Scale bar = 20µm.

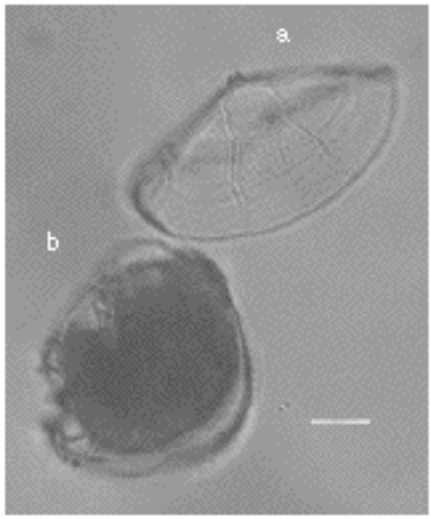


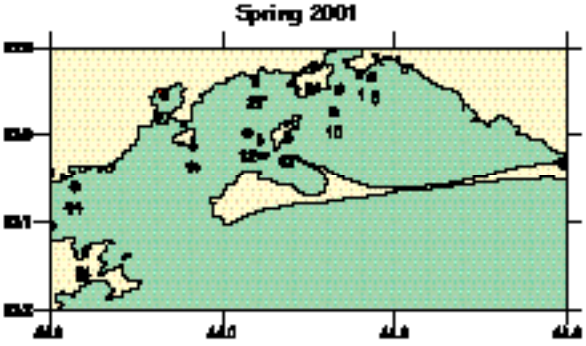
Figure 1. Dissociated cell: (a) epitheca in dorsal view, (b) hypotheca with cyst (LM, bright field). Scale bar = 20µm.

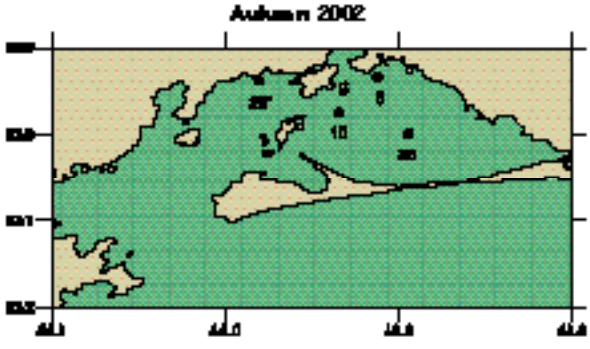
Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 187). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	mean ± SD	n
TL	72.0 - 145	92 ± 20	30
D	66.5 - 165	87 ± 18	30
EL	30.0 - 65	45 ± 11	22
HL	30.0 - 80	51 ± 13	22





Division Dinophyta Plate 91
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Protoperidinium mariaelebouriae (Paulsen) Balech, 1974

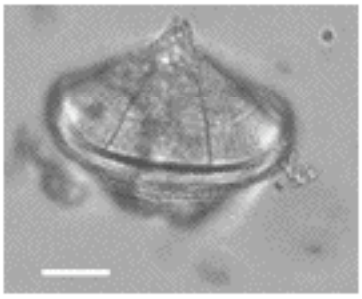


Figure 1. Cell in dorsal view (LM, bright field, cleared material). Scale bar = 20µm.

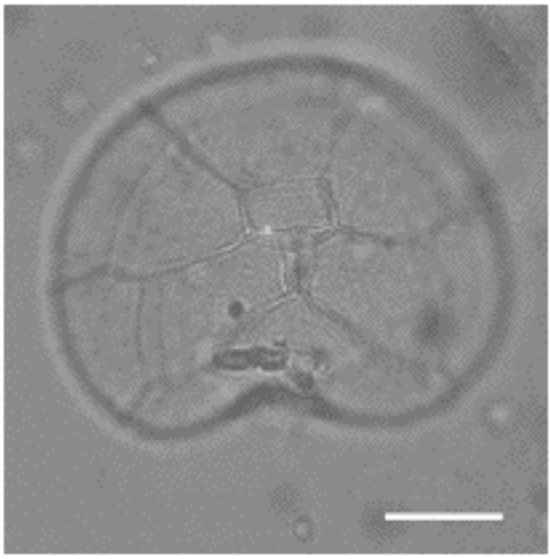


Figure 2. Apical view (LM, bright field, cleared material). Scale bar = 20µm.

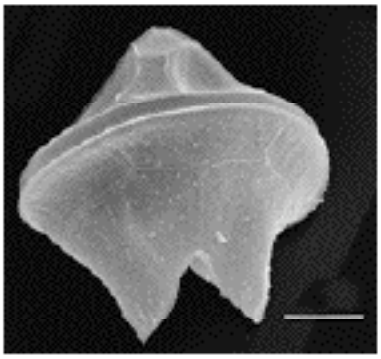


Figure 3. Cell in dorsal view (SEM). Scale bar = 20µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 178). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)		
TL	100	-	112
D	90	-	94

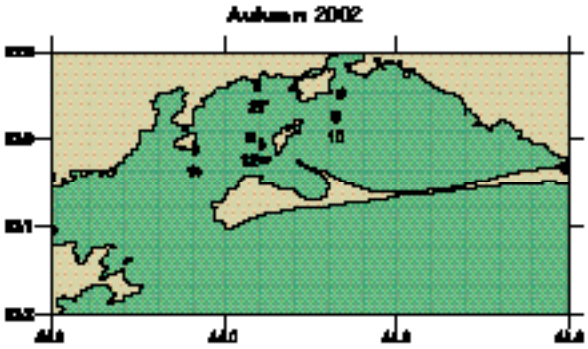


Plate 92

Division Dinophyta
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Protoperidinium oblongum (Aurivillius) Parke & Dodge, 1976

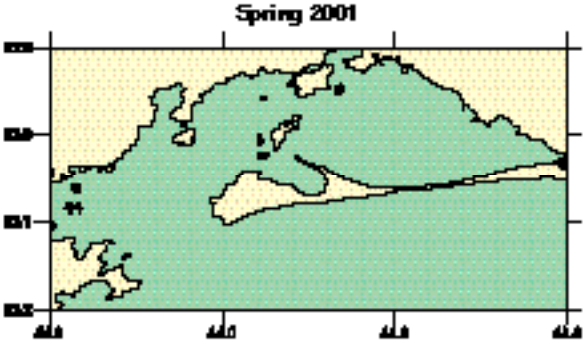
Figure 1. Cell in dorsal view (LM, bright field). Scale bar = 20µm.

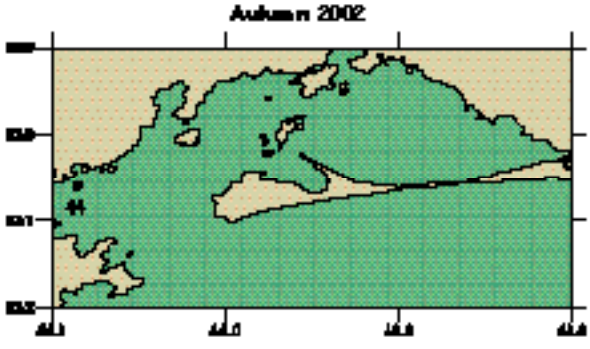
Taxonomic source

Steindinger, K. A. & Tangen, K. (1997). Dinoflagellates (p. 541). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	n
TL	155	1
D	90	1





Division Dinophyta Plate 93
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Protoperidinium parviventer Balech, 1978

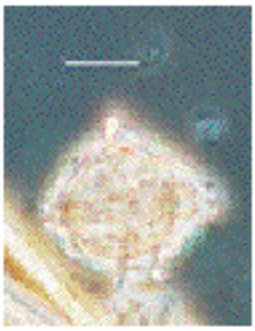


Figure 1. Cell in ventral view (LM, phase contrast). Scale bar = 20µm.

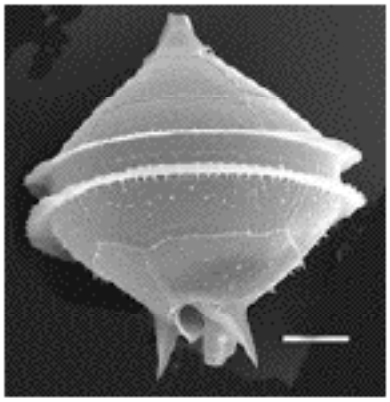


Figure 2. Cell in dorsal view (SEM). Scale bar = 10µm.

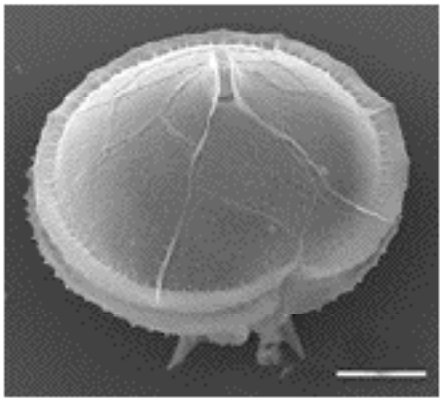


Figure 3. Epitheca in ventral view (SEM). Scale bar = 10µm.

Taxonomic source

Balech, E. (1988). Los dinoflagellates del Atlantico Sudoccidental (p. 105).
Publ. Espec. Inst. Esp. Oceanogr., n. 1. Ministerio de Agricultura Pesca y Alimentacion, Madrid.

Morphometrics

	range (µm)			mean ± SD			n
TL	40	-	57	49	±	5	11
D	40	-	55	47	±	4	11
EL	15	-	26	23	±	4	11
HL	15	-	26	18	±	4	11

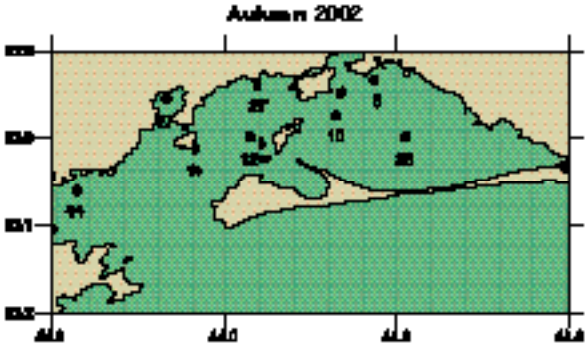


Plate 94

Division Dinophyta
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Protoperidinium cf. *pentagonum* (Gran) Balech, 1974

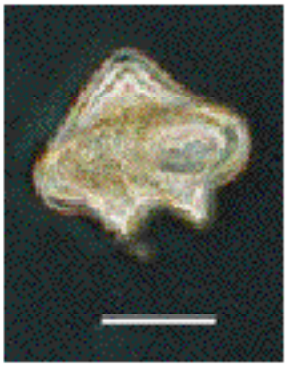


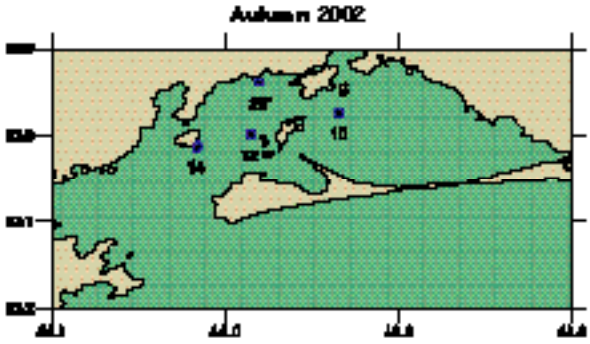
Figure 1. Cell in ventral view (LM, phase contrast). Scale bar = 50µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 188). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	mean ± SD	n
TL	80 - 90	85 ± 7	2
D	95 - 100	98 ± 4	2
EL	30		1
HL	50		1



Division Dinophyta
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Plate 95

Protoperidinium simulum (Paulsen) Balech, 1974

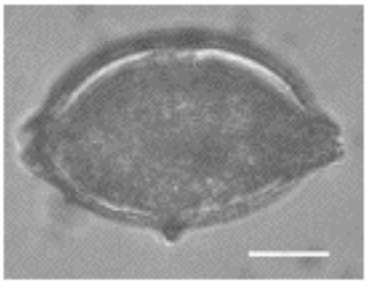


Figure 1. Cell in dorsal view (LM, bright field). Scale bar = 20µm.

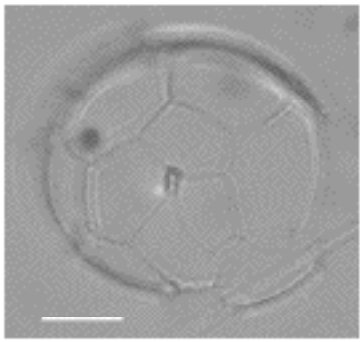


Figure 2. Apical view (LM, bright field). Scale bar = 20µm.

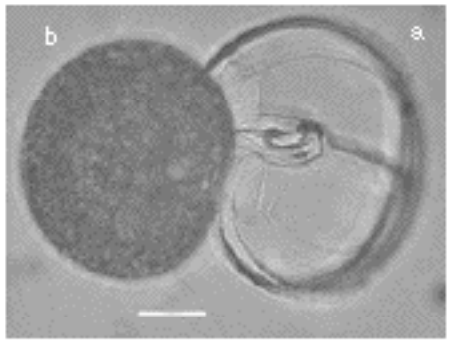


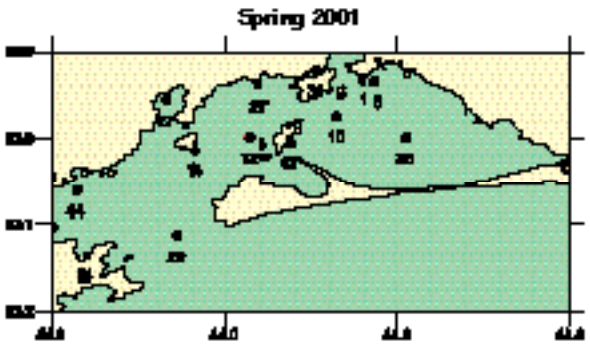
Figure 3. Dissociated cell: (a) hypotheca and (b) protoplasm encysted (LM, bright field). Scale bar = 20µm.

Taxonomic source

Balech, E. (1988). Los dinoflagelados del Atlantico Sudoccidental (p. 112). Publ. Espec. Inst. Esp. Oceanogr., n. 1. Ministerio de Agricultura Pesca y Alimentacion, Madrid.

Morphometrics

	range (µm)	mean ± SD	n
AA	35 - 89.1	64 ± 12	26
TA	43 - 99.0	70 ± 14	26



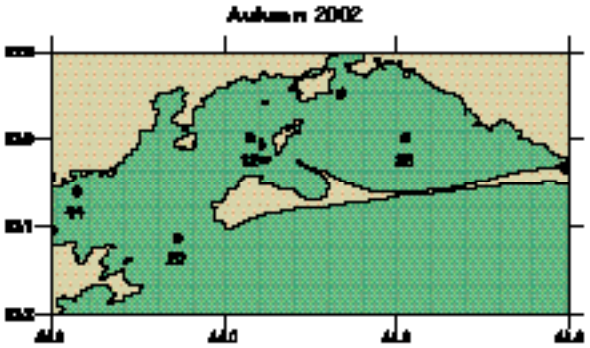


Plate 96

Division Dinophyta
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Protoperidinium steinii (Jörgensen) Balech, 1974

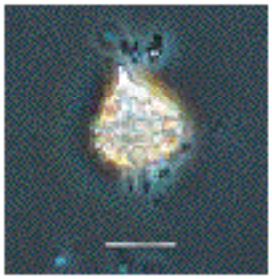


Figure 1. Cell in ventral view (LM, phase contrast). Scale bar = 20µm.

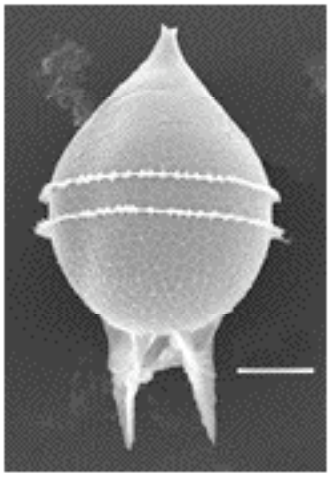


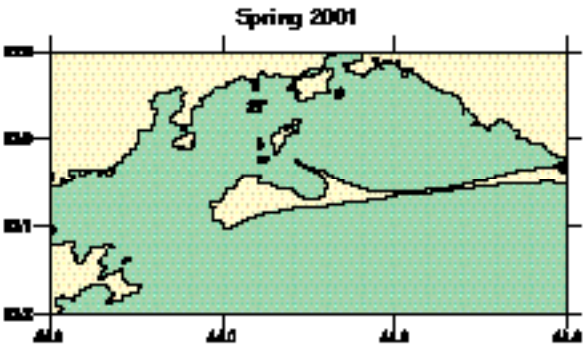
Figure 1. Cell in dorsal view (SEM). Scale bar = 10µm.

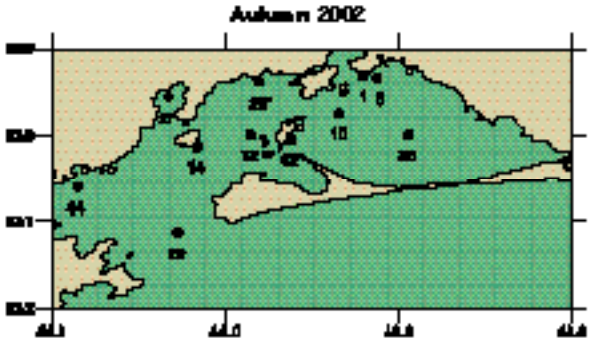
Taxonomic source

Dodge, J. D. 1982. Marine Dinoflagellates of the British Isles (p. 303). Her Majesty's Stationery Office, London

Morphometrics

	range (µm)	mean ± SD	n
TL	27 - 70	40 ± 8	61
D	20 - 55	28 ± 7	61
EL	13 - 30	21 ± 3	59
HL	8 - 25	12 ± 3	59





Division Dinophyta Plate 97
Class Dinophyceae
Order Peridiniales
Family Congruentidiaceae

Protoperidinium symmetricum (Halim) Balech, 1974

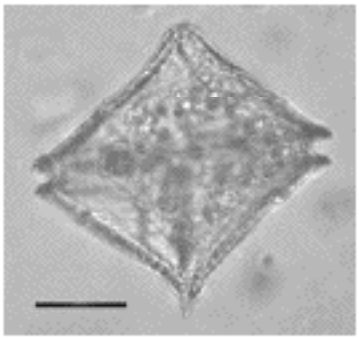


Figure 1. Cell in lateral view (LM, bright field). Scale bar = 20µm.

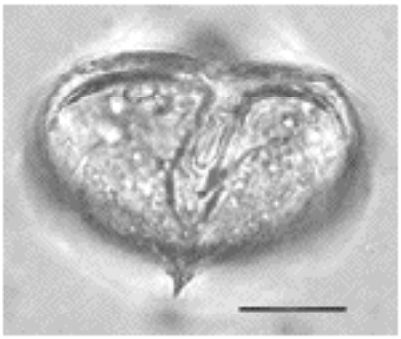


Figure 2. Detail of sulcus (LM, bright field). Scale bar = 20µm.

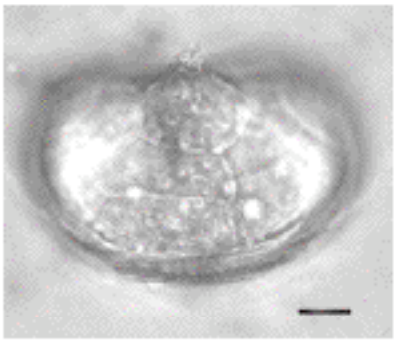


Figure 3. Epitheca in dorsal view (LM, bright field). Scale bar = 10µm.

Taxonomic source

Balech, E. (1988). Los Dinoflagelados del Atlantico Sudoccidental (p. 189). Publ. Espec. Inst. Esp. Oceanogr., n. 1. Ministerio de Agricultura Pesca y Alimentacion, Madrid.

Morphometrics

	range (µm)	n
TL	65	1
D	60	1

Plate 98

Division Dinophyta
Class Dinophyceae
Order Dinophysiales
Family Dinophysiaceae

Dinophysis acuminata Claparède & Lachmann, 1859

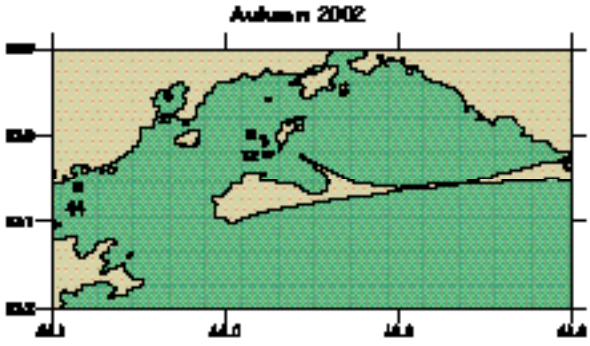
Figure 1. Cell in left lateral view (LM, phase contrast). Scale bar = 20µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 44). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	mean ± SD	n
TL	40 - 60	45 ± 8	6
CD	10 - 25	17 ± 6	6
LD	25 - 40	32 ± 6	5
R1-R2	5 - 50	17 ± 22	4
R2-R3	2 - 15	9 ± 5	4



Division Dinophyta
Class Dinophyceae
Order Dinophysiales
Family Dinophysiaceae

Plate 99

Dinophysis caudata Saville-Kent, 1881

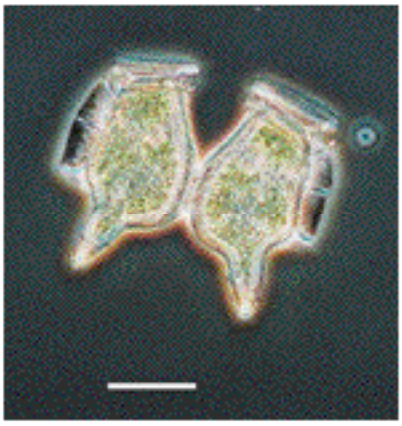


Figure 1. Cell in dorsal view (LM, phase contrast). Scale bar = 50µm.

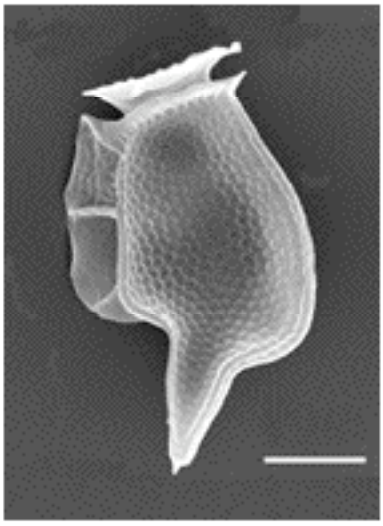


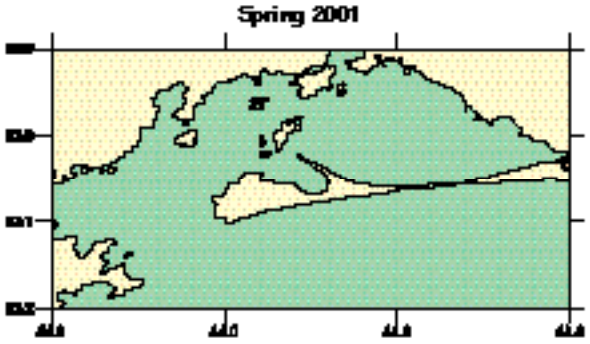
Figure 1. Cell in left lateral view (SEM). Scale bar = 10µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 45). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)	mean ± SD	n
TL	45 - 115	88 ± 15	32
CD	15 - 55	34 ± 11	32
LD	8 - 65	47 ± 10	32
R1-R2	6 - 15	12 ± 3	12
R2-R3	5 - 20	13 ± 4	12



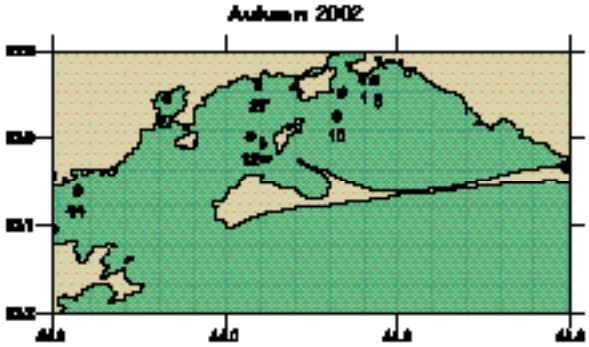


Plate 100

Division Dinophyta
Class Dinophyceae
Order Prorocentrales
Family Prorocentraceae

Prorocentrum gracile Schütt, 1895

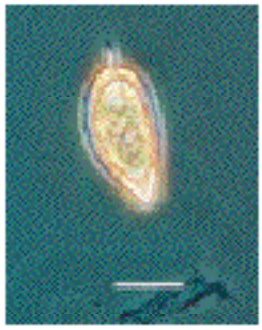


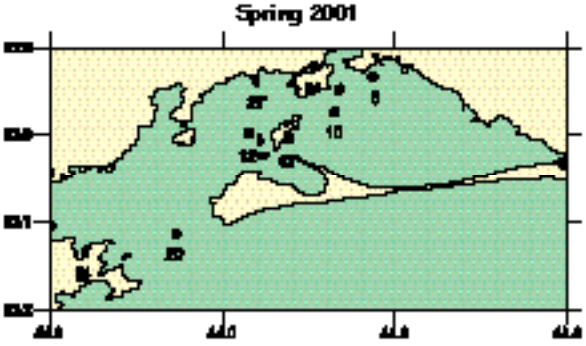
Figure 1. Cell in valve view (LM, phase contrast). Scale bar = 20µm.

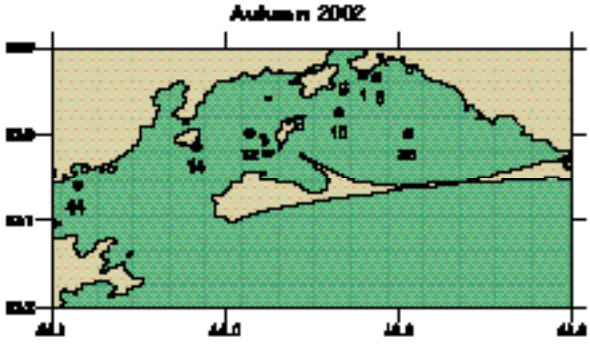
Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 33). Her Majesty's Stationery Office, London.

Morphometrics

	range (µm)			mean ± SD			n
TL	22.5	-	60	48	±	9	16
LD	10.0	-	27	22	±	4	16
SL	2.5	-	10	7	±	2	16





Division Dinophyta
Class Dinophyceae
Order Prorocentrales
Family Prorocentraceae

Plate 101

Prorocentrum mexicanum Tafall, 1942

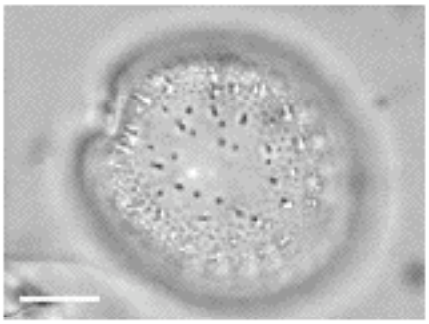


Figure 1. Cell in valve view, showing rows of pores (LM, bright field, cleared material). Scale bar = 10µm.

Taxonomic source

Steindinger, K. A. & Tangen, K. (1997). Dinoflagellates (p. 424). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)			mean ± SD			n
TL	35	-	50	40	±	4	28
LD	30	-	45	34	±	3	28

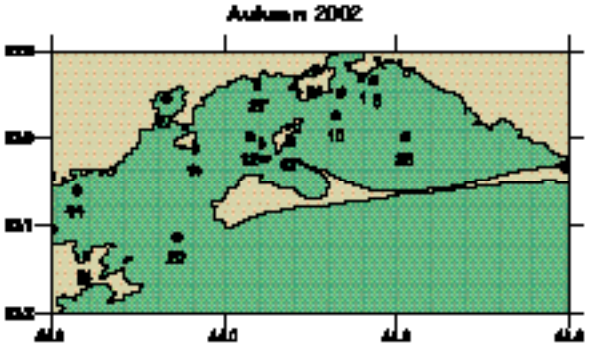


Plate 102

Division Dinophyta
Class Noctiluciphyceae
Order Noctilucales
Family Noctilucaceae

Noctiluca scintillans (Macartney) Kofoid & Swezy, 1921

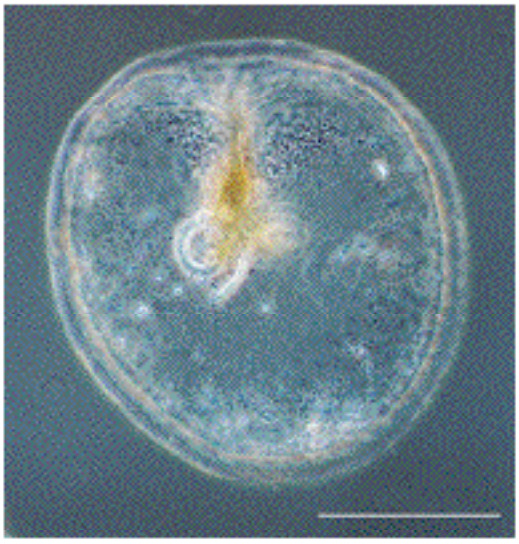


Figure 1. Cell in general view, showing tentacle (LM, phase contrast). Scale bar = 100µm.

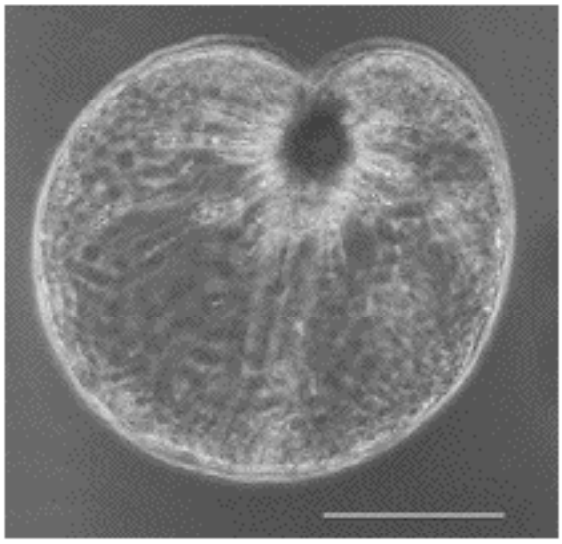
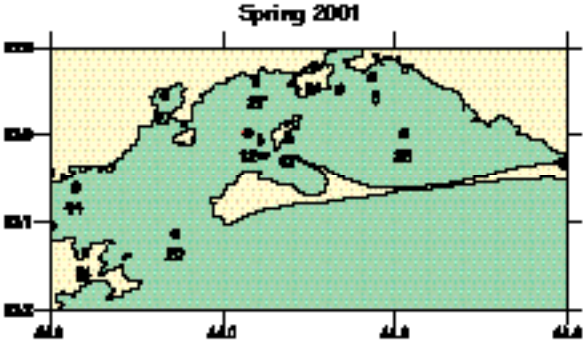
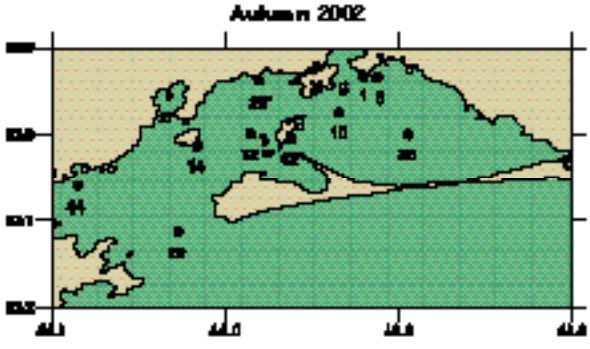


Figure 2. Cell in general view (LM, phase contrast). Scale bar = 100µm.

Taxonomic source

Dodge, J. D. (1982). Marine Dinoflagellates of the British Isles (p. 135). Her Majesty's Stationery Office, London.





SILICOFLAGELLATES

Plate 103

Division Chromophyta
Class Dictyochophyceae
Order Dictyochales
Family Dictyochaceae

Dictyocha fibula Ehrenberg, 1839

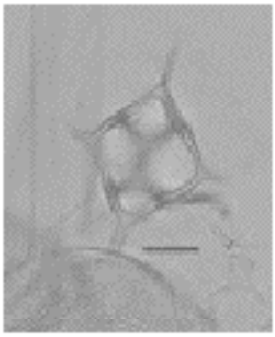


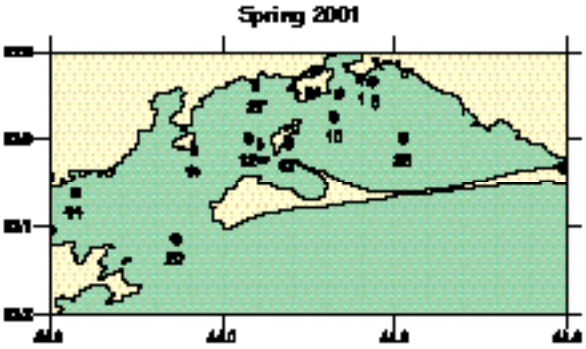
Figure 1. Skeleton with four protruding spines and four "windows" (LM, bright field, water mount). Scale bar = 10µm.

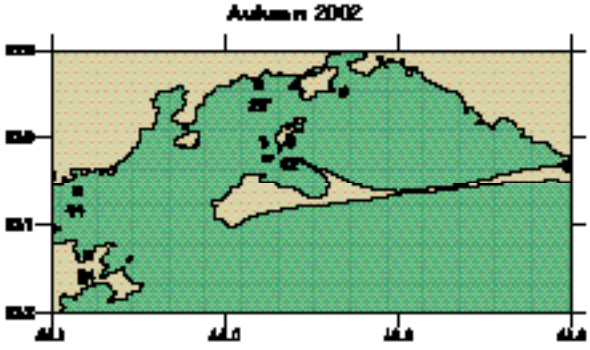
Taxonomic source

Throndsen, J. (1997). The planktonic marine flagellates (p. 632). In: Identifying marine phytoplankton. Tomas, C. R. (ed). Academic Press, San Diego.

Morphometrics

	range (µm)	mean ± SD	n
TL	30 - 60	43 ± 7	40
D	20 - 35	28 ± 4	40





APPENDIX 1

Complete list of the phytoplankton taxa found in Sepetiba Bay.

spring Bottle Net

autumn Bottle Net

A - stands for abundant: taxon that reached more than the average cell number in a given sample.
D - stands for dominant: taxon that reached more than 50% of the cell number in a given sample.

spring

sampling sites	1	6	10	12	14	20	24	27	29	31	38	42	44
DIVISION BACILLARIOPHYTA													
Class Coscinodiscophyceae													
Order Thalassiosirales													
Family Thalassiosiraceae													
Thalassiosira punctigera (Castracane) Hasle													
Thalassiosira rotula Meunier													
Thalassiosira sp.1													
Thalassiosira sp.2													
Thalassiosira sp.3													
Thalassiosira sp.4													
Thalassiosira sp.5													
Family Skeletonemataceae													
Detonula pumila (Castracane) Schütt													
Skeletonema costatum (Greville) Cleve													
Family Stephanodiscaceae													
Cyclotella litoralis Lange & Syvertsen + *C. stylorum* Brightwell													
Order Melosirales													
Family Stephanopyxidaceae													
Stephanopyxis turris (Greville & Arnott) Ralfs													
Order Paraliales													
Family Paraliaceae													
Paralia sulcata (Ehrenberg) Cleve													
Order Coscinodiscales													
Family Coscinodiscaceae													
Coscinodiscus cf. *centralis* Ehrenberg													
Coscinodiscus granii Gough													
Coscinodiscus cf. *perforatus* Ehrenberg													
Coscinodiscus wailesii Gran & Angst													
Coscinodiscus gigas Gough + *C.* cf. *asteromphalus* Ehrenberg	A	A	A		A				A	A		A	
Coscinodiscus sp.													
Family Heliopeltaceae													
Actinoptychus senarius (Ehrenberg) Ehrenberg													
Order Asterolamprales													
Family Asterolampraceae													
Asteromphalus cf. *flabellatus* (Brébisson) Greville													
Order Triceratiales													
Family Triceratiaceae													
Odontella mobiliensis (Bailey) Grunow													
Odontella regia (Schultze) Simonsen													
Odontella sinensis (Greville) Grunow	A		A	A	A				A	A		A	A
Triceratium favus Ehrenberg													
Order Biddulphiales													
Family Biddulphiaceae													
Biddulphia rhombus (Ehrenberg) W. Smith													
Order Hemiaulales													
Family Hemiaulaceae													
Cerataulina cf. *bicornis* (Ehrenberg) Hasle													
Cerataulina cf. *dentata* Hasle				A									
Cerataulina pelagica (Cleve) Hendey													
Cerataulina sp.													
Climacodium frauenfeldianum Grunow													
Eucampia cornuta (Cleve) Grunow													
Eucampia zodiacus Ehrenberg													
Hemiaulus hauckii Grunow													
Hemiaulus membranaceus Cleve													
Hemiaulus sinensis Greville													
Family Bellerocheaceae													
Bellerochea horologicalis von Stosch													
Order Lithodesmiales													
Family Lithodesmiaceae													
Ditylum brightwellii (T. West) Grunow													
Helicotheca tamesis (Shrubsole) Ricard													
Lithodesmium undulatum Ehrenberg													
Order Corethrales													
Family Corethraceae													
Corethron criophilum Castracane													
Corethron sp.													

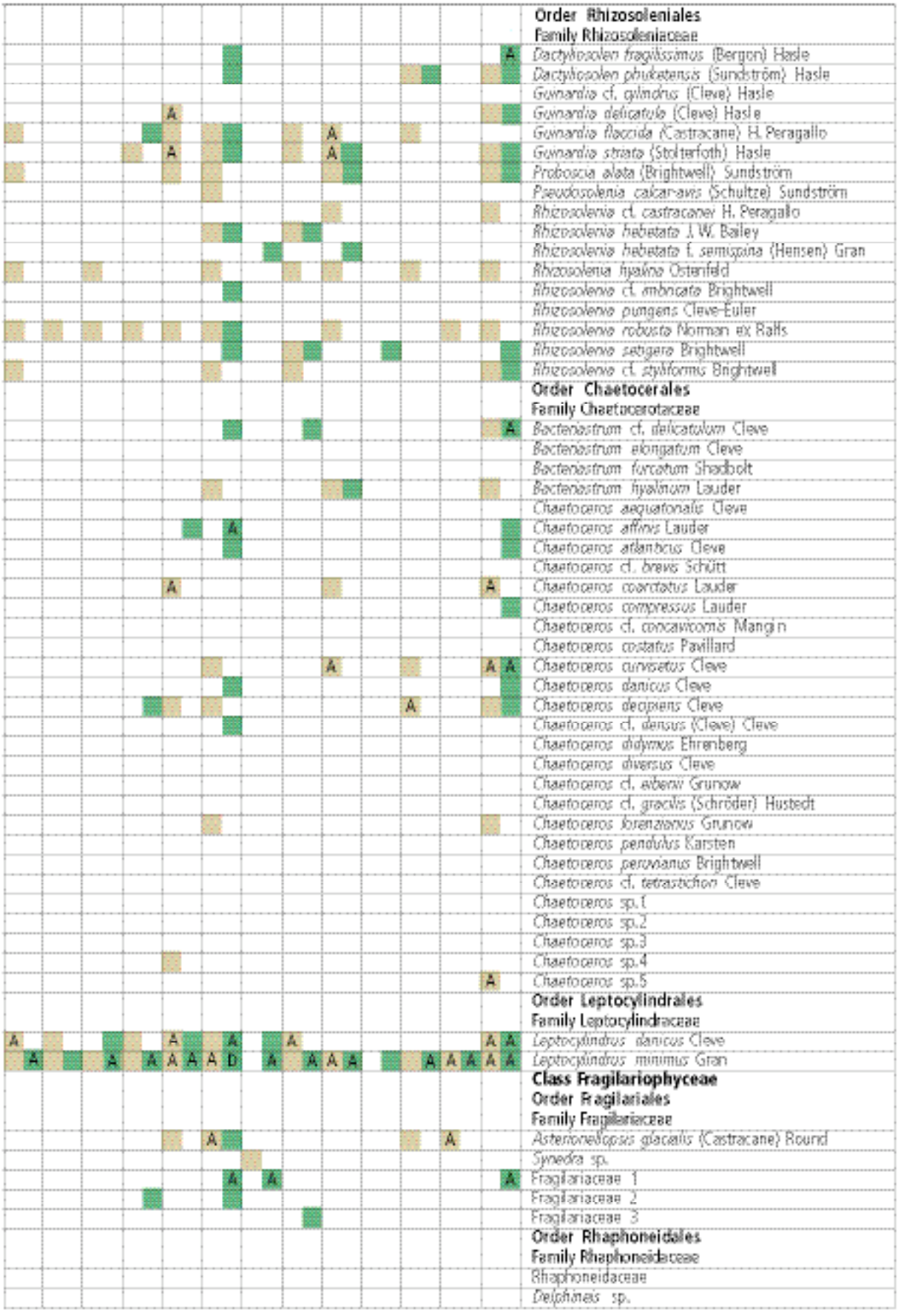
autumn

1	6	10	12	14	20	24	27	29	31	38	42	44	sampling sites
													DIVISION BACILLARIOPHYTA
													Class Coscinodiscophyceae
													Order Thalassiosirales
													Family Thalassiosiraceae
	▫	▫			▫	■	▫	■			▫				▫				*Thalassiosira punctigera* (Castracane) Hasle
▫										▫									*Thalassiosira rotula* Meunier
						■													*Thalassiosira* sp.1
						■													*Thalassiosira* sp.2
						■				■								■	*Thalassiosira* sp.3
						■													*Thalassiosira* sp.4
												▫							*Thalassiosira* sp.5
																			Family Skeletonemataceae
																			Detonula pumila (Castracane) Schütt
								A				A							*Skeletonema costatum* (Greville) Cleve
																			Family Stephanodiscaceae
▫	▫	▫	▫	A	A	A	A	A	▫	▫	▫	A							*Cyclotella litoralis* Lange & Syvertsen + *C.stylorum* Brightwell
																			Order Melosirales
																			Family Stephanopyxidaceae
																			Stephanopyxis turris (Greville & Arnott) Ralfs
																			Order Paraliales
																			Family Paraliaceae
▫				■	▫		▫	A	▫	■	▫	■	▫			▫				*Paralia sulcata* (Ehrenberg) Cleve
																			Order Coscinodiscales
																			Family Coscinodiscaceae
																			Coscinodiscus cf. *centralis* Ehrenberg
																		▫	*Coscinodiscus granii* Gough
				■														■	*Coscinodiscus* cf. *perforatus* Ehrenberg
					▫			▫	■										*Coscinodiscus wailesii* Gran & Angst
▫	▫	▫			▫	▫	▫	▫		▫	▫	▫	▫						*Coscinodiscus gigas* Gough + *C.cf. asteromphalus* Ehrenberg
									■										*Coscinodiscus* sp.
																			Family Heliopeltaceae
											▫								*Actinoptychus senarius* (Ehrenberg) Ehrenberg
																			Order Asterolamprales
																			Family Asterolamproaceae
▫													▫						*Asteromphalus* cf. *flabellatus* (Brébisson) Greville
																			Order Triceratiales
																			Family Triceratiaceae
						■		A	■	▫		▫							*Odontella mobiliensis* (Bailey) Grunow
																			Odontella regia (Schultze) Simonsen
▫			▫		▫			▫											*Odontella sinensis* (Greville) Grunow
							▫												*Triceratium favus* Ehrenberg
																			Order Biddulphiales
																			Family Biddulphiaceae
																			Biddulphia rhombus (Ehrenberg) W. Smith
																			Order Hemiaulales
																			Family Hemiaulaceae
							■				■							■	*Cerataulina* cf. *bicornis* (Ehrenberg) Hasle
																			Cerataulina cf. *dentata* Hasle
▫			▫	▫	A	■	▫	A			■	▫			▫		A	■	*Cerataulina pelagica* (Cleve) Hendey
								■											*Cerataulina* sp.
						▫		■									▫	■	*Climacodium frauenfeldianum* Grunow
					▫	▫											▫		*Eucampia cornuta* (Cleve) Grunow
																			Eucampia zodiacus Ehrenberg
				■		▫											▫	A	*Hemiaulus hauckii* Grunow
		▫	▫	▫	▫		▫	▫									A		*Hemiaulus membranaceus* Cleve
			▫	▫	▫	■	▫	▫									A		*Hemiaulus sinensis* Greville
																			Family Bellerocheaceae
								■											*Bellerochea horologicalis* von Stosch
																			Order Lithodesmiales
																			Family Lithodesmiaceae
																			Ditylum brightwellii (T.West) Grunow
	▫			▫			▫												*Helicotheca tamesis* (Shrubsole) Ricard
																			Lithodesmium undulatum Ehrenberg
																			Order Corethrales
																			Family Corethraceae
					▫		■	■										■	*Corethron criophilum* Castracane
▫		▫			▫			▫									▫		*Corethron* sp.

spring

sampling sites	1	6	10	12	14	20	24	27	29	31	38	42	44
Order Rhizosoleniales													
Family Rhizosoleniaceae													
Dactyliosolen fragilissimus (Bergon) Hasle	A			A		A		A					A
Dactyliosolen phuketensis (Sundström) Hasle	A			A A	A A	A	A D	A A	A A	A		A	A A
Guinardia cf. *cylindrus* (Cleve) Hasle													
Guinardia delicatula (Cleve) Hasle				A	A			A	A	A	A	A A	
Guinardia flaccida (Castracane) H. Peragallo									A				
Guinardia striata (Stolterfoth) Hasle					A								
Proboscia alata (Brightwell) Sundström													
Pseudosolenia calcar-avis (Schultze) Sundström												A	
Rhizosolenia cf. *castracanei* H. Peragallo													
Rhizosolenia hebetata Bailey													
Rhizosolenia hebetata f. *semispina* (Hensen) Gran													
Rhizosolenia hyalina Ostenfeld													
Rhizosolenia cf. *imbricata* Brigthwell													
Rhizosolenia pungens Cleve-Euler												A	
Rhizosolenia robusta Norman ex Ralfs													
Rhizosolenia setigera Brightwell											A	A	
Rhizosolenia cf. *styliformis* Brightwell													
Order Chaetocerales													
Family Chaetocerotaceae													
Bacteriastrum cf. *delicatulum* Cleve				A									
Bacteriastrum elongatum Cleve													
Bacteriastrum furcatum Shadbolt													
Bacteriastrum hyalinum Lauder										A			A A
Chaetoceros aequatorialis Cleve													
Chaetoceros affinis Lauder													A
Chaetoceros atlanticus Cleve													
Chaetoceros cf. *brevis* Schütt					A	A			A	A	A		
Chaetoceros coarctatus Lauder													
Chaetoceros compressus Lauder													
Chaetoceros cf. *concavicornis* Mangin													A
Chaetoceros costatus Pavillard													
Chaetoceros curvisetus Cleve		A	A		A	A A	A		A	A	A		A
Chaetoceros danicus Cleve										A			
Chaetoceros decipiens Cleve									A				A
Chaetoceros cf. *densus* Cleve													
Chaetoceros didymus Ehrenberg		A	A		A	A			A	A	A	A	
Chaetoceros diversus Cleve													
Chaetoceros cf. *elbenii* Grunow													
Chaetoceros cf. *gracilis* (Schröder) Hustedt													
Chaetoceros lorenzianus Grunow													
Chaetoceros pendulus Karsten													
Chaetoceros peruvianus Brightwell					A	A	A						
Chaetoceros cf. *tetrastichon* Cleve													
Chaetoceros sp.1	A	A	A	A	A	A			A	A		A	
Chaetoceros sp.2													
Chaetoceros sp.3													
Chaetoceros sp.4													
Chaetoceros sp.5													
Order Leptocylindrales													
Family Leptocylindraceae													
Leptocylindrus danicus Cleve	A			A A		A A	A A	A A	A A	A A		A A	A
Leptocylindrus minimus Gran	A A	A A	A	A A	A	A	A	A	A	A A		A	A A
Class Fragilariophyceae													
Order Fragilariales													
Family Fragilariaceae													
Asterionellopsis glacialis (Castracane) Round					A	A							
Synedra sp.													
Fragilariaceae 1													
Fragilariaceae 2													
Fragilariaceae 3													
Order Rhaphoneidales													
Family Rhaphoneidaceae													
Rhaphoneidaceae					A				A				
Delphineis sp.													

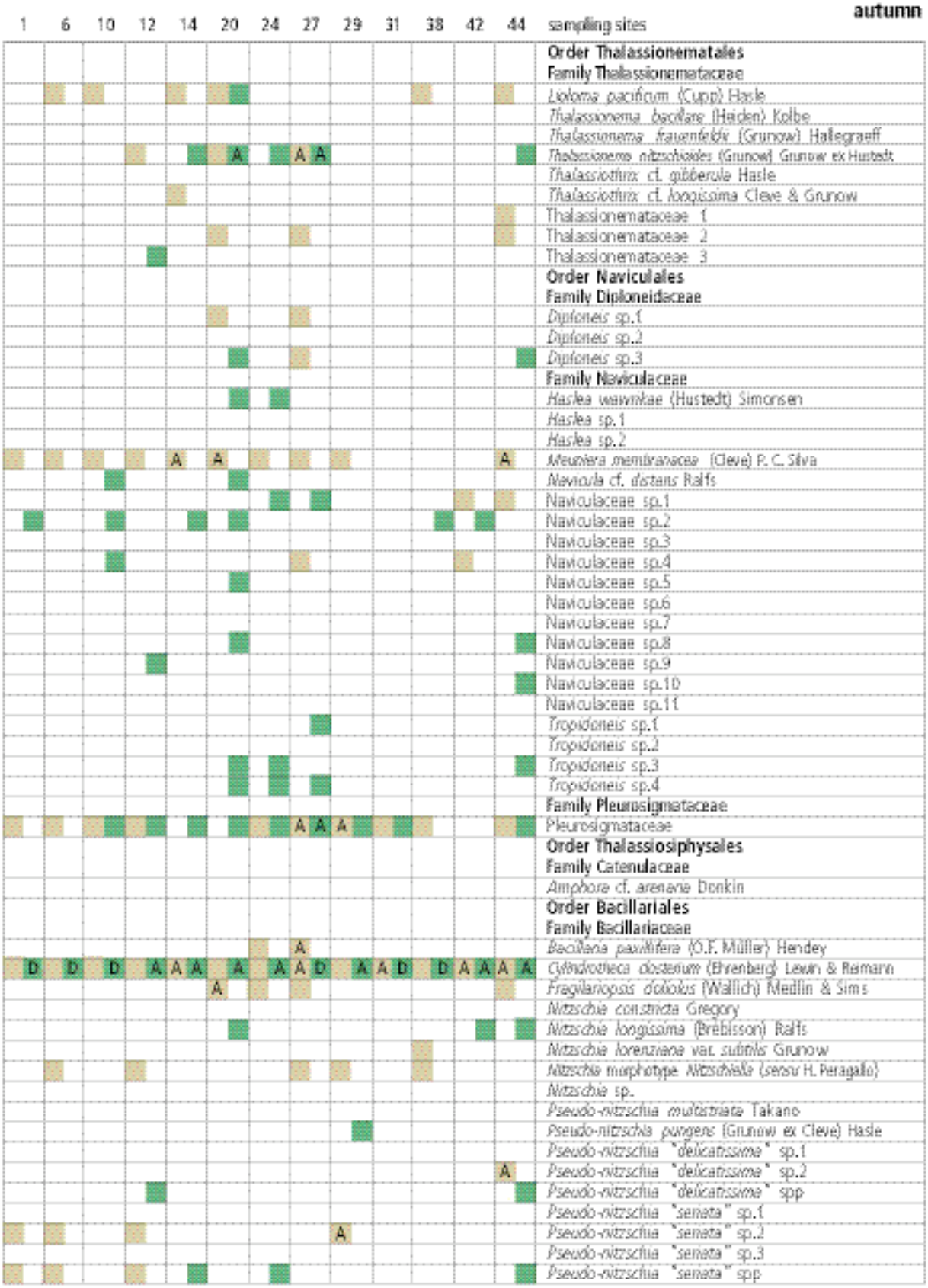
autumn


1
6
10
12
14
20
24
27
29
31
38
42
44
sampling sites
Order Rhizosoleniales
Family Rhizosoleniaceae
Dactyliosolen fragilissimus (Bergon) Hasle
Dactyliosolen phuketensis (Sundström) Hasle
Guinardia cf. cylindrus (Cleve) Hasle
Guinardia delicatula (Cleve) Hasle
Guinardia flaccida (Castracane) H. Peragallo
Guinardia striata (Stolterfoth) Hasle
Proboscia alata (Brightwell) Sundström
Pseudosolenia calcar-avis (Schultze) Sundström
Rhizosolenia cf. castracanei H. Peragallo
Rhizosolenia hebetata J.W. Bailey
Rhizosolenia hebetata f. semispina (Hensen) Gran
Rhizosolenia hyalina Ostenfeld
Rhizosolenia cf. imbricata Brightwell
Rhizosolenia pungens Cleve-Euler
Rhizosolenia robusta Norman ex Ralfs
Rhizosolenia setigera Brightwell
Rhizosolenia cf. styliformis Brightwell
Order Chaetocerales
Family Chaetocerotaceae
Bacteriastrum cf. delicatulum Cleve
Bacteriastrum elongatum Cleve
Bacteriastrum furcatum Shadbolt
Bacteriastrum hyalinum Lauder
Chaetoceros aequatorialis Cleve
Chaetoceros affinis Lauder
Chaetoceros atlanticus Cleve
Chaetoceros cf. brevis Schütt
Chaetoceros coarctatus Lauder
Chaetoceros compressus Lauder
Chaetoceros cf. concavicornis Mangin
Chaetoceros costatus Pavillard
Chaetoceros curvisetus Cleve
Chaetoceros danicus Cleve
Chaetoceros decipiens Cleve
Chaetoceros cf. densus (Cleve) Cleve
Chaetoceros didymus Ehrenberg
Chaetoceros diversus Cleve
Chaetoceros cf. eibenii Grunow
Chaetoceros cf. gracilis (Schröder) Hustedt
Chaetoceros lorenzianus Grunow
Chaetoceros pendulus Karsten
Chaetoceros peruvianus Brightwell
Chaetoceros cf. tetrastichon Cleve
Chaetoceros sp.1
Chaetoceros sp.2
Chaetoceros sp.3
Chaetoceros sp.4
Chaetoceros sp.5
Order Leptocylindrales
Family Leptocylindraceae
Leptocylindrus danicus Cleve
Leptocylindrus minimus Gran
Class Fragilariophyceae
Order Fragilariales
Family Fragilariaceae
Asterionellopsis glacialis (Castracane) Round
Synedra sp.
Fragilariaceae 1
Fragilariaceae 2
Fragilariaceae 3
Order Rhaphoneidales
Family Rhaphoneidaceae
Rhaphoneidaceae
Delphineis sp.

spring

sampling sites	1	6	10	12	14	20	24	27	29	31	38	42	44
Order Thalassionematales													
Family Thalassionemataceae													
Lioloma pacificum (Cupp) Hasle		A	A	A	A	A	A		A	A	A	A	
Thalassionema bacillare (Heiden) Kolbe													
Thalassionema frauenfeldii (Grunow) Hallegraeff													
Thalassionema nitzschioides (Grunow) Grunow ex Hustedt		A											
Thalassiothrix cf. *gibberula* Hasle													
Thalassiothrix cf. *longissima* Cleve & Grunow													
Thalassionemataceae 1		A	A	A					A		A	A	
Thalassionemataceae 2													
Thalassionemataceae 3													
Order Naviculales													
Family Diploneidaceae													
Diploneis sp.1													
Diploneis sp.2													
Diploneis sp.3													
Family Naviculaceae													
Hasleа wawrikae (Hustedt) Simonsen					A		A		A	A		A	
Hasleа sp.1													
Hasleа sp.2													
Meuniera membranacea (Cleve) P.C. Silva													
Navicula cf. *distans* (W. Smith) Ralfs													
Naviculaceae sp.1													
Naviculaceae sp.2													
Naviculaceae sp.3													
Naviculaceae sp.4													
Naviculaceae sp.5													
Naviculaceae sp.6													
Naviculaceae sp.7													
Naviculaceae sp.8													
Naviculaceae sp.9													
Naviculaceae sp.10													
Naviculaceae sp.11													
Tropidoneis sp.1													
Tropidoneis sp.2													
Tropidoneis sp.3													
Tropidoneis sp.4													
Family Pleurosigmataceae													
Pleurosigmataceae							A						
Order Thalassiosiphysales													
Family Catenulaceae													
Amphora cf. *arenaria* Donkin													
Order Bacillariales													
Family Bacillariaceae													
Bacillaria paxillifera (O.F. Müller) Hendey					A	A	A						
Cylindrotheca closterium (Ehrenberg) Lewin & Reimann	A	A	A	A	A	A	A	A	A	A	A	A	A
Fragilariopsis doliolus (Wallich) Medlin & Sims													
Nitzschia constricta Gregory													
Nitzschia longissima (Brébisson) Ralfs													
Nitzschia lorenziana var. *subtilis* Grunow													
Nitzschia morphotype *Nitzschiella* Lange-H. Peragallo													
Nitzschia sp.													
Pseudo-nitzschia multistriata Takano													
Pseudo-nitzschia pungens (Grunow ex Cleve) Hasle													
Pseudo-nitzschia "delicatissima" sp.1					A							A	
Pseudo-nitzschia "delicatissima" sp.2		A			A	A				A	A	A	
Pseudo-nitzschia "delicatissima" spp											A	A	
Pseudo-nitzschia "seriata" sp.1													
Pseudo-nitzschia "seriata" sp.2		A			A	A							
Pseudo-nitzschia "seriata" sp.3													
Pseudo-nitzschia "seriata" spp	A		A	A			A		A			A	

autumn


1
6
10
12
14
20
24
27
29
31
38
42
44
sampling sites
Order Thalassionematales
Family Thalassionemataceae
Lioloma pacificum (Cupp) Hasle
Thalassionema bacillare (Heiden) Kolbe
Thalassionema frauenfeldii (Grunow) Hallegraeff
Thalassionema nitzschioides (Grunow) Grunow ex Hustedt
Thalassiothrix cf. gibberula Hasle
Thalassiothrix cf. longissima Cleve & Grunow
Thalassionemataceae 1
Thalassionemataceae 2
Thalassionemataceae 3
Order Naviculales
Family Diploneidaceae
Diploneis sp.1
Diploneis sp.2
Diploneis sp.3
Family Naviculaceae
Haslea wawrikae (Hustedt) Simonsen
Haslea sp.1
Haslea sp.2
Meuniera membranacea (Cleve) P. C. Silva
Navicula cf. distans Ralfs
Naviculaceae sp.1
Naviculaceae sp.2
Naviculaceae sp.3
Naviculaceae sp.4
Naviculaceae sp.5
Naviculaceae sp.6
Naviculaceae sp.7
Naviculaceae sp.8
Naviculaceae sp.9
Naviculaceae sp.10
Naviculaceae sp.11
Tropidoneis sp.1
Tropidoneis sp.2
Tropidoneis sp.3
Tropidoneis sp.4
Family Pleurosigmataceae
Pleurosigmataceae
Order Thalassiosiphysales
Family Catenulaceae
Amphora cf. arenaria Donkin
Order Bacillariales
Family Bacillariaceae
Bacillaria paxillifera (O.F. Müller) Hendey
Cylindrotheca closterium (Ehrenberg) Lewin & Reimann
Fragilariopsis doliolus (Wallich) Medlin & Sims
Nitzschia constricta Gregory
Nitzschia longissima (Brébisson) Ralfs
Nitzschia lorenziana var. subtilis Grunow
Nitzschia morphotype Nitzschiella (sensu H. Peragallo)
Nitzschia sp.
Pseudo-nitzschia multistriata Takano
Pseudo-nitzschia pungens (Grunow ex Cleve) Hasle
Pseudo-nitzschia "delicatissima" sp.1
Pseudo-nitzschia "delicatissima" sp.2
Pseudo-nitzschia "delicatissima" spp
Pseudo-nitzschia "seriata" sp.1
Pseudo-nitzschia "seriata" sp.2
Pseudo-nitzschia "seriata" sp.3
Pseudo-nitzschia "seriata" spp

spring

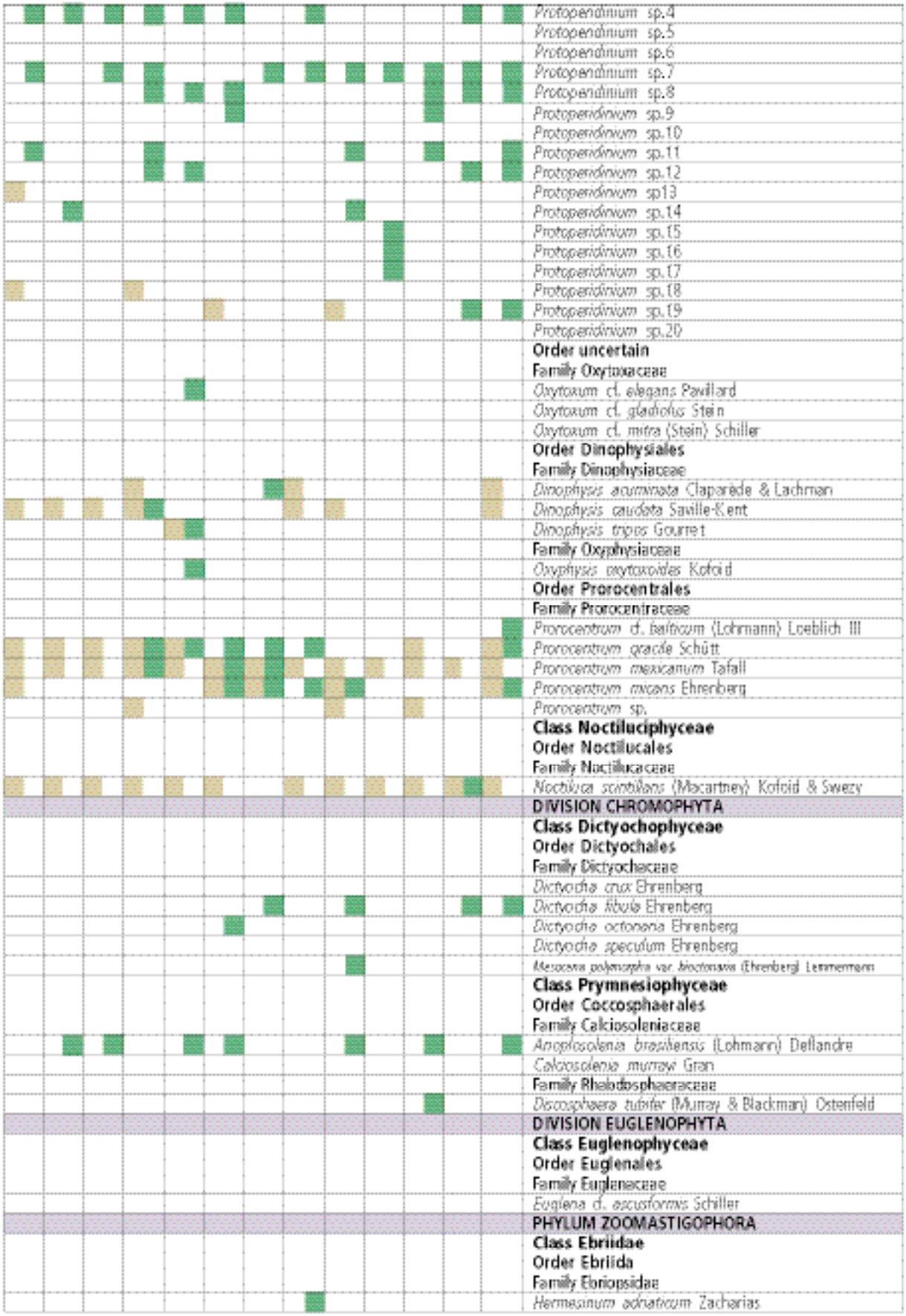
sampling sites	1	6	10	12	14	20	24	27	29	31	38	42	44
DIVISION DINOFLAGELLATA													
Class Dinophyceae													
Order Gymnodiniales													
Family Gymnodiniaceae													
Gymnodinium sp.1													
Gymnodinium sp.2													
Gyrodinium sp.1													
Gyrodinium sp.2													
Gyrodinium sp.3													
Order Gonyaulacales													
Family Gonyaulacaceae													
Gonyaulax spinifera (Claparède & Lachmann) Diesing													
Gonyaulax sp.1													
Gonyaulax sp.2													
Gonyaulax sp.3													
Family Ceratocoryaceae													
cf. *Ceratocorys*													
Family Ceratiaceae													
Ceratium breve (Ostenfeld & Schmidt) Schröder													
Ceratium cf. *candelabrum* (Ehrenberg) Stein													
Ceratium cf. *contortum* var. *kastenii* (Pavillard) Sournia													
Ceratium cf. *contrarium* (Gourret) Pavillard													
Ceratium deflexum (Kofoid) Jörgensen													
Ceratium furca (Ehrenberg) Claparède & Lachmann													
Ceratium fusus (Ehrenberg) Dujardin													
Ceratium gibberum Gourret													
Ceratium cf. *hexacanthum* Gourret													
Ceratium horridum (Cleve) Gran					A								A A
Ceratium humile Jörgensen													
Ceratium cf. *lunula* (Schimper) Jörgensen													
Ceratium macroceros (Ehrenberg) VanHöffen													
Ceratium massiliense (Gourret) Jörgensen													A
Ceratium cf. *trichoceros* (Ehrenberg) Kofoid													
Ceratium cf. *tripos* (O.F.Müller) Nitzsch													
Ceratium sp.1													
Ceratium sp.2													
Ceratium sp.3													
Family Goniodomaceae													
Alexandrium cf. *fraterculus* (Balech) Balech					A								
Alexandrium sp.													
Family Pyrocystaceae													
Pyrocystis cf. *fusiformis* (W.Thompson) Murray													
Pyrocystis lunula (Schütt) Schütt													
Order Peridiniales													
Family Peridiniaceae													
Scrippsiella spinifera Honsell & Cabrini	A	A	A	A	A	A	A						A A
Scrippsiella trochoidea (Stein) Loeblich III		A	A	A	A	A	A						
Family Congruentidiaceae													
Protoperidinium cf. *conicum* (Gran) Balech													
Protoperidinium cf. *crassipes* (Kofoid) Balech													
Protoperidinium cf. *depressum* (Bailey) Balech													
Protoperidinium divergens (Ehrenberg) Balech													
Protoperidinium cf. *elegans* (Cleve) Balech													
Protoperidinium cf. *grande* (Kofoid) Balech													
Protoperidinium cf. *hirobis* Abé													
Protoperidinium cf. *inflatum* Okamura													
Protoperidinium leonis (Pavillard) Balech													
Protoperidinium mangelethoumae (Paulsen) Balech													
Protoperidinium oblongum (Aurivillius) Parke & Dodge													
Protoperidinium cf. *oceanicum* (VanHöffen) Balech													
Protoperidinium cf. *ovatum* Pouchet													
Protoperidinium pannventer Balech													
Protoperidinium cf. *pellucidum* Bergh													
Protoperidinium cf. *pentagonum* (Gran) Balech													
Protoperidinium roseum (Paulsen) Paulsen													
Protoperidinium simulum (Paulsen) Balech					A								
Protoperidinium steinii (Jörgensen) Balech													
Protoperidinium symmetricum (Halim) Balech													
Protoperidinium variegatum Peters													
Protoperidinium sp.1													
Protoperidinium sp.2													
Protoperidinium sp.3													

autumn

1	6	10	12	14	20	24	27	29	31	38	42	44	sampling sites
													DIVISION DINOFLAGELLATA
													Class Dinophyceae
													Order Gymnodiniales
													Family Gymnodiniaceae
	■												*Gymnodinium* sp.1
													Gymnodinium sp.2
	■												*Gyrodinium* sp.1
													Gyrodinium sp.2
								■					*Gyrodinium* sp.3
													Order Gonyaulacales
													Family Gonyaulacaceae
		■											*Gonyaulax spinifera* (Claparède & Lachmann) Diesing
		▫											*Gonyaulax* sp.1
									■				*Gonyaulax* sp.2
									■				*Gonyaulax* sp.3
													Family Ceratocoryaceae
				■									cf. *Ceratocorys*
													Family Ceratiaceae
		▫		▫	▫		■		▫		▫		*Ceratium breve* (Ostenfeld & Schmidt) Schröder
	▫											▫	*Ceratium* cf. *candelabrum* (Ehrenberg) Stein
													Ceratium contortum var. *karstenii* (Pavillard) Sournia
	▫			▫									*Ceratium* cf. *contrarium* (Gourret) Pavillard
													Ceratium deflexum (Kofoid) Jörgensen
A	A	A	■	A	A	■	■	A	■	▫	▫	A	*Ceratium furca* (Ehrenberg) Claparède & Lachmann
A	■	A	■	A	■	■	■	A	■	▫	▫	■	*Ceratium fusus* (Ehrenberg) Dujardin
												▫	*Ceratium gibberum* Gourret
					▫								*Ceratium* cf. *hexacanthum* Gourret
▫	▫	▫	▫	▫	A	▫	▫	▫		▫		A	*Ceratium horridum* (Cleve) Gran
			▫										*Ceratium humile* Jörgensen
													Ceratium cf. *lunula* (Schimper) Jörgensen
						■							*Ceratium macroceros* (Ehrenberg) Vanhöffen
					▫							▫	*Ceratium massiliense* (Gourret) Jörgensen
							▫					▫	*Ceratium* cf. *trichoceros* (Ehrenberg) Kofoid
													Ceratium cf. *tripos* (O.F.Müller) Nitzsch
												■	*Ceratium* sp.1
													Ceratium sp.2
						▫	▫					▫	*Ceratium* sp.3
													Family Goniodomaceae
			▫	▫				▫				▫	*Alexandrium* cf. *fraterculus* (Balech) Balech
												■	*Alexandrium* sp.
													Family Pyrocystaceae
													Pyrocystis cf. *fusiformis* (W.Thompson) Murray
					▫								*Pyrocystis lunula* (Schütt) Schütt
													Order Peridiniales
													Family Peridiniaceae
A	A	D	A	D		D	D	A	A	A	A	A	A	A		A	A	A		D	A	A			A	*Scrippsiella spinifera* Horsell & Cabrini
	A	A	A	A		A	D	A	A	A	A	A	A	A			A			A	A			A	A	*Scrippsiella trochoidea* (Stein) Loeblich III
																										Family Congruentidiaceae
				■		■																				*Protoperidinium* cf. *conicum* (Gran) Balech
												■														*Protoperidinium* cf. *crassipes* (Kofoid) Balech
▫		▫	■	▫		▫		▫	▫		▫		▫													*Protoperidinium* cf. *depressum* (Bailey) Balech
				■																						*Protoperidinium divergens* (Ehrenberg) Balech
																										Protoperidinium cf. *elegans* (Cleve) Balech
▫																										*Protoperidinium* cf. *grande* (Kofoid) Balech
			■																							*Protoperidinium* cf. *hirobis* Abé
		■																								*Protoperidinium* cf. *inflatum* Okamura
	▫		▫					▫			▫															*Protoperidinium leonis* (Pavillard) Balech
																										Protoperidinium mariaelebourae (Paulsen) Balech
												▫														*Protoperidinium oblongum* (Aurivillius) Parke & Dodge
																										Protoperidinium cf. *oceanicum* (VanHöffen) Balech
	■																									*Protoperidinium* cf. *ovatum* Pouchet
	▫	▫	▫	▫				▫	▫		▫		▫													*Protoperidinium perriventer* Balech
										■																*Protoperidinium* cf. *pellucidum* Bergh
		▫	▫	▫				▫																		*Protoperidinium* cf. *pentagonum* (Gran) Balech
	■																									*Protoperidinium roseum* (Paulsen) Paulsen
			▫		▫						▫													A		*Protoperidinium simulum* (Paulsen) Balech
A		A	■	A	A		▫			▫	A		▫													*Protoperidinium steinii* (Jörgensen) Balech
																										Protoperidinium symmetricum (Halim) Balech
																								■		*Protoperidinium variegatum* Peters
																										Protoperidinium sp.1
▫		▫	▫	▫		▫	▫	▫	▫	▫	▫	▫														*Protoperidinium* sp.2
																										Protoperidinium sp.3

spring	sampling sites	1	6	10	12	14	20	24	27	29	31	38	42	44
Protoperidinium sp.4														
Protoperidinium sp.5														
Protoperidinium sp.6														
Protoperidinium sp.7														
Protoperidinium sp.8														
Protoperidinium sp.9														
Protoperidinium sp.10														
Protoperidinium sp.11			A											
Protoperidinium sp.12														
Protoperidinium sp.13														
Protoperidinium sp.14														
Protoperidinium sp.15														
Protoperidinium sp.16														
Protoperidinium sp.17														
Protoperidinium sp.18														
Protoperidinium sp.19														
Protoperidinium sp.20														
Order uncertain														
Family Oxytoxaceae														
Oxytoxum cf. *elegans* Pavillard														
Oxytoxum cf. *gladiolus* Stein														
Oxytoxum cf. *mitra* (Stein) Schiller														
Order Dinophysiales														
Family Dinophysiaceae														
Dinophysis acuminata Claparède & Lachman														
Dinophysis caudata Saville-Kent														
Dinophysis tripos Gourret														
Family Oxyphysiaceae														
Oxyphysis oxytoxoides Kofoid														
Order Prorocentrales														
Family Prorocentraceae														
Prorocentrum cf. *balticum* (Lohmann) Loeblich III														
Prorocentrum gracile Schütt														
Prorocentrum mexicanum Tafall														
Prorocentrum micans Ehrenberg														
Prorocentrum sp.														
Class Noctiluciphyceae														
Order Noctilucales														
Family Noctilucaceae														
Noctiluca scintillans (Macartney) Kofoid & Swezy														
DIVISION CHROMOPHYTA														
Class Dictyochophyceae														
Order Dictyochales														
Family Dictyochaceae														
Dictyocha crux Ehrenberg														
Dictyocha fibula Ehrenberg														
Dictyocha octonaria Ehrenberg														
Dictyocha speculum Ehrenberg														
Mesocena polymorpha var. *bicotonis* (Ehrenberg) Lemmermann														
Class Prymnesiophyceae														
Order Coccosphaerales														
Family Calciosoleniaceae														
Anoplosolenia brasiliensis (Lohmann) Deflandre														
Calciosolenia murrayi Gran in Murray & Hjort														
Family Rhabdosphaeraceae														
Discosphaera tubifer (Murray & Blackman) Ostenfeld														
DIVISION EUGLENOPHYTA														
Class Euglenophyceae														
Order Euglenales														
Family Euglenaceae														
Euglena cf. *acusformis* Schiller														
PHYLUM ZOOMASTIGOPHORA														
Class Ebriidae														
Order Ebriida														
Family Ebriopsidae														
Hermesinum adriaticum Zacharias														

autumn


1
6
10
12
14
20
24
27
29
31
38
42
44
sampling sites
Protopendinium sp.4
Protopendinium sp.5
Protopendinium sp.6
Protopendinium sp.7
Protopendinium sp.8
Protopendinium sp.9
Protopendinium sp.10
Protopendinium sp.11
Protopendinium sp.12
Protopendinium sp13
Protopendinium sp.14
Protopendinium sp.15
Protopendinium sp.16
Protopendinium sp.17
Protopendinium sp.18
Protopendinium sp.19
Protopendinium sp.20
Order uncertain
Family Oxytoxaceae
Oxytoxum cf. elegans Pavillard
Oxytoxum cf. gladiolus Stein
Oxytoxum cf. mitra (Stein) Schiller
Order Dinophysiales
Family Dinophysiaceae
Dinophysis acuminata Claparède & Lachman
Dinophysis caudata Saville-Kent
Dinophysis tripos Gourret
Family Oxyphysiaceae
Oxyphysis oxytoxoides Kofoid
Order Prorocentrales
Family Prorocentraceae
Prorocentrum cf. balticum (Lohmann) Loeblich III
Prorocentrum gracile Schütt
Prorocentrum mexicanum Tafall
Prorocentrum micans Ehrenberg
Prorocentrum sp.
Class Noctiluciphyceae
Order Noctilucales
Family Noctilucaceae
Noctiluca scintillans (Macartney) Kofoid & Swezy
DIVISION CHROMOPHYTA
Class Dictyochophyceae
Order Dictyochales
Family Dictyochaceae
Dictyocha crux Ehrenberg
Dictyocha fibula Ehrenberg
Dictyocha octonaria Ehrenberg
Dictyocha speculum Ehrenberg
Mesocena polymorpha var. bioctonaria (Ehrenberg) Lemmermann
Class Prymnesiophyceae
Order Coccosphaerales
Family Calciosoleniaceae
Anoplosolenia brasiliensis (Lohmann) Deflandre
Calciosolenia murrayi Gran
Family Rhabdosphaeraceae
Discosphaera tubifer (Murray & Blackman) Ostenfeld
DIVISION EUGLENOPHYTA
Class Euglenophyceae
Order Euglenales
Family Euglenaceae
Euglena cf. oxyuriformis Schiller
PHYLUM ZOOMASTIGOPHORA
Class Ebriidae
Order Ebriida
Family Ebriopsidae
Hermesinum adriaticum Zacharias

More Information?

Programme Coordination Unit
Global Ballast Water Management Programme
International Maritime Organization
4 Albert Embankment
London SE1 7SR United Kingdom

Tel: +44 (0)20 7587 3247 or 3251
Fax: +44 (0)20 7587 3261
Web: http://globallast.imo.org

Denise Rivera Tenenbaum
Departamento de Biologia Marinha
Instituto de Biologia
Universidade Federal do Rio de Janeiro
CCS-A Ilha do Fundão, 21944-970
Rio de Janeiro - RJ, Brasil

Tel: +55 (21) 25626305
Fax: +55 (21) 22802394
deniser@biologia.ufrj.br